

Long-time dynamics for the magnetic Schrödinger equation on the torus

Julien LECHAUX

September 2, 2026

Abstract

We investigate time-dependent semiclassical measures associated with the magnetic Schrödinger flow on the flat two-dimensional torus in long-time regimes. Under a geometric nonvanishing assumption on an effective magnetic force, we identify two thresholds and prove a dynamical form of quantum unique ergodicity for such systems. If the time scale $\tau_h \gg h^{-1/2}$, then the configuration marginal is the normalized Lebesgue measure on $\mathbb{T}^2$. In addition, if $\tau_h \gg h^{-1}$, then the momentum marginal is, conditionally on each regular level set, the normalized invariant measure on that curve.

Contents

1	Introduction	2
1.1	The semiclassical context	4
1.2	Relation to previous works	8
1.3	Strategy of the proof	11
1.4	Structure of the paper	14
2	Magnetic field and quantization	15
2.1	Magnetic fields on $\mathbb{T}^2$	15
2.2	Magnetic translations and magnetically periodic functions	16
2.3	Torus-adapted magnetic Weyl quantization	17
2.4	Magnetic operators	18
3	Time-dependent magnetic semiclassical measures	20
3.1	Time-dependent quantum limits on configuration space	20
3.2	Magnetic Wigner distributions and phase-space lifts	21
3.3	Decomposition of invariant measures	31
4	A two-microlocal procedure to analyze periodic orbits	33
4.1	A microlocal splitting around $\Lambda^\perp$	33
4.2	The weakly-resonant part: vanishing of Λ -modes	40
4.3	The resonant part: a two-microlocal lift	48
4.4	Effective dynamics for the two-microlocal lift	55

5	Regularity of configuration-space quantum limits	61
5.1	Reduction to the compact two-microlocal contribution	63
5.2	A geometric criterion for equidistribution	64
5.3	Application to the magnetic Laplacian	68
6	Beyond the Heisenberg time	70
A	Appendix	75
A.1	Functional calculus	75
A.1.1	The Helffer–Sjöstrand formula	78
A.2	Disintegration of Radon measures	79

1 Introduction

This paper studies long-time equidistribution phenomena for magnetic Schrödinger evolutions on the two-dimensional torus $\mathbb{T}^2 = \mathbb{R}^2/\mathbb{Z}^2$. Throughout this paper, we work within a semiclassical framework. More precisely, we consider a sequence $(h_n)_{n \in \mathbb{N}}$ of positive semiclassical parameters such that $h_n \rightarrow 0^+$. A *magnetic field* on $\mathbb{T}^2$ is a smooth real-valued closed 2-form

$$B(x) dx_1 \wedge dx_2,$$

which we identify with the function $B \in \mathcal{C}^\infty(\mathbb{T}^2, \mathbb{R})$. We assume that its total flux satisfies

$$\widehat{B}_0 := \int_{\mathbb{T}^2} B(x) dx_1 dx_2 \in 2\pi\mathbb{Z} \setminus \{0\}. \quad (1.1)$$

We focus on magnetic Schrödinger-type equations of the form

$$\begin{cases} ih_n \partial_t u_n(t, x) = \widehat{H}_{h_n} u_n(t, x), \\ u_n(t = 0, x) = u_n^{(0)}(x), \end{cases} \quad (t, x) \in \mathbb{R} \times \mathbb{T}^2, \quad (1.2)$$

where the initial data are assumed to be normalized, i.e., $\|u_n^{(0)}\|_{L^2(\mathbb{T}^2, L)} = 1$. The operator $\widehat{H}_{h_n}$ is a semiclassical magnetic Hamiltonian of the form

$$\widehat{H}_{h_n} = \text{Op}_{h_n}^B(H(\xi) + h_n R_{h_n}(x, \xi)), \quad (1.3)$$

where $\text{Op}_{h_n}^B$ denotes the toral magnetic Weyl quantization recalled in Section 2. Despite the notation, this quantization depends only on the magnetic flux $\widehat{B}_0$. The precise assumptions on the Hamiltonian H and on the perturbation R_{h_n} are stated in Section 2.4. In particular, H is a real-valued elliptic Hamiltonian depending only on ξ , and $\mathcal{R}$ will denote the principal symbol of R_{h_n} as $h_n \rightarrow 0$ and it will depend both on x and ξ . These assumptions are precisely those needed to ensure that the operator $\widehat{H}_{h_n}$ has the appropriate functional-analytic properties and is self-adjoint; see Proposition 2.3. Due to the magnetic field, the operator $\widehat{H}_{h_n}$ acts on the magnetic Hilbert space $L^2(\mathbb{T}^2, L)$ of twisted L^2 -functions associated with the flux condition (1.1) (see Section 2.2).

The magnetic Laplacian is the main motivating example. If A^{per} is a smooth real-valued periodic vector potential and V is a smooth real-valued potential, then the magnetic Laplacian $\mathcal{L}^B$ associated with

$$B(x) = \widehat{B}_0 + dA^{\text{per}}(x), \quad (1.4)$$

is included in the class of operators (1.3), namely, one has

$$h_n^2(\mathcal{L}^B + V) = \text{Op}_{h_n}^B \left(|\xi - h_n A^{\text{per}}(x)|^2 + h_n^2 V(x) \right), \quad (1.5)$$

where $|\cdot|$ denotes the Euclidean norm on $\mathbb{R}^2$.

A standing assumption throughout the paper is that there exists a compact energy window $[E_1, E_2] \subset (0, +\infty)$ containing no critical value of the principal Hamiltonian H :

$$\forall \xi \in \Omega_{E_1, E_2} := \{\xi \in \mathbb{R}^2 : E_1 \leq H(\xi) \leq E_2\}, \quad \nabla H(\xi) \neq 0. \quad (1.6)$$

For every $E \in [E_1, E_2]$, we denote by

$$\mathbb{S}_E := \{\xi \in \mathbb{R}^2 : H(\xi) = E\} \quad (1.7)$$

the corresponding energy level. By the ellipticity of H and (1.6), each $\mathbb{S}_E$ is a smooth compact one-dimensional submanifold of $\mathbb{R}^2$. We further assume that $\mathbb{S}_E$ is connected for every $E \in [E_1, E_2]$ so that each $\mathbb{S}_E$ is diffeomorphic to the circle $\mathbb{S}^1$. We also assume that the initial data are spectrally localized in this window:

$$u_n^{(0)} = \mathbb{1}_{[E_1, E_2]} \left(\widehat{H}_{h_n} \right) u_n^{(0)} + o_{L^2(\mathbb{T}^2, L)}(1) \quad \text{as } h_n \rightarrow 0^+. \quad (1.8)$$

Our first result is stated under a simple sufficient condition.

Theorem 1.1. *Suppose that the above assumptions hold and that the Hessian of H is positive definite on Ω_{E_1, E_2} . Then, there exists an explicit constant (see Corollary 5.6) $C_{H, \widehat{B}_0, E_1, E_2} > 0$, such that, if*

$$\|\partial_x \mathcal{R}\|_{L^\infty(\mathbb{T}^2 \times \Omega_{E_1, E_2})} \leq C_{H, \widehat{B}_0, E_1, E_2}, \quad (1.9)$$

if $(u_n)_{n \in \mathbb{N}}$ is a sequence of normalized solutions to (1.2) satisfying (1.8), and if $(\tau_{h_n})_{n \in \mathbb{N}}$ is a positive sequence satisfying

$$\tau_{h_n} \gg h_n^{-1/2},$$

then, for every $\psi \in \mathcal{C}_c^\infty(\mathbb{R})$ and every $a \in \mathcal{C}^\infty(\mathbb{T}^2)$, one has

$$\lim_{n \rightarrow +\infty} \int_{\mathbb{R}} \psi(t) \int_{\mathbb{T}^2} a(x) |u_n(t\tau_{h_n}, x)|^2 dx dt = \left(\int_{\mathbb{R}} \psi(t) dt \right) \left(\int_{\mathbb{T}^2} a(x) dx \right). \quad (1.10)$$

Equivalently, if one introduces the positive Radon measures

$$\nu_{h_n}(\tau_{h_n}) := |u_n(t\tau_{h_n}, x)|^2 dx dt \quad \text{on } \mathbb{T}^2 \times \mathbb{R}, \quad (1.11)$$

then Theorem 1.1 says that

$$\nu_{h_n}(\tau_{h_n}) \rightharpoonup dx \otimes dt \quad \text{in } \mathcal{D}'(\mathbb{T}^2 \times \mathbb{R}). \quad (1.12)$$

In particular, for almost every time t , the configuration-space marginal associated with any accumulation point of (1.11) is the uniform measure dx . The threshold $h_n^{-1/2}$ is essential. In general, for observation scales $\tau_{h_n} \lesssim h_n^{-1/2}$, one cannot expect a statement of the form (1.10) for arbitrary initial data. In those regimes, nontrivial concentration mechanisms may persist near periodic directions of

the underlying classical flow. A precise description of these shorter scales is given later in Section 4. The smallness assumption on $\partial_x \mathcal{R}$ is a convenient sufficient condition ensuring the validity of the effective non-vanishing condition used in the general theorem. The latter condition is geometric in nature and will be stated precisely in Theorem 1.3. In the case of the magnetic Laplacian, it takes a simple form and is satisfied, for instance, as soon as the magnetic field is positive, namely $B > 0$ on $\mathbb{T}^2$.

Theorem 1.2 (Magnetic Laplacian). *Assume that $B > 0$ and let $(v_n^{(0)})_{n \in \mathbb{N}}$ be a normalized sequence of initial data. We consider the non-semiclassical magnetic Schrödinger evolution*

$$v_n(t) := e^{-it(\mathcal{L}^B + V)} v_n^{(0)}. \quad (1.13)$$

Assume that the initial data are spectrally localized in the energy window $[E_1, E_2]$ in the sense of (1.8), with $H(\xi) = |\xi|^2$. Then, for every $\psi \in \mathcal{C}_c^\infty(\mathbb{R})$ and every $a \in \mathcal{C}^\infty(\mathbb{T}^2)$,

$$\lim_{n \rightarrow +\infty} \int_{\mathbb{R}} \psi(t) \int_{\mathbb{T}^2} a(x) \left| e^{-it(\mathcal{L}^B + V)} v_n^{(0)}(x) \right|^2 dx dt = \left(\int_{\mathbb{R}} \psi(t) dt \right) \left(\int_{\mathbb{T}^2} a(x) dx \right). \quad (1.14)$$

This theorem corresponds to the time scale $\tau_{h_n} = h_n^{-1}$ which is natural in this setting. Indeed, the evolution (1.13) solves the non-semiclassical magnetic Schrödinger equation

$$i\partial_t v_n = (\mathcal{L}^B + V)v_n.$$

It is related to the semiclassical equation by setting

$$u_n(t) := v_n(h_n t) = e^{-ih_n t(\mathcal{L}^B + V)} v_n^{(0)}.$$

Then u_n satisfies

$$ih_n \partial_t u_n = h_n^2 (\mathcal{L}^B + V) u_n.$$

By (1.5), this equation belongs to the class (1.2) and, taking $\tau_{h_n} = h_n^{-1}$, one has

$$u_n(t\tau_{h_n}) = v_n(t).$$

The time scale $\tau_{h_n} = h_n^{-1}$ is the Heisenberg scale in the semiclassical normalization used here. It lies far beyond the Ehrenfest regime, which in general corresponds to small logarithmic times of order $|\log h_n|$ for semiclassical Egorov estimates. We shall discuss other long observation regimes, including

$$h_n^{-1/2} \ll \tau_{h_n} \ll h_n^{-1} \quad \text{and} \quad h_n^{-1} \ll \tau_{h_n},$$

later in the article.

1.1 The semiclassical context

As mentioned above, Theorem 1.1 follows from a more general result, stated in terms of semiclassical measures. We now describe this framework and state the main structure theorem of the article.

To simplify notation, we shall henceforth write h instead of h_n , and u_h instead of u_n . All limiting statements are understood along the fixed sequence $(h_n)_{n \in \mathbb{N}}$, possibly after extraction of

a subsequence. Let $(\tau_h)_{h \rightarrow 0^+}$ be a sequence of positive observation times. In order to describe the long-time behavior of the solutions, we first consider the configuration-space distributions

$$\nu_h(\tau_h) := |u_h(t\tau_h, x)|^2 dx dt \quad \text{on } \mathbb{T}^2 \times \mathbb{R}. \quad (1.15)$$

The compactness arguments recalled in Section 3 show that every accumulation point of $(\nu_h(\tau_h))_{h \rightarrow 0^+}$ is a positive Radon measure ν on $\mathbb{T}^2 \times \mathbb{R}$. Such limits describe the macroscopic distribution of the position densities along the rescaled evolution $t \mapsto u_h(t\tau_h)$.

The semiclassical strategy consists in lifting these configuration-space distributions to phase space. In the present magnetic setting, this is achieved through time-dependent magnetic Wigner distributions that generalize (1.15) and that are defined by

$$W_h^B(\tau_h) : a \in \mathcal{C}_c^\infty(\mathbb{R} \times T^*\mathbb{T}^2) \mapsto \int_{\mathbb{R}} \langle \text{Op}_h^B(a(t))u_h(t\tau_h), u_h(t\tau_h) \rangle_{L^2(\mathbb{T}^2, L)} dt, \quad (1.16)$$

where $(u_h)_{h \rightarrow 0^+}$ is a family of solutions to (1.2).

More precisely, Section 3 shows that, after extraction of a subsequence, one obtains after taking $h \rightarrow 0^+$ a positive Radon measure

$$W^B(dt, dx, d\xi) = \nu_t^B(dx, d\xi) \otimes dt \quad \text{on } T^*\mathbb{T}^2 \times \mathbb{R}, \quad (1.17)$$

where $(\nu_t^B)_{t \in \mathbb{R}}$ is a measurable family of probability measures on $T^*\mathbb{T}^2$, defined for almost every t . The family $(\nu_t^B)_{t \in \mathbb{R}}$ will be referred to as a time-dependent magnetic semiclassical measure. Moreover, under the spectral localization assumption (1.8), one has

$$\text{supp } \nu_t^B \subset \mathbb{T}^2 \times \Omega_{E_1, E_2} \quad \text{for a.e. } t \in \mathbb{R}. \quad (1.18)$$

The corresponding configuration-space limit is recovered by projection:

$$\nu_t := (\pi_x)_* \nu_t^B, \quad \pi_x : (x, \xi) \in T^*\mathbb{T}^2 \mapsto x.$$

In particular, the corresponding configuration-space distributions satisfy

$$\nu_h(\tau_h) \rightharpoonup \nu_t(dx) \otimes dt \quad \text{in } \mathcal{D}'(\mathbb{T}^2 \times \mathbb{R}).$$

When $\tau_h \rightarrow +\infty$, the phase-space measures ν_t^B are, for almost every t , invariant under the Hamiltonian flow generated by the principal symbol $H(\xi)$,

$$\varphi_H^s(x, \xi) = (x + s\nabla H(\xi), \xi). \quad (1.19)$$

This invariance alone does not exclude concentration near the periodic trajectories of φ_H^s . To rule out these remaining concentration phenomena, we introduce a non-vanishing condition along the periodic directions of the classical flow. Periodic directions are indexed by primitive rank-one submodules $\Lambda \subset \mathbb{Z}^2$. We denote by $\mathcal{L}_1$ the set of such submodules. For $\Lambda \in \mathcal{L}_1$, let $\mathbf{e}_\Lambda$ be a primitive generator of Λ and set

$$L_\Lambda := |\mathbf{e}_\Lambda|.$$

The resonant set associated with Λ is

$$E_{\Lambda^\perp \setminus \{0\}} := \{\xi \in \mathbb{R}^2 : \nabla H(\xi) \in \Lambda^\perp \setminus \{0\}\}.$$

Along these directions, the flow φ_H^s is periodic. The two-microlocal analysis developed in Section 4.4 gives rise to the effective coefficient

$$G_{\widehat{B}_0, \Lambda}(x, \xi) := \text{Hess}(H)(\xi) \frac{\mathfrak{e}_\Lambda}{L_\Lambda} \cdot \left(\widehat{B}_0 \nabla H(\xi)^\perp + \partial_x \mathcal{I}_\Lambda(\mathcal{R})(x, \xi) \right), \quad (1.20)$$

where we use the convention $(v_1, v_2)^\perp = (-v_2, v_1)$, and where $\mathcal{I}_\Lambda(\mathcal{R})$ denotes the Λ -average of $\mathcal{R}$, defined in Section 3.3. Roughly speaking, it consists in keeping only the Fourier coefficients of $\mathcal{R}$ lying in Λ . For instance, in the case of the magnetic Laplacian, one has

$$H(\xi) = |\xi|^2, \quad \mathcal{R}(x, \xi) = -2\xi \cdot A^{\text{per}}(x),$$

and, on the resonant set $E_{\Lambda^\perp \setminus \{0\}}$, the effective coefficient reduces to

$$G_{\widehat{B}_0, \Lambda}(x, \xi) = -4 \left(\xi \cdot \frac{\mathfrak{e}_\Lambda^\perp}{L_\Lambda} \right) \mathcal{I}_\Lambda(B)(x).$$

Thus, in this particular case, the non-vanishing of $G_{\widehat{B}_0, \Lambda}$ is directly related to the non-vanishing of the magnetic field averaged along the corresponding periodic direction. The main result of this article shows that the magnetic subprincipal structure removes any concentration phenomena under a suitable non-vanishing condition. It also yields a complete description of the limiting measures on the long time scales considered here.

Theorem 1.3. *Assume that the standing assumptions on H and R_h stated above hold. More precisely, assume that the ellipticity of H , the regular energy condition (1.6) and the spectral localization condition (1.8) hold. Suppose also that for every $\Lambda \in \mathcal{L}_1$ and every $(x, \xi) \in \mathbb{T}^2 \times (E_{\Lambda^\perp \setminus \{0\}} \cap \Omega_{E_1, E_2})$,*

$$G_{\widehat{B}_0, \Lambda}(x, \xi) \neq 0. \quad (1.21)$$

Let $W^B(dt, dx, d\xi) = \nu_t^B(dx, d\xi) \otimes dt$ be any time-dependent magnetic semiclassical measure associated with the rescaled evolution $(u_h(t\tau_h))_{h \rightarrow 0^+}$ as in (1.16), and assume that $\tau_h \gg h^{-1/2}$. Then, there exists a measurable family of probability measures $(\lambda_t)_{t \in \mathbb{R}}$ on Ω_{E_1, E_2} such that

$$\nu_t^B(dx, d\xi) = dx \otimes \lambda_t(d\xi) \quad \text{for a.e. } t. \quad (1.22)$$

Moreover, up to another extraction, there exists a probability measure λ_0 on Ω_{E_1, E_2} associated to the initial data in the sense that

$$\forall a \in \mathcal{C}_c^\infty(\mathbb{R}^2), \quad \left\langle \text{Op}_h^B(a) u_h^{(0)}, u_h^{(0)} \right\rangle_{L^2(\mathbb{T}^2, L)} \xrightarrow{h \rightarrow 0^+} \int_{\mathbb{R}^2} a(\xi) \lambda_0(d\xi) \quad (1.23)$$

and such that the following finer description of λ_t holds.

- (i) **Intermediate long times.** *If $h^{-1/2} \ll \tau_h \ll h^{-1}$, then one has $\lambda_t = \lambda_0$. More precisely, ν_t^B is of the form*

$$\nu_t^B(dx, d\xi) = dx \otimes \lambda_0(d\xi) \quad \text{for a.e. } t.$$

- (ii) **The Heisenberg scale.** *Assume that $\tau_h = h^{-1}$. Then one has*

$$\nu_t^B(dx, d\xi) = dx \otimes (\Upsilon_H^t)_* \lambda_0(d\xi),$$

where Υ_H^t is the flow on momentum space generated by

$$\tilde{X}_H := -\widehat{B}_0 \nabla H(\xi)^\perp \cdot \partial_\xi. \quad (1.24)$$

(iii) **Beyond the Heisenberg scale.** If $\tau_h \gg h^{-1}$, then letting $\sigma^B := H_* \lambda_0$, one has

$$\nu_t^B(dx, d\xi) = dx \otimes \int_{E_1}^{E_2} \mu_E(d\xi) \sigma^B(dE) \quad \text{for a.e. } t, \quad (1.25)$$

where μ_E denotes the unique Υ_H^t -invariant probability measure on $\mathbb{S}_E$. Moreover, one has for every $f \in C_c^\infty(\mathbb{R})$

$$\int_{E_1}^{E_2} f(E) \sigma^B(dE) = \int_{\mathbb{R}^2} f(H(\xi)) \lambda_0(d\xi) = \lim_{h \rightarrow 0^+} \left\langle f(\hat{H}_h) u_h^{(0)}, u_h^{(0)} \right\rangle_{L^2(\mathbb{T}^2, L)}.$$

The spectral localization condition (1.8) can be read at the level of the initial momentum measure. Namely, it implies that

$$\lambda_0(\Omega_{E_1, E_2}) = 1, \quad \text{supp}(\lambda_0) \subset \Omega_{E_1, E_2},$$

and therefore the initial energy distribution $\sigma^B = H_* \lambda_0$ is a probability measure supported in $[E_1, E_2]$. Theorem 1.3 reveals two distinct long-time thresholds. The first threshold occurs at the scale $h^{-1/2}$: once $\tau_h \gg h^{-1/2}$, the configuration variable becomes equidistributed. At scales $\tau_h \leq h^{-1/2}$, such an equidistribution statement fails. Concentration effects may persist near periodic directions of the classical flow, and the limiting measures can be described in terms of the measure arising from the initial data. This shorter-time analysis is carried out in Section 4. The second threshold occurs at the scale h^{-1} : once $\tau_h \gg h^{-1}$, the remaining momentum dynamics equidistributes along the regular energy curves of H . The non-vanishing condition (1.21) expresses an escape mechanism for the effective dynamics near each periodic direction. For each fixed Λ , this condition yields a positive lower bound on $|G_{\hat{B}_0, \Lambda}|$ on the relevant compact resonant set. It can be viewed as a directional curvature condition: on the resonant set associated with Λ , the leading term of $G_{\hat{B}_0, \Lambda}$ involves the Hessian of H in the direction $\mathbf{e}_\Lambda$. This is why the positive definiteness assumption on $\text{Hess}(H)$, together with the smallness assumption (1.9) on $\partial_x \mathcal{R}$, implies (1.21): the term proportional to $\hat{B}_0$ in (1.20) dominates the contribution involving $\partial_x \mathcal{I}_\Lambda(\mathcal{R})$. This mechanism is explained more precisely in Section 5. Therefore, Theorem 1.1 follows from Theorem 1.3. Likewise, Theorem 1.2 is obtained from Theorem 1.3 in the magnetic Laplacian case.

Corollary 1.4 (Eigenfunctions). *Assume the standing assumptions on H and R_h . Let $E \in (0, +\infty)$ be a regular value of H , and assume that, for every $\Lambda \in \mathcal{L}_1$ and every $(x, \xi) \in \mathbb{T}^2 \times (E_{\Lambda^\perp \setminus \{0\}} \cap \mathbb{S}_E)$, one has*

$$G_{\hat{B}_0, \Lambda}(x, \xi) \neq 0.$$

Let $(u_h)_{h \rightarrow 0^+}$ be a normalized family of eigenfunctions of $\hat{H}_h$,

$$\hat{H}_h u_h = E_h u_h, \quad E_h \rightarrow E.$$

Then, for every $a \in C_c^\infty(T^*\mathbb{T}^2)$, one has

$$\lim_{h \rightarrow 0^+} \left\langle \text{Op}_h^B(a) u_h, u_h \right\rangle_{L^2(\mathbb{T}^2, L)} = \int_{\mathbb{T}^2 \times \mathbb{S}_E} a(x, \xi) dx \mu_E(d\xi),$$

where μ_E is uniformly equidistributed on $\mathbb{S}_E$. In particular, for every $a \in C^\infty(\mathbb{T}^2)$, one has

$$\lim_{h \rightarrow 0^+} \int_{\mathbb{T}^2} a(x) |u_h(x)|^2 dx = \int_{\mathbb{T}^2} a(x) dx.$$

This stationary consequence may be viewed as a quantum unique ergodicity-type statement in a magnetic and completely integrable setting.

1.2 Relation to previous works

In summary, the above results describe regularization and equidistribution phenomena for quantum systems associated with completely integrable classical dynamics, where the relevant rigidity mechanism is created by a magnetic subprincipal effect. Such results pertain to the broader field where one aims at describing the long time dynamics of Schrödinger equations depending on the underlying classical Hamiltonian dynamics. We will now briefly describe other results in that direction.

Semiclassical measures and quantum ergodicity

The Quantum Ergodicity Theorem of Šnirelman [48], Zelditch [49] and Colin de Verdière [16] asserts that, on a compact Riemannian manifold whose geodesic flow is ergodic, a density-one subsequence of Laplace eigenfunctions equidistributes in phase space. The stronger Quantum Unique Ergodicity conjecture of Rudnick–Sarnak [47] predicts that, in negatively curved settings, no subsequence extraction should be necessary.

Several major developments have clarified the possible structure of semiclassical measures in chaotic geometries. Lindenstrauss proved quantum unique ergodicity for Hecke eigenfunctions on arithmetic hyperbolic surfaces [35]. In the Anosov setting, Anantharaman [1] and Anantharaman–Nonnenmacher [6] established positive entropy bounds for semiclassical measures, excluding in particular concentration on a single closed geodesic. Hassell showed, by constructing ergodic billiards that are not quantum uniquely ergodic, that ergodicity of the classical dynamics does not imply quantum unique ergodicity in full generality [24]. More recently, Dyatlov and Jin proved that every semiclassical measure on a compact hyperbolic surface has full support [19], a result extended to negatively curved surfaces of variable curvature by Dyatlov, Jin and Nonnenmacher [20]. Related higher-dimensional support results were recently obtained by Kim and Miller [32].

The situation considered in the present paper is of very different nature. The principal Hamiltonian depends only on the momentum variable,

$$H = H(\xi),$$

and therefore generates a completely integrable flow on $T^*\mathbb{T}^2$. In such a setting, one cannot rely on chaoticity of the classical flow to force equidistribution. Nevertheless, even in the non-magnetic case, the flat torus exhibits remarkable regularity properties for quantum limits. Indeed, the classical estimates of Cooke [17] and Zygmund [52] imply uniform L^4 -bounds for normalized eigenfunctions of the Laplacian on $\mathbb{T}^2$. As a consequence, if $(u_{\lambda_n})_{n \in \mathbb{N}}$ is a sequence of normalized Laplace eigenfunctions on $\mathbb{T}^2$, then every accumulation point of the probability densities

$$|u_{\lambda_n}(x)|^2 dx$$

is absolutely continuous with respect to the Lebesgue measure and has a density in $L^2(\mathbb{T}^2)$. Jakobson [30] refined this result by classifying the configuration-space quantum limits on $\mathbb{T}^2$: their densities are trigonometric polynomials whose Fourier supports satisfy additional geometric constraints. He also obtained absolute continuity results for quantum limits on flat tori in arbitrary dimension. In a related pseudo-integrable setting, Marklof and Rudnick [43] proved that, for rational polygons, a density-one subsequence of eigenfunctions becomes uniformly distributed in configuration

space. These stationary results show that integrable or pseudo-integrable geometries may already impose nontrivial regularity on configuration-space limits. The present article is concerned with a dynamical and magnetic approach to these phenomena.

Long-time semiclassical dynamics, observability and two-microlocal analysis

A second line of work, much closer in spirit to the present article, concerns time-dependent semiclassical measures for Schrödinger evolutions observed on long time scales. Macià initiated a systematic study of such limits on compact manifolds and on flat tori [37, 38]. In the torus case, he showed that the limiting dynamics is not only determined by the initial semiclassical measure: one must keep track of additional information concentrated near resonant frequencies. He also proved that limits of configuration-space densities for the Schrödinger flow are absolutely continuous with respect to Lebesgue measure. This program was further developed by Anantharaman and Macià [4, 5]. They obtained a detailed description of semiclassical measures for the Schrödinger equation on flat tori and established strong regularity properties of the associated configuration-space limits. As a by-product, their analysis also yielded observability consequences for the Schrödinger evolution.

These questions are also closely related to observability and control problems. The seminal work of Bardos, Lebeau and Rauch [8] concerned the wave equation. For Schrödinger equations, Lebeau [34] established exact controllability results in a general geometric framework while, in periodic geometries, observability phenomena already appear in the work of Jaffard [29]. In a non-semiclassical formulation, Burq and Zworski [13] studied the equation

$$i\partial_t u = (-\Delta + V)u \quad \text{on } \mathbb{T}^2,$$

with $V \in C^\infty(\mathbb{T}^2)$, and proved that any nonempty open subset controls, in the L^2 -sense, both stationary and dynamical Schrödinger solutions. Bourgain, Burq and Zworski [11] later extended this analysis to potentials $V \in L^2(\mathbb{T}^2)$ on two-dimensional flat tori, including both rational and irrational tori.

A systematic framework for long-time semiclassical dynamics in completely integrable systems on flat tori $\mathbb{T}^d$ was developed by Anantharaman, Fermanian-Kammerer and Macià [2]. They consider semiclassical evolutions with principal Hamiltonian $H(\xi)$ and study time-dependent semiclassical measures associated with the rescaled evolution $t \mapsto u_h(t\tau_h)$. Their analysis reveals a critical time scale

$$\tau_h = h^{-1},$$

at which two-microlocal effects become decisive. Below this threshold, concentration phenomena near resonant directions may persist while at and beyond it, one obtains stronger regularity properties in the configuration variable under suitable nondegeneracy assumptions on d^2H . Their results also provide observability consequences in this integrable framework. The two-microlocal viewpoint underlying these works has a longer history in semiclassical analysis. Early constructions of two-microlocal semiclassical measures appear in the thesis of Fermanian-Kammerer [21], in the work of Nier [45] and in Fermanian-Kammerer [22]. Related two-microlocal methods were used by Anantharaman, Léautaud and Macià [3] for the Schrödinger equation on the disk which is another example of an integrable system but with a singularity at the boundary. Again, they obtained a detailed structural description of Wigner measures associated with the billiard dynamics and used it to derive observability results. Related long-time concentration, quantum-limit, and observability phenomena on manifolds with periodic geodesic flow were investigated by Macià and Rivi re on Zoll manifolds [39] and on spheres [41]. The work of Macià and Rivi re [40] is especially close

to the present paper. They study stationary and time-dependent solutions to strong semiclassical perturbations of the free Schrödinger equation on the two-dimensional flat torus. By performing a second microlocalization adapted to the perturbative scale, they describe refined concentration and nonconcentration mechanisms. In particular, they prove that sufficiently accurate quasimodes may concentrate only on the critical set of the geodesic average of the perturbation.

The present article belongs to this family of results on long-time dynamics in integrable geometries, but the magnetic structure produces a genuinely different effective behavior. The key new feature is the appearance of an earlier critical time scale,

$$\tau_h = h^{-1/2},$$

at which the magnetic contribution already enforces spatial equidistribution. At longer times, and in particular beyond the Heisenberg scale

$$\tau_h = h^{-1},$$

the remaining momentum distribution also becomes rigid along the regular energy curves of H .

Magnetic semiclassical problems

The magnetic effect at work in the present work is most directly related to the recent article of Morin and Rivière [44]. They study high-energy eigenfunctions of magnetic Schrödinger operators on $\mathbb{T}^2$, acting on sections of a magnetic line bundle. Under the assumption $B > 0$, they prove a quantum unique ergodicity result: every sequence of high-energy eigenfunctions equidistributes in phase space, despite the complete integrability of the underlying principal dynamics.

The present article may be viewed as a dynamical counterpart to this stationary result. We study long-time Schrödinger evolutions on the same magnetic geometric background, but for the much broader class of semiclassical Hamiltonians

$$ih\partial_t u_h = \text{Op}_h^B(H(\xi) + hR_h(x, \xi))u_h. \quad (1.26)$$

Here the operator acts on sections of a magnetic line bundle over $\mathbb{T}^2$ (see Section 2), the principal Hamiltonian $H(\xi)$ remains completely integrable, and the subprincipal perturbation R_h is not necessarily assumed to be of magnetic origin. However, in the magnetic Laplacian case introduced in (1.5), this perturbation is precisely generated by the periodic part of the magnetic potential A^{per} , and therefore encodes the nonconstant component dA^{per} of the magnetic field. The twisted structure of our torus nevertheless enters the effective long-time dynamics and is responsible for the equidistribution mechanisms proved below.

The non-vanishing condition appearing in Theorem 1.3,

$$G_{\widehat{B}_0, \Lambda}(x, \xi) \neq 0,$$

is the analogue, in this more general dynamical setting, of the geometric positivity condition used by Morin and Rivière which itself already appeared in the works of Glass and Han-Kwan on Vlasov equations [23]. In the particular case of the magnetic Laplacian, where $H(\xi) = |\xi|^2$ and the subprincipal contribution comes from the magnetic potential, this condition implies the positivity of the magnetic field averaged along periodic geodesics. Thus the mechanism used in the present work extends to general integrable Hamiltonians and general subprincipal perturbations the magnetic

geometric condition underlying the stationary result of [44]. See also the recent work of Le Balc'h, Niu and Sun [33] for observability results on (1.26) under related geometric conditions.

Other magnetic semiclassical regimes have been studied in different geometric and asymptotic settings. In contrast with the regime considered in the present article, these works concern a strong magnetic field which, in the normalization used here, corresponds to taking $\widehat{B}_0$ of order h^{-1} . The magnetic effect therefore enters the principal classical dynamics rather than appearing as a subprincipal correction. For instance, magnetic Laplacians on hyperbolic surfaces with constant magnetic field are related to earlier work of Zelditch [50]; more recently, Charles and Lefeuvre [15] obtained a finer description of the associated semiclassical defect measures. Their analysis separates three regimes: a flexible low-energy regime, a quantum unique ergodicity result at the critical energy level, and a Shnirelman-type equidistribution theorem in the high-energy regime.

In the Euclidean setting, Boil and Vũ Ngọc [10] studied the long-time propagation of generalized coherent states for the magnetic Laplacian on $\mathbb{R}^2$, again in the strong magnetic field regime. For low-energy initial states, they obtain a precise description of the Schrödinger evolution up to times of order h^{-1} , without performing any time averaging. At this time scale, an initially localized coherent state may split into several coherent states, each following the averaged guiding-center dynamics with a different effective speed.

1.3 Strategy of the proof

We now outline the main ideas of the proof. We work with the time-dependent magnetic Wigner distributions and semiclassical measures introduced in (1.16). For a given observation time scale τ_h , we consider a subsequential limit

$$W^B(dt, dx, d\xi) = \nu_t^B(dx, d\xi) \otimes dt,$$

where $(\nu_t^B)_{t \in \mathbb{R}}$ is a measurable family of probability measures on $T^*\mathbb{T}^2$ defined for a.e. t .

Propagation constraints and dependence on the time scale. A first step is to derive the dynamical constraints satisfied by ν_t^B and to understand how they depend on τ_h . This is obtained from a commutator identity. For any time-independent test symbol $a(x, \xi)$, one has

$$\frac{d}{dt} \langle \text{Op}_h^B(a) u_h(t\tau_h), u_h(t\tau_h) \rangle = \tau_h \frac{i}{h} \langle [\widehat{H}_h, \text{Op}_h^B(a)] u_h(t\tau_h), u_h(t\tau_h) \rangle. \quad (1.27)$$

Using the symbolic commutator expansion in the magnetic calculus, one has

$$\frac{i}{h} [\widehat{H}_h, \text{Op}_h^B(a)] = \text{Op}_h^B(\{H, a\}) + h \text{Op}_h^B(\{R_h, a\}) + \mathcal{O}(h). \quad (1.28)$$

Hence, combining (1.27) and (1.28), one obtains

$$\begin{aligned} \frac{1}{\tau_h} \frac{d}{dt} \langle \text{Op}_h^B(a) u_h(t\tau_h), u_h(t\tau_h) \rangle &= \langle \text{Op}_h^B(\{H, a\}) u_h(t\tau_h), u_h(t\tau_h) \rangle \\ &\quad + h \langle \text{Op}_h^B(\{R_h, a\}) u_h(t\tau_h), u_h(t\tau_h) \rangle + \mathcal{O}(h). \end{aligned}$$

The relative size of τ_h determines which terms survive in the limit. For instance, in the long-time regime $\tau_h \rightarrow +\infty$, the left-hand side formally vanishes at the level of semiclassical limits, while the terms of size h disappear, leading to invariance under the Hamiltonian flow φ_H^s defined in (1.19).

Resonant versus non-resonant directions. The principal symbol $H(\xi)$ depends only on ξ and generates a completely integrable Hamiltonian flow on $T^*\mathbb{T}^2$, which was defined in (1.19). For fixed ξ , the corresponding motion on $\mathbb{T}^2$ is the straight-line flow with velocity $\nabla H(\xi)$. Recall that we are working on regular energy levels, so that $\nabla H \neq 0$. Depending on the arithmetic properties of $\nabla H(\xi)$, this trajectory is either dense or periodic. This dichotomy is reflected at the level of semiclassical measures and is at the core of our rigidity argument.

In the non-resonant (dense) case, invariance under the Hamiltonian flow forces the measure $\nu_t^B(dx, d\xi)$ to be of the form $dx \otimes \lambda_t(d\xi)$ along these directions.

Periodic directions are parametrized by primitive rank-one submodules $\Lambda \in \mathcal{L}_1$. For the long-time regimes $\tau_h \gg h^{-1/2}$ relevant to our main theorem, the critical concentration phenomenon occurs in neighborhoods of size $h^{1/2}$ of the corresponding resonant sets

$$E_{\Lambda^\perp \setminus \{0\}} = \{\xi \in \mathbb{R}^2 : \nabla H(\xi) \in \Lambda^\perp \setminus \{0\}\}.$$

To separate the two contributions, we introduce a cutoff $\chi_{h,\Lambda}(\xi)$ supported in such a neighborhood and decompose an observable as

$$a = a \chi_{h,\Lambda} + a(1 - \chi_{h,\Lambda}).$$

The second term corresponds to the region that stays away from the resonant set at the relevant $h^{1/2}$ -scale. In this region, the propagation identities coming from the invariance under φ_H^s force again the corresponding contribution to be the Lebesgue measure in the x -variable. Hence this part does not contribute to the non-uniform component of the limiting measure in the configuration variable. In other words, only the first term can capture the possible concentration near periodic directions and requires a finer analysis.

Two-microlocal analysis near periodic directions and the first threshold. To analyze this remaining contribution, we perform a two-microlocal localization near each resonant set, in the spirit of [21, 22, 2, 3, 40, 44]. The idea is to enrich the usual phase-space description by an additional variable η , which measures the transverse distance to the resonant set at the critical $h^{1/2}$ -scale. This lift separates genuine concentration along periodic directions from the part already controlled by the ordinary semiclassical measure. This new contribution is referred to as the two-microlocal measure along Λ .

The resulting two-microlocal measures satisfy different dynamical properties according to the size of the observation time τ_h . For $\tau_h \ll h^{-1/2}$, the magnetic contribution is not yet visible at the two-microlocal scale, and nontrivial periodic concentration may persist, see Lemma 4.14. At the critical scale

$$\tau_h = h^{-1/2},$$

the lifted measures are transported by an effective vector field in the extended variables (x, ξ, η) ,

$$X_\Lambda = \eta \frac{\mathbf{e}_\Lambda}{L_\Lambda} \cdot \partial_x - G_{\widehat{B}_0, \Lambda}(x, \xi) \partial_\eta,$$

where $G_{\widehat{B}_0, \Lambda}$ is the coefficient introduced in (1.20).

In the supercritical regime

$$\tau_h \gg h^{-1/2},$$

this transport relation becomes an invariance property under the flow generated by X_Λ . The non-vanishing condition imposed in Theorem 1.3 forces the trajectories of this effective flow to escape in

the η -direction. Since this two-microlocal component has finite mass, such an invariance is possible only if this component vanishes. Consequently, concentration in $h^{1/2}$ -neighborhoods of $E_{\Lambda^\perp \setminus \{0\}}$ cannot contribute to a non-uniform configuration-space limit. Gathering all these observations, this yields

$$(\pi_x)_* \nu_t^B = dx \quad \text{for a.e. } t,$$

and therefore

$$\nu_t^B(dx, d\xi) = dx \otimes \lambda_t(d\xi) \quad \text{for a.e. } t,$$

for some measurable family $(\lambda_t)_{t \in \mathbb{R}}$ of probability measures on $\mathbb{R}^2$.

Momentum dynamics and the Heisenberg threshold. Once the configuration marginal is determined, the remaining long-time behavior is entirely encoded in the momentum marginal

$$\lambda_t = (\pi_\xi)_* \nu_t^B, \quad \pi_\xi : (x, \xi) \in T^*\mathbb{T}^2 \mapsto \xi.$$

To study this marginal, it is enough to test the commutator identity (1.28) against symbols depending only on the momentum variable, namely $b = b(\xi)$. Since

$$\{H, b\} = 0,$$

the leading classical transport disappears in (1.27). Expanding the commutator identity (1.28) to the next order reveals a magnetic contribution proportional to the flux $\widehat{B}_0$. Moreover, using the spatial equidistribution obtained above allows us to describe the propagation/invariance of the momentum marginal and to see that it comes from the magnetic correction in the commutator expansion.

For intermediate time scales

$$h^{-1/2} \ll \tau_h \ll h^{-1},$$

this correction is still negligible in the limit. One therefore obtains that

$$t \mapsto \int_{\mathbb{R}^2} b(\xi) \lambda_t(d\xi)$$

is constant for every test function b , which shows that λ_t is independent of t . At the critical scale

$$\tau_h = h^{-1},$$

the magnetic correction contributes at leading order and induces a genuine transport dynamics in the momentum variable. More precisely, the family $(\lambda_t)_{t \in \mathbb{R}}$ is transported by the flow Υ_H^t generated by the vector field defined in (1.24)

$$\widetilde{X}_H = -\widehat{B}_0 (\partial_{\xi_1} H \partial_{\xi_2} - \partial_{\xi_2} H \partial_{\xi_1}).$$

Equivalently, one has

$$\lambda_t = (\Upsilon_H^t)_* \lambda_0.$$

For longer time scales

$$\tau_h \gg h^{-1},$$

the same commutator argument yields invariance of λ_t under the flow Υ_H^s . Moreover, conservation of the energy distribution along the quantum evolution implies that the energy marginal

$$\sigma_t^B := H_* \lambda_t$$

is independent of t . More precisely, one has

$$\sigma_t^B = H_* \lambda_t = H_* \lambda_0 =: \sigma^B \quad \text{for a.e. } t.$$

Hence, one may disintegrate λ_t with respect to the energy variable

$$\lambda_t(d\xi) = \int_{E_1}^{E_2} \lambda_{t,E}(d\xi) \sigma^B(dE),$$

where $\lambda_{t,E}$ is a probability measure on

$$\mathbb{S}_E = \{\xi \in \mathbb{R}^2 : H(\xi) = E\}.$$

The invariance of λ_t under Υ_H^s implies that, for almost every E , the conditional measure $\lambda_{t,E}$ is invariant under the induced flow on $\mathbb{S}_E$. Since each $\mathbb{S}_E$ is a smooth connected closed curve and since $\nabla H \neq 0$, this invariant probability measure is unique. Denoting it by μ_E , one obtains

$$\lambda_t(d\xi) = \int_{E_1}^{E_2} \mu_E(d\xi) \sigma^B(dE).$$

In the particular case $H(\xi) = |\xi|^2$, the measures μ_E coincide with the normalized arc-length measures on the circles $\mathbb{S}_E$.

Summary. The proof therefore relies on two complementary mechanisms. The first one, which becomes effective at the scale

$$\tau_h = h^{-1/2},$$

eliminates concentration near periodic directions and enforces equidistribution in the configuration variable. The second one, which appears at the scale

$$\tau_h = h^{-1},$$

governs the remaining momentum dynamics and yields rigidity along the regular energy curves of H . Together, these two mechanisms give the structure of the time-dependent semiclassical measures stated in the main theorem.

1.4 Structure of the paper

Section 2 recalls the magnetic line bundle framework on $\mathbb{T}^2$, the torus-adapted magnetic Weyl quantization and the class of semiclassical operators considered in the paper. Section 3 constructs the time-dependent magnetic Wigner distributions and proves the first propagation properties of the associated semiclassical measures, following the work of Macià [37, 38] that we adapt in a magnetic setting. Section 4 develops the resonant decomposition and the two-microlocal analysis near periodic directions. Section 5 uses this analysis to prove the configuration-space equidistribution statement (1.22), hence the first part of Theorem 1.3 and Theorem 1.1. Section 6 analyzes the momentum marginal at and beyond the time scale h^{-1} , proving Theorem 1.3 (i)–(iii). Finally, Appendix A.1 gathers the functional-calculus and self-adjointness results needed for the magnetic pseudodifferential operators, while Appendix A.2 recalls the disintegration theorem for Radon measures used throughout the paper.

Acknowledgements

The author thanks Gabriel Rivière for many helpful comments throughout this project. The author acknowledges the support of the PRC grant ADYCT (ANR-20-CE40-0017) and of the ANR project La Gabare (ANR-25-CE40-7296).

2 Magnetic field and quantization

This section gathers the basic geometric and pseudodifferential tools needed throughout the paper. It is essentially a review of the torus-adapted framework introduced by Morin–Rivière [44] (itself rooted in the magnetic pseudodifferential calculus of Măntoiu–Purice [42] and the geometric quantization viewpoint of Charles [14]). For the reader’s convenience, we recall: (i) the flux quantization condition ensuring that magnetic translations define a consistent notion of magnetic periodicity on $\mathbb{T}^2$; (ii) the associated Hilbert space $L^2(\mathbb{T}^2, L)$; and (iii) the corresponding torus-adapted magnetic Weyl quantization Op_h^B acting on large symbol classes $\mathcal{S}^m(T^*\mathbb{T}^2)$. Our main results rely only on the structural properties of this calculus (boundedness, adjointness, composition and commutator expansions) proved in [44] and which we will use as a black box in the analysis of time-dependent semiclassical measures in the next sections.

2.1 Magnetic fields on $\mathbb{T}^2$

A magnetic field on a configuration space is a real closed 2-form (the curvature of a unitary connection on a complex line bundle). On the flat two-torus $\mathbb{T}^2 = \mathbb{R}^2/\mathbb{Z}^2$, once the orientation (and volume form) $dx_1 \wedge dx_2$ is fixed, any magnetic field can be identified with a smooth real-valued function

$$B \in C^\infty(\mathbb{T}^2; \mathbb{R}), \quad \mathbf{B} := B(x)dx_1 \wedge dx_2.$$

We denote by

$$\widehat{B}_0 := \int_{\mathbb{T}^2} B(x)dx_1 \wedge dx_2$$

the total magnetic flux, and we impose the condition

$$\widehat{B}_0 \in 2\pi\mathbb{Z} \setminus \{0\}. \quad (2.1)$$

This is the usual flux quantization condition ensuring the existence of a compatible line bundle structure (equivalently, a global magnetic Schrödinger operator acting on twisted periodic functions); see the discussion in [44, Section 2]. On $\mathbb{T}^2$, the zero-mean part $B - \widehat{B}_0$ is exact. Thus one may choose a periodic vector potential A^{per} such that

$$B(x) - \widehat{B}_0 = dA^{\text{per}}(x). \quad (2.2)$$

Following Morin–Rivière, we fix once and for all the decomposition

$$A = A^0 + A^{\text{per}}, \quad A^0(x_1, x_2) = \frac{\widehat{B}_0}{2}(-x_2, x_1), \quad A^{\text{per}} \text{ is } \mathbb{Z}^2\text{-periodic}. \quad (2.3)$$

Thus, one has

$$B(x) = dA(x).$$

The periodic part A^{per} is not unique and may be changed by a periodic gauge $A^{\text{per}} \mapsto A^{\text{per}} + \nabla\varphi$, $\varphi \in \mathcal{C}^\infty(\mathbb{T}^2; \mathbb{R})$, yielding unitarily equivalent operators. Let us finally stress that, in the abstract semiclassical model studied in this article, only the flux $\widehat{B}_0$ enters the torus-adapted magnetic Weyl quantization Op_h^B and the operators $\widehat{H}_h$ introduced in (1.3). The full magnetic field B appears only when this abstract framework is specialized to the magnetic Laplacian example (1.5).

2.2 Magnetic translations and magnetically periodic functions

A key feature of magnetic-type operators is that they commute with magnetic (rather than usual) translations defined, for $m \in \mathbb{Z}^2$, by

$$(T_m^B u)(x) := \exp\left(i \frac{\widehat{B}_0}{2} (m \wedge x + m_1 m_2)\right) u(x - m), \quad m \wedge x := m_1 x_2 - m_2 x_1, \quad (2.4)$$

as operators on $L_{\text{loc}}^2(\mathbb{R}^2)$. They form a *projective representation* of $\mathbb{Z}^2$ on $L_{\text{loc}}^2(\mathbb{R}^2)$, meaning that

$$T_m^B T_{m'}^B = \sigma(m, m') T_{m+m'}^B, \quad m, m' \in \mathbb{Z}^2,$$

for a phase factor $\sigma(m, m') \in \mathbb{S}^1$. The map $\sigma : \mathbb{Z}^2 \times \mathbb{Z}^2 \rightarrow \mathbb{S}^1$ is called the associated cocycle; see [44] for details. In the present case, a direct computation gives

$$\sigma(m, m') = \exp\left(-i \widehat{B}_0 m'_1 m_2\right),$$

and therefore

$$T_m^B T_{m'}^B = \exp\left(-i \widehat{B}_0 m'_1 m_2\right) T_{m+m'}^B, \quad m, m' \in \mathbb{Z}^2.$$

The corresponding commutator cocycle is given by

$$\frac{\sigma(m, m')}{\sigma(m', m)} = \exp\left(i \widehat{B}_0 m \wedge m'\right).$$

It is trivial, equivalently the magnetic translations commute, if and only if (2.1) holds. In particular, under (2.1), the notion of magnetic periodicity is consistent, meaning that the magnetic translations commute with each other (see [44]). We then define the space of *twisted periodic* smooth functions by

$$\mathcal{C}^\infty(\mathbb{T}^2, L) := \{u \in \mathcal{C}^\infty(\mathbb{R}^2, \mathbb{C}) : T_m^B u = u \text{ for all } m \in \mathbb{Z}^2\}. \quad (2.5)$$

Although we keep using B in the notation T_m^B , the definition of these magnetic translations, and therefore of $\mathcal{C}^\infty(\mathbb{T}^2, L)$, depends only on the flux $\widehat{B}_0$, not on the full magnetic field B . Similarly, we define $L^2(\mathbb{T}^2, L)$ and Sobolev spaces $\mathcal{H}^s(\mathbb{T}^2, L)$ as the subspaces of $L_{\text{loc}}^2(\mathbb{R}^2)$ and $\mathcal{H}_{\text{loc}}^s(\mathbb{R}^2)$ invariant under all T_m^B , namely

$$L^2(\mathbb{T}^2, L) = \{u \in L_{\text{loc}}^2(\mathbb{R}^2) : T_m^B u = u \quad \forall m \in \mathbb{Z}^2\}, \quad (2.6)$$

$$\mathcal{H}^s(\mathbb{T}^2, L) = \{u \in H_{\text{loc}}^s(\mathbb{R}^2) : T_m^B u = u \quad \forall m \in \mathbb{Z}^2\}. \quad (2.7)$$

These are the natural “twisted” analogs of the usual Sobolev spaces on $\mathbb{T}^2$. They provide the natural functional setting for the operators appearing in the introduction.

For any integer $s \in \mathbb{N}$, one may equivalently define $\mathcal{H}^s(\mathbb{T}^2, L)$ as the completion of $\mathcal{C}^\infty(\mathbb{T}^2, L)$ for the norm built out of the magnetic derivative. More precisely, with A^0 as in (2.3), set $\partial^A = (\partial_{x_1} - iA_1^0, \partial_{x_2} - iA_2^0)$, and

$$\|u\|_{H^s} := \left(\sum_{|\alpha| \leq s} \int_{\mathbb{T}^2} |(\partial^A)^\alpha u(x)|^2 dx \right)^{\frac{1}{2}}, \quad s \in \mathbb{N}. \quad (2.8)$$

For general, possibly non-integer, $s \geq 0$, we refer to Appendix A.1 for another definition of $\mathcal{H}^s(\mathbb{T}^2; L)$.

Remark 2.1. *Geometrically, $\mathcal{C}^\infty(\mathbb{T}^2, L)$ can be identified with smooth sections of a complex line bundle $L \rightarrow \mathbb{T}^2$ with curvature $\widehat{B}_0 dx_1 \wedge dx_2$. Yet, the reader may only keep in mind the concrete realization in (2.5) which is the point of view adopted all along the article.*

2.3 Torus-adapted magnetic Weyl quantization

In order to quantize observables while preserving the magnetic periodicity, we use the torus-adapted magnetic Weyl calculus of [44], inspired by the magnetic calculus on $\mathbb{R}^d$ of Măntoiu–Purice and collaborators [42, 27, 25, 28]. The construction relies on the magnetic Weyl operators $\mathcal{W}_h^B(\eta, \zeta)$, defined for $\eta \in 2\pi\mathbb{Z}^2$ and $\zeta \in \mathbb{R}^2$ by

$$\mathcal{W}_h^B(\eta, \zeta)u(x) := \exp\left(ih \frac{\widehat{B}_0}{2} \zeta \wedge x\right) \exp\left(i\eta \cdot \left(x + \frac{h\zeta}{2}\right)\right) u(x + h\zeta), \quad u \in L^2(\mathbb{T}^2, L), \quad (2.9)$$

which commute with all magnetic translations T_m^B and therefore act on $L^2(\mathbb{T}^2, L)$. They satisfy the adjoint relation $\mathcal{W}_h^B(\eta, \zeta)^* = \mathcal{W}_h^B(-\eta, -\zeta)$ and an explicit composition law (a twisted Heisenberg relation); see [44, §3.1].

Given a smooth symbol $a = a(x, \xi)$, periodic in x , we define its Fourier transform

$$\mathcal{F}(a)(\eta, \zeta) := \int_{\mathbb{T}^2 \times \mathbb{R}^2} a(x, \xi) e^{-i\eta \cdot x} e^{-i\zeta \cdot \xi} dx d\xi,$$

and we set

$$\text{Op}_h^B(a) := \frac{1}{(2\pi)^2} \sum_{\eta \in 2\pi\mathbb{Z}^2} \int_{\mathbb{R}^2} \mathcal{F}(a)(\eta, \zeta) \mathcal{W}_h^B(\eta, \zeta) d\zeta, \quad (2.10)$$

initially for $a \in \mathcal{C}_c^\infty(\mathbb{T}^2 \times \mathbb{R}^2)$ and then extended to the symbol classes used below.

We work with the large symbol class

$$\mathcal{S}^m(T^*\mathbb{T}^2) := \{a \in \mathcal{C}^\infty(\mathbb{T}^2 \times \mathbb{R}^2) : |\partial_x^\alpha \partial_\xi^\beta a(x, \xi)| \leq C_{\alpha\beta} \langle \xi \rangle^m\},$$

which is stable under the operations required in this article. This is the toral analogue of the class $S(\langle \xi \rangle^m)$ from [51, Chapter 4]. In the sequel, we shall also use h -dependent symbols $a_h \in \mathcal{S}^m(T^*\mathbb{T}^2)$. Unless otherwise stated, this always means that the estimates defining $\mathcal{S}^m(T^*\mathbb{T}^2)$ hold uniformly with respect to $h \in (0, 1]$. The associated magnetic calculus enjoys the usual structural properties (boundedness, adjoint, composition and commutator expansions). In particular, for $a \in \mathcal{S}^m$, the formal adjoint is given by [44, Proposition 3.4]

$$\text{Op}_h^B(a)^* = \text{Op}_h^B(\bar{a}) \quad \text{on } L^2(\mathbb{T}^2, L). \quad (2.11)$$

We shall also use the following composition formula for the magnetic Weyl quantization, established in [44, Theorem 3.5]: for every $a \in \mathcal{S}^{m_1}$ and $b \in \mathcal{S}^{m_2}$,

$$\text{Op}_h^B(a)\text{Op}_h^B(b) = \text{Op}_h^B(a \star_h b), \quad (2.12)$$

where $a \star_h b \in \mathcal{S}^{m_1+m_2}$. Moreover, [44, Theorem 3.5] provides an explicit formula for the expansion of $a \star_h b$ at any order $N \geq 1$:

$$a \star_h b(x, \xi) = \sum_{n=0}^{N-1} \frac{1}{n!} \left(\frac{ih}{2} \right)^n \Omega_{hB}(D_x, D_\xi, D_{x'}, D_{\xi'})^n a(x, \xi) b(x', \xi')|_{(x, \xi) = (x', \xi')} + \mathcal{O}_{\mathcal{S}^{m_1+m_2}}(h^N), \quad (2.13)$$

where $\Omega_{hB}(\eta, \zeta, \eta', \zeta') = \eta' \cdot \zeta - \eta \cdot \zeta' + h \widehat{B}_0 \zeta' \wedge \zeta$ and where we use the notation $D := -i\partial$. Applying (2.13) with $N = 1$, one obtains

$$a \star_h b = ab + \mathcal{O}_{\mathcal{S}^{m_1+m_2}}(h). \quad (2.14)$$

In order to study magnetic semiclassical measures, we will use the following commutator identity:

$$a \star_h b - b \star_h a = \frac{h}{i} \{a, b\} + \frac{\widehat{B}_0 h^2}{i} (\partial_{\xi_2} a \partial_{\xi_1} b - \partial_{\xi_1} a \partial_{\xi_2} b) + \mathcal{O}_{\mathcal{S}^{m_1+m_2}}(h^3). \quad (2.15)$$

This identity is the basic input for the propagation and invariance properties of time-dependent semiclassical measures in the next section. In what follows, we will mainly use the expansion (2.15) at order 3, and we will use the notation $\mathcal{O}_{\mathcal{S}^k}(h^n)$ to say that the remaining term can be expressed as $h^n r_h$, where r_h belongs uniformly in h to the symbol class $\mathcal{S}^k(T^*\mathbb{T}^2)$. Throughout the paper, we shall repeatedly use the following rescaling formula. As observed in [44, Remark 3.8], for every $\delta > 0$, one has

$$\text{Op}_h^B(a(x, \xi)) = \text{Op}_{\delta h}^B(a(x, \delta^{-1}\xi)). \quad (2.16)$$

In order to study the limit as the semiclassical parameter $h \rightarrow 0^+$, we shall also repeatedly use the Calderón–Vaillancourt theorem [44, Theorem 3.7], which ensures that the quantization of a symbol in $\mathcal{S}^0(\mathbb{T}^2 \times \mathbb{R}^2)$ defines a bounded operator on $L^2(\mathbb{T}^2, L)$.

Theorem 2.2 (Calderón–Vaillancourt). *There exist constants $C_0 > 0$ and $N_0 \in \mathbb{N}$ such that, for every $a \in \mathcal{S}^0(\mathbb{T}^2 \times \mathbb{R}^2)$, the operator $\text{Op}_h^B(a)$ is bounded on $L^2(\mathbb{T}^2, L)$ and satisfies*

$$\forall h \in (0, 1], \quad \|\text{Op}_h^B(a)\|_{L^2 \rightarrow L^2} \leq C_0 \sum_{|\alpha|+|\beta| \leq N_0} h^{|\beta|} \|\partial_x^\alpha \partial_\xi^\beta a\|_\infty. \quad (2.17)$$

2.4 Magnetic operators

As already described in the introduction, the rest of this article will be dedicated to the study of solutions to Schrödinger-type equations of the form

$$ih\partial_t(u_h) = \widehat{H}_h u_h, \quad (2.18)$$

where the operator $\widehat{H}_h$ is the magnetic quantization of specific symbols. Let $H \in \mathcal{C}^\infty(\mathbb{R}^2, \mathbb{R})$ be a general Hamiltonian depending only on the ξ -variable, belonging to the symbol class $\mathcal{S}^m(\mathbb{R}^2)$ and elliptic of order m . This means that there exist $M > 0$ and $c_0 > 0$ such that

$$\forall |\xi| \geq M, \quad c_0 \langle \xi \rangle^m \leq H(\xi). \quad (2.19)$$

Throughout the paper, we fix an energy window $[E_1, E_2]$, with $0 < E_1 < E_2$, containing no critical values of H . Equivalently, setting $\Omega_{E_1, E_2} := \{\xi \in \mathbb{R}^2 : E_1 \leq H(\xi) \leq E_2\}$, we assume that

$$\nabla H(\xi) \neq 0, \quad \forall \xi \in \Omega_{E_1, E_2}. \quad (2.20)$$

For $E \in [E_1, E_2]$, we denote by $\mathbb{S}_E := \{\xi \in \mathbb{R}^2 : H(\xi) = E\}$ the corresponding energy layer. We also assume that each $\mathbb{S}_E$ is compact and connected. Therefore, $\mathbb{S}_E$ is a smooth compact connected one-dimensional submanifold of $\mathbb{R}^2$, hence

$$\mathbb{S}_E \simeq \mathbb{S}^1. \quad (2.21)$$

Let also R_h belonging to $\mathcal{S}^m(\mathbb{T}^2 \times \mathbb{R}^2)$ uniformly in $h \in (0, 1]$ and real valued. More precisely, we suppose that R_h is of the form

$$R_h(x, \xi) = \mathcal{R}(x, \xi) + \mathcal{O}_{\mathcal{S}^m}(h). \quad (2.22)$$

We will consider magnetic operators denoted by $\hat{H}_h$ which are of the form

$$\hat{H}_h = \text{Op}_h^B(H(\xi) + hR_h(x, \xi)). \quad (2.23)$$

Since it is the quantization of a real-valued symbol and its principal symbol H is elliptic, the operator $\hat{H}_h$ enjoys the strong properties recalled in the following proposition.

Proposition 2.3 (Self-adjointness). *Let $m > 1$, assume that $H \in \mathcal{S}^m(\mathbb{R}^2)$ is real-valued and elliptic of order m , and that $R_h \in \mathcal{S}^m(\mathbb{T}^* \mathbb{T}^2)$ is real-valued and of the form (2.22). Then, for $h > 0$ small enough, the operator*

$$\hat{H}_h := \text{Op}_h^B(H(\xi) + hR_h(x, \xi))$$

is essentially self-adjoint on $C^\infty(\mathbb{T}^2, L)$. Its closure (still denoted $\hat{H}_h$) is self-adjoint on $L^2(\mathbb{T}^2, L)$ with domain $\mathcal{H}^m(\mathbb{T}^2, L)$ and has compact resolvent.

We refer to Appendix A for a reminder of the standard arguments leading to this result in the torus magnetic setting. By Stone's theorem [46, Theorem VIII.7], since $\hat{H}_h$ is self-adjoint, it generates a strongly continuous one-parameter unitary group

$$U_h(t) := \exp\left(-\frac{i}{h}t\hat{H}_h\right), \quad t \in \mathbb{R}, \quad (2.24)$$

so that the Cauchy problem $ih\partial_t u_h = \hat{H}_h u_h$ is globally well-posed in $L^2(\mathbb{T}^2, L)$ and satisfies $\|u_h(t)\|_{L^2(\mathbb{T}^2, L)} = \|u_h(0)\|_{L^2(\mathbb{T}^2, L)}$. The operator $\hat{H}_h$ can be viewed as a twisted counterpart of the classical operator $\text{Op}_h^W(H(\xi) + hR_h(x, \xi))$ on the (non-twisted) torus, where Op_h^W denotes the usual Weyl quantization.

Under (2.1), Morin–Rivière considered the magnetic Laplacian,

$$\mathcal{L}^B u := (-i\partial_{x_1} - A_1(x))^2 u + (-i\partial_{x_2} - A_2(x))^2 u, \quad u \in H^2(\mathbb{T}^2, L), \quad (2.25)$$

where $A(x) = (A_1(x), A_2(x))$ is defined by (2.3). This is exactly the magnetic Laplacian for the magnetic field $B = \hat{B}_0 + dA^{\text{per}}$ and in that case, the magnetic Weyl quantization produces

$$h^2 \mathcal{L}^B = \text{Op}_h^B(|\xi|^2 + h\xi \cdot c_1(x) + h^2 c_0(x)), \quad (2.26)$$

for some smooth real-valued coefficients $c_1 : \mathbb{T}^2 \rightarrow \mathbb{R}^2$ and $c_0 : \mathbb{T}^2 \rightarrow \mathbb{R}$; see [44, §3.3].

Remark 2.4. *For a constant magnetic field, i.e. $B(x) = \hat{B}_0$, the magnetic quantization produces*

$$h^2 \mathcal{L}^{\hat{B}_0} = \text{Op}_h^B(\|\xi\|^2). \quad (2.27)$$

3 Time-dependent magnetic semiclassical measures

This section introduces the space–time objects used throughout the paper to describe the high-frequency behavior of solutions to (2.18) observed on a macroscopic time scale τ_h . Our presentation follows the standard semiclassical framework for time-dependent Wigner distributions and their disintegration properties [37, 5, 2], but all pseudodifferential objects are defined here through the torus-adapted magnetic Weyl quantization recalled in Section 2. We also record the basic propagation and invariance properties satisfied by the limiting measures, depending on the asymptotic behavior of the time scale τ_h .

3.1 Time-dependent quantum limits on configuration space

Let $h \rightarrow 0^+$ and let $(u_h)_{h \rightarrow 0^+} \in \mathcal{C}(\mathbb{R}; \mathcal{H}^m(\mathbb{T}^2, L)) \cap \mathcal{C}^1(\mathbb{R}; L^2(\mathbb{T}^2, L))$ solve

$$\begin{cases} ih\partial_t u_h(t, x) = \hat{H}_h u_h(t, x), \\ u_h(t = 0, x) = u_h^{(0)}(x), \end{cases} \quad (3.1)$$

with the normalization

$$\|u_h^{(0)}\|_{L^2(\mathbb{T}^2, L)} = 1.$$

By unitarity, for every $t \in \mathbb{R}$, one has $\|u_h(t)\|_{L^2(\mathbb{T}^2, L)} = 1$. Fix a time scale $(\tau_h)_{h \rightarrow 0^+} > 0$. We say that a (positive) Radon measure ν on $\mathbb{R} \times \mathbb{T}^2$ is a *time-dependent quantum limit* associated with the rescaled family $t \mapsto u_h(t\tau_h)$ if for every test function $a \in \mathcal{C}_c^\infty(\mathbb{R} \times \mathbb{T}^2, \mathbb{C})$,

$$\int_{\mathbb{R} \times \mathbb{T}^2} a(t, x) |u_h(t\tau_h, x)|^2 dx dt \xrightarrow{h \rightarrow 0^+} \int_{\mathbb{R} \times \mathbb{T}^2} a(t, x) \nu(dt, dx). \quad (3.2)$$

Using the disintegration theorem A.5 for Radon measures and the fact that the sequence $(u_h(t\tau_h))_{h \rightarrow 0^+}$ is normalized in L^2 , there exist a positive Radon measure ω on $\mathbb{R}$ and a ω -measurable family of probability measures $(\nu_t)_{t \in \mathbb{R}}$ on $\mathbb{T}^2$ such that

$$\nu(dt, dx) = \nu_t(dx) \otimes \omega(dt). \quad (3.3)$$

In particular, for all $\psi \in \mathcal{C}_c^\infty(\mathbb{R})$ and $b \in \mathcal{C}^\infty(\mathbb{T}^2)$,

$$\int_{\mathbb{R} \times \mathbb{T}^2} \psi(t) b(x) \nu(dt, dx) = \int_{\mathbb{R}} \psi(t) \left(\int_{\mathbb{T}^2} b(x) \nu_t(dx) \right) \omega(dt). \quad (3.4)$$

Moreover, choosing $b \equiv 1$ in (3.4) and using the normalization $\|u_h(t\tau_h)\|_{L^2(\mathbb{T}^2, L)} = 1$ for all t , we obtain for every $\psi \in \mathcal{C}_c^\infty(\mathbb{R})$

$$\int_{\mathbb{R}} \psi(t) \omega(dt) = \int_{\mathbb{R} \times \mathbb{T}^2} \psi(t) \nu(dt, dx) = \lim_{h \rightarrow 0^+} \int_{\mathbb{R}} \psi(t) dt,$$

and therefore $\omega(dt) = dt$. In particular, (3.3) becomes

$$\nu(dt, dx) = \nu_t(dx) \otimes dt. \quad (3.5)$$

We denote by $\mathcal{N}(\hat{H}_h, \tau_h)$ the set of all such configuration-space limits arising from solutions to (3.1).

Remark 3.1. Since $\|u_h(t\tau_h)\|_{L^2(\mathbb{T}^2, L)} = 1$ for all t and since test functions are compactly supported in time, the family of measures

$$|u_h(t\tau_h, x)|^2 dx dt$$

is bounded in the space of Radon measures on $\mathbb{R} \times \mathbb{T}^2$. In particular, $\mathcal{N}(\hat{H}_h, \tau_h)$ is nonempty. Along any sequence $h \rightarrow 0^+$, one can extract a subsequence such that (3.2) and (3.5) hold for some ν .

For later use, we also record the semiclassical objects associated with the initial data. Since $\|u_h^{(0)}\|_{L^2(\mathbb{T}^2, L)} = 1$, one may extract a subsequence such that

$$\int_{\mathbb{T}^2} b(x) |u_h^{(0)}(x)|^2 dx \longrightarrow \int_{\mathbb{T}^2} b(x) \nu_0(dx), \quad \forall b \in C^\infty(\mathbb{T}^2), \quad (3.6)$$

for some probability measure ν_0 on $\mathbb{T}^2$.

The goal of this article is to describe elements of the set $\mathcal{N}(\hat{H}_h, \tau_h)$ depending on the time scale τ_h and the initial data used to generate it. In order to do this, we will now lift the measure to the phase space using the Weyl magnetic quantization introduced in [44] and recalled in Section 2.

3.2 Magnetic Wigner distributions and phase-space lifts

To lift the previous limits to phase space, we use the magnetic Weyl quantization. For $a \in \mathcal{C}_c^\infty(\mathbb{R} \times T^*\mathbb{T}^2)$, we define the *time-dependent magnetic Wigner distribution* by

$$\langle W_h^B(\tau_h), a \rangle := \int_{\mathbb{R}} \langle \text{Op}_h^B(a(t)) u_h(t\tau_h), u_h(t\tau_h) \rangle_{L^2(\mathbb{T}^2, L)} dt. \quad (3.7)$$

The next lemma collects basic compactness and positivity properties.

Lemma 3.2. *Let $(u_h)_{h \rightarrow 0^+}$ solve (3.1). Then, the following holds:*

1. *the family $(W_h^B(\tau_h))_{h \rightarrow 0^+}$ is bounded in $\mathcal{D}'(\mathbb{R} \times T^*\mathbb{T}^2)$;*
2. *any accumulation point W^B of $(W_h^B(\tau_h))_{h \rightarrow 0^+}$ is a nonnegative Radon measure on $\mathbb{R} \times T^*\mathbb{T}^2$.*

Proof. Boundedness follows from the Calderón–Vaillancourt Theorem 2.2 in the torus-adapted magnetic calculus; see [44, §3.4]. Indeed, for a supported in t on a compact set $[t_1, t_2] \subset \mathbb{R}$, this theorem implies that

$$|\langle W_h^B(\tau_h), a \rangle| \leq C_0 |t_2 - t_1| \sum_{|\alpha|+|\beta| \leq N_0} \sup_{t \in [t_1, t_2]} \|\partial_x^\alpha \partial_\xi^\beta a(t)\|_\infty,$$

uniformly for h small and where C_0 and N_0 are constants independent of the symbol a . Positivity is obtained from the easy Gårding's inequality: let $a \in \mathcal{C}_c^\infty(\mathbb{R} \times T^*\mathbb{T}^2)$ be nonnegative and let $[t_1, t_2] \subset \mathbb{R}$ be a compact interval containing the support of a in the time variable. For $\varepsilon > 0$, define for each $t \in [t_1, t_2]$

$$a_\varepsilon(t, x, \xi) := \sqrt{a(t, x, \xi) + \varepsilon}.$$

Then $(a_\varepsilon(t))_{t \in [t_1, t_2]}$ is a bounded family in $\mathcal{S}^0$. Using the adjointness (2.11) and composition (2.12) rules in the magnetic calculus,

$$\|\text{Op}_h^B(a_\varepsilon) u_h(t\tau_h)\|_{L^2}^2 = \langle \text{Op}_h^B(a_\varepsilon)^* \text{Op}_h^B(a_\varepsilon) u_h(t\tau_h), u_h(t\tau_h) \rangle = \langle \text{Op}_h^B(a + \varepsilon) u_h(t\tau_h), u_h(t\tau_h) \rangle + \mathcal{O}_\varepsilon(h),$$

uniformly for $t \in [t_1, t_2]$. Hence, for every nonnegative $a \in \mathcal{C}_c^\infty(\mathbb{R} \times T^*\mathbb{T}^2)$ and for every $\varepsilon > 0$, $\langle W^B, a \rangle \geq -|t_2 - t_1|\varepsilon$, which allows us to deduce that the limit distribution W^B is nonnegative, hence a measure. $\square$

The previous lemma only yields the existence of a nonnegative phase-space measure. Under the spectral localization assumption on the initial data, one can further localize its support in the energy variable.

Lemma 3.3 (Support of the limit measure). *Suppose that the initial data $(u_h^{(0)})_{h \rightarrow 0^+}$ satisfy the following spectral localization assumption:*

$$\lim_{h \rightarrow 0^+} \|\mathbb{1}_{[E_1, E_2]}(\widehat{H}_h)u_h^{(0)} - u_h^{(0)}\|_{L^2} = 0. \quad (3.8)$$

Then, the measure W^B is supported in $\mathbb{R} \times \mathbb{T}^2 \times \Omega_{E_1, E_2}$.

Proof. Let $(u_h)_{h \rightarrow 0^+}$ be a solution to (3.1), i.e.

$$u_h(t) = e^{-it\widehat{H}_h/h}u_h^{(0)}.$$

Since the pseudodifferential calculus developed in Section 2 is only available for smooth functions, we first need to show that the spectral projection assumption (3.8) can be written in terms of smooth functions. This will then allow us to apply the functional calculus developed in the Appendix A, in particular Corollary A.4. We first show that for $\kappa \in \mathcal{C}_c^\infty(\mathbb{R})$ satisfying $0 \leq \kappa \leq 1$ and $\kappa \equiv 1$ on $[E_1, E_2]$ assumption (3.8) implies that

$$\|(\text{Id} - \kappa(\widehat{H}_h))u_h^{(0)}\|_{L^2} \xrightarrow{h \rightarrow 0^+} 0. \quad (3.9)$$

Indeed, since $\kappa \equiv 1$ on $[E_1, E_2]$, one has $(1 - \kappa)\mathbb{1}_{[E_1, E_2]} = 0$, hence, by the spectral theorem applied to the self-adjoint operator $\widehat{H}_h$, one has

$$(\text{Id} - \kappa(\widehat{H}_h))\mathbb{1}_{[E_1, E_2]}(\widehat{H}_h) = 0.$$

Therefore, for any u , one gets

$$(\text{Id} - \kappa(\widehat{H}_h))u = (\text{Id} - \kappa(\widehat{H}_h))(\text{Id} - \mathbb{1}_{[E_1, E_2]}(\widehat{H}_h))u.$$

Moreover, $0 \leq \kappa \leq 1$ implies $\|\text{Id} - \kappa(\widehat{H}_h)\|_{L^2 \rightarrow L^2} \leq 1$. Hence using (3.8) leads to (3.9). By unitarity of the propagator, the following also holds

$$\|(\text{Id} - \kappa(\widehat{H}_h))u_h(t)\|_{L^2} = \|(\text{Id} - \kappa(\widehat{H}_h))u_h^{(0)}\|_{L^2}. \quad (3.10)$$

Let $a \in \mathcal{C}_c^\infty(\mathbb{T}^2 \times \mathbb{R}^2)$ such that $\text{supp}(a) \subset \mathbb{T}^2 \times \{\xi \in \mathbb{R}^2 \mid H(\xi) \notin [E_1, E_2]\}$ and set $\mathcal{K} := \{H(\xi) : (x, \xi) \in \text{supp}(a)\}$. To prove the lemma, it is enough to show that

$$\langle \text{Op}_h^B(a)u_h(t), u_h(t) \rangle \xrightarrow{h \rightarrow 0^+} 0 \quad (3.11)$$

uniformly in $t \in \mathbb{R}$. Indeed, for every $\psi \in \mathcal{C}_c^\infty(\mathbb{R})$, one then has

$$\langle W_h^B(\tau_h), \psi(t)a(x, \xi) \rangle = \int_{\mathbb{R}} \psi(t) \langle \text{Op}_h^B(a)u_h(t\tau_h), u_h(t\tau_h) \rangle dt \xrightarrow{h \rightarrow 0^+} 0.$$

As this is valid for any a with the above support properties, it implies that W^B does not charge the region

$$\mathbb{R} \times \mathbb{T}^2 \times \{\xi \in \mathbb{R}^2 : H(\xi) \notin [E_1, E_2]\}.$$

We now prove (3.11). Since $\text{supp}(a)$ is compact and H is continuous, $\mathcal{K}$ is a compact subset of $\mathbb{R} \setminus [E_1, E_2]$. Moreover, by the support assumption on a , one has $\mathcal{K} \cap [E_1, E_2] = \emptyset$. Hence, the distance is positive:

$$d := \text{dist}(\mathcal{K}, [E_1, E_2]) > 0.$$

Choose $\kappa \in \mathcal{C}_c^\infty(\mathbb{R})$ such that

$$0 \leq \kappa \leq 1, \quad \kappa \equiv 1 \text{ on } [E_1, E_2], \quad \text{supp}(\kappa) \subset (E_1 - \frac{d}{3}, E_2 + \frac{d}{3}).$$

Next choose $\varphi \in \mathcal{C}_c^\infty(\mathbb{R})$ such that

$$\varphi \equiv 1 \text{ on a neighbourhood of } \mathcal{K}, \quad \text{supp}(\varphi) \subset (-\infty, E_1 - \frac{d}{2}] \cup [E_2 + \frac{d}{2}, \infty).$$

Since $\text{supp}(\varphi) \cap \text{supp}(\kappa) = \emptyset$, one has $\varphi\kappa \equiv 0$. Therefore, by the spectral theorem,

$$\varphi(\hat{H}_h)\kappa(\hat{H}_h) = 0.$$

Moreover, since $\varphi \equiv 1$ near $\mathcal{K}$, we have the pointwise identity

$$a(x, \xi) = a(x, \xi) \varphi(H(\xi)) \quad \text{for all } \xi \in \mathbb{R}^2. \quad (3.12)$$

We also have

$$\varphi(\hat{H}_h)u_h(t) = \varphi(\hat{H}_h)(\text{Id} - \kappa(\hat{H}_h))u_h(t).$$

Since (3.10) holds and $\varphi(\hat{H}_h)$ is bounded on L^2 , one has

$$\|\varphi(\hat{H}_h)u_h(t)\|_{L^2} \leq \|\varphi(\hat{H}_h)\|_{\mathcal{L}(L^2)} \|(\text{Id} - \kappa(\hat{H}_h))u_h(t)\|_{L^2} = \mathcal{O}\left(\|(\text{Id} - \kappa(\hat{H}_h))u_h^{(0)}\|_{L^2}\right), \quad (3.13)$$

uniformly in $t \in \mathbb{R}$. By (3.9), the right-hand side tends to 0 as $h \rightarrow 0^+$. One gets

$$\|\varphi(\hat{H}_h)u_h(t)\|_{L^2} \xrightarrow{h \rightarrow 0} 0. \quad (3.14)$$

Using (3.12), the symbolic composition in the magnetic calculus (2.14) and the Calderón–Vaillancourt Theorem 2.2, one has

$$\text{Op}_h^B(a) = \text{Op}_h^B(a) \text{Op}_h^B(\varphi \circ H) + \mathcal{O}_{L^2 \rightarrow L^2}(h). \quad (3.15)$$

On the other hand, applying Corollary A.4, one gets

$$\text{Op}_h^B(\varphi \circ H) = \varphi(\hat{H}_h) + \mathcal{O}_{L^2 \rightarrow L^2}(h). \quad (3.16)$$

Combining (3.15)–(3.16) gives

$$\langle \text{Op}_h^B(a)u_h(t), u_h(t) \rangle = \langle \text{Op}_h^B(a) \varphi(\hat{H}_h)u_h(t), u_h(t) \rangle + \mathcal{O}(h).$$

By Calderón–Vaillancourt, $\text{Op}_h^B(a)$ is uniformly bounded on L^2 (because $a \in \mathcal{S}^0$ due to its compact support). We infer by Cauchy–Schwarz that

$$|\langle \text{Op}_h^B(a)u_h(t), u_h(t) \rangle| \leq C \|\varphi(\hat{H}_h)u_h(t)\|_{L^2} + \mathcal{O}(h) \xrightarrow{h \rightarrow 0} 0$$

thanks to (3.14). This proves (3.11) and concludes the proof of the lemma. $\square$

Now that the phase-space distribution W^B has been constructed, we make explicit its link with the configuration-space limits ν_t introduced above. In particular, the next lemma shows that W^B admits a disintegration with respect to time,

$$W^B(dt, dx, d\xi) = \nu_t^B(dx, d\xi) \otimes dt,$$

where $(\nu_t^B)_{t \in \mathbb{R}}$ is a measurable family of probability measures on $T^*\mathbb{T}^2$. We denote by

$$\pi_x : (x, \xi) \in \mathbb{T}^2 \times \mathbb{R}^2 \mapsto x, \quad (3.17)$$

the canonical projection onto the configuration space. Then, the following lemma also identifies the projection of ν_t^B onto $\mathbb{T}^2$ with the original time-dependent quantum limit ν_t .

Lemma 3.4 (Time disintegration and projection). *Let $(u_h)_{h \rightarrow 0+}$ solve (3.1) such that the initial data are spectrally localized with respect to the magnetic operator $\widehat{H}_h$, in the sense of (3.8). Then any accumulation point W^B in $\mathcal{D}'(\mathbb{R} \times T^*\mathbb{T}^2)$ of $W_h^B(\tau_h)$ is of the form*

$$W^B(dt, dx, d\xi) = \nu_t^B(dx, d\xi) \otimes dt, \quad (3.18)$$

where $(\nu_t^B)_{t \in \mathbb{R}}$ is a measurable family of probability measures on $\mathbb{T}^2 \times \Omega_{E_1, E_2}$. Moreover, for almost every $t \in \mathbb{R}$, the projection onto the base equals the configuration-space limit:

$$(\pi_x)_* \nu_t^B = \nu_t.$$

Proof. Let

$$\tilde{\pi}_t : (t, x, \xi) \in \mathbb{R} \times T^*\mathbb{T}^2 \mapsto t \quad (3.19)$$

be the projection onto the time variable and define ω^B the time marginal of W^B as follows

$$\omega^B = (\tilde{\pi}_t)_* W^B. \quad (3.20)$$

From the ellipticity property (2.19) of the Hamiltonian H , the set $\mathbb{T}^2 \times H^{-1}([E_1, E_2])$ is compact. Since W^B is a Radon measure, hence a σ -finite measure, for every $I \subset \mathbb{R}$ compact, using Lemma 3.3 one has

$$\omega^B(I) = W^B(I \times \mathbb{T}^2 \times H^{-1}([E_1, E_2])) < \infty. \quad (3.21)$$

Hence, the marginal ω^B is a σ -finite measure and applying the disintegration Theorem A.5 with respect to ω^B leads to $W^B(dt, dx, d\xi) = \nu_t^B(dx, d\xi) \omega^B(dt)$ with $(\nu_t^B)_t$ a ω^B -measurable family of probability measures on $\mathbb{T}^2 \times \Omega_{E_1, E_2}$. We now identify ω^B and rule out a loss of mass at infinity in the momentum variable. Fix $\kappa \in \mathcal{C}_c^\infty(\mathbb{R})$ such that $\kappa \equiv 1$ on a neighborhood of $[E_1, E_2]$. By the functional calculus in the magnetic pseudodifferential calculus (see Corollary A.4), one has

$$\text{Op}_h^B(\kappa(H)) = \kappa(\widehat{H}_h) + \mathcal{O}(h) \quad \text{in } \mathcal{L}(L^2). \quad (3.22)$$

Moreover, the spectral localization assumption (3.9) implies that

$$\kappa(\widehat{H}_h) u_h^{(0)} = u_h^{(0)} + o_{L^2}(1).$$

Since $\kappa(\widehat{H}_h)$ commutes with the propagator, it follows that

$$\kappa(\widehat{H}_h) u_h(t\tau_h) = u_h(t\tau_h) + o_{L^2}(1), \quad (3.23)$$

uniformly with respect to $t \in \mathbb{R}$. Let $\zeta \in \mathcal{C}_c^\infty(\mathbb{R})$ satisfy

$$0 \leq \zeta \leq 1, \quad \zeta \equiv 1 \text{ on } [-1, 1], \quad \zeta \equiv 0 \text{ on } \mathbb{R} \setminus [-2, 2],$$

and, for $R > 0$, set

$$\zeta_R(\xi) := \zeta\left(\frac{|\xi|}{R}\right).$$

Since H is elliptic and κ is compactly supported, the symbol $\kappa(H(\xi))$ is compactly supported in ξ . Hence, there exists $R_0 > 0$ such that, for every $R \geq R_0$,

$$(1 - \zeta_R(\xi))\kappa(H(\xi)) = 0. \quad (3.24)$$

Using (3.22), the composition formula (2.14), and the Calderón–Vaillancourt Theorem, we obtain, for every fixed $R \geq R_0$,

$$\begin{aligned} \text{Op}_h^B(1 - \zeta_R)\kappa(\widehat{H}_h) &= \text{Op}_h^B(1 - \zeta_R)\text{Op}_h^B(\kappa(H)) + \mathcal{O}_{L^2 \rightarrow L^2}(h) \\ &= \text{Op}_h^B\left((1 - \zeta_R)\kappa(H)\right) + \mathcal{O}_{L^2 \rightarrow L^2}(h) \\ &= \mathcal{O}_{L^2 \rightarrow L^2}(h). \end{aligned}$$

Together with (3.23), this yields

$$\sup_{t \in \mathbb{R}} \left| \left\langle \text{Op}_h^B(1 - \zeta_R)u_h(t\tau_h), u_h(t\tau_h) \right\rangle_{L^2} \right| = o(1) \quad (3.25)$$

as $h \rightarrow 0^+$, for every fixed $R \geq R_0$. Now fix $\psi \in \mathcal{C}_c^\infty(\mathbb{R})$. Since

$$1 = \zeta_R(\xi) + (1 - \zeta_R(\xi)),$$

the normalization of $u_h(t\tau_h)$ and (3.25) imply that

$$\begin{aligned} \langle W_h^B(\tau_h), \psi \times \zeta_R \rangle &= \int_{\mathbb{R}} \psi(t) \left\langle \text{Op}_h^B(\zeta_R)u_h(t\tau_h), u_h(t\tau_h) \right\rangle_{L^2} dt \\ &= \int_{\mathbb{R}} \psi(t) dt + o(1). \end{aligned}$$

Passing to the limit along the subsequence defining W^B , we obtain

$$\int_{\mathbb{R} \times T^* \times \mathbb{T}^2} \psi(t) \zeta_R(\xi) W^B(dt, dx, d\xi) = \int_{\mathbb{R}} \psi(t) dt. \quad (3.26)$$

For R sufficiently large, $\zeta_R \equiv 1$ on Ω_{E_1, E_2} . Since W^B is supported in $\mathbb{R} \times \mathbb{T}^2 \times \Omega_{E_1, E_2}$, the left-hand side of (3.26) is equal to

$$\int_{\mathbb{R}} \psi(t) \omega^B(dt).$$

Therefore,

$$\forall \psi \in \mathcal{C}_c^\infty(\mathbb{R}), \quad \int_{\mathbb{R}} \psi(t) \omega^B(dt) = \int_{\mathbb{R}} \psi(t) dt,$$

which leads to

$$\omega^B = dt.$$

Since ν_t^B is a probability measure for ω^B -almost every t , and since $\omega^B = dt$, we finally obtain

$$W^B(dt, dx, d\xi) = \nu_t^B(dx, d\xi) \otimes dt, \quad (3.27)$$

where $(\nu_t^B)_{t \in \mathbb{R}}$ is a measurable family of probability measures on $\mathbb{T}^2 \times \Omega_{E_1, E_2}$, defined for almost every $t \in \mathbb{R}$.

To prove that $(\pi_x)_* \nu_t^B = \nu_t$, fix $\psi \in \mathcal{C}_c^\infty(\mathbb{R})$ and $b \in \mathcal{C}^\infty(\mathbb{T}^2)$. Then one has

$$\int_{\mathbb{R}} \psi(t) \langle (\pi_x)_* \nu_t^B, b \rangle dt = \int_{\mathbb{R}} \psi(t) \left[\int_{T^*\mathbb{T}^2} b(x) \nu_t^B(dx, d\xi) \right] dt.$$

Insert the energy cut-off $\kappa(H(\xi))$, which is equal to 1 on the support of ν_t^B :

$$\begin{aligned} \int_{\mathbb{R}} \psi(t) \left[\int_{T^*\mathbb{T}^2} b(x) \nu_t^B(dx, d\xi) \right] dt &= \int_{\mathbb{R}} \psi(t) \left[\int_{T^*\mathbb{T}^2} b(x) \kappa(H(\xi)) \nu_t^B(dx, d\xi) \right] dt \\ &= \lim_{h \rightarrow 0^+} \int_{\mathbb{R}} \psi(t) \langle \text{Op}_h^B(b(x) \kappa(H(\xi))) u_h(t\tau_h), u_h(t\tau_h) \rangle_{L^2} dt. \end{aligned}$$

Using (3.22) and the composition formula at first order (2.14) together with the Calderón-Vaillancourt Theorem, one has

$$\begin{aligned} &\int_{\mathbb{R}} \psi(t) \langle \text{Op}_h^B(b(x) \kappa(H(\xi))) u_h(t\tau_h), u_h(t\tau_h) \rangle_{L^2} dt \\ &= \int_{\mathbb{R}} \psi(t) \langle b(x) \kappa(\widehat{H}_h) e^{-i \frac{t\tau_h}{h} \widehat{H}_h} u_h^{(0)}, e^{-i \frac{t\tau_h}{h} \widehat{H}_h} u_h^{(0)} \rangle_{L^2} dt + \mathcal{O}(h). \end{aligned}$$

By unitarity of the propagator (2.24) and the property (3.9) on the spectral localization, we can replace $\kappa(\widehat{H}_h) u_h^{(0)}$ by $u_h^{(0)}$ in the limit as $h \rightarrow 0^+$. Thus,

$$\int_{\mathbb{R}} \psi(t) \left[\int_{T^*\mathbb{T}^2} b(x) \nu_t^B(dx, d\xi) \right] dt = \lim_{h \rightarrow 0^+} \int_{\mathbb{R}} \psi(t) \langle b(x) u_h(t\tau_h), u_h(t\tau_h) \rangle_{L^2} dt.$$

By the definition of the time-dependent quantum limits $(\nu_t)_{t \in \mathbb{R}}$, the right-hand side is equal to

$$\int_{\mathbb{R}} \psi(t) \left[\int_{\mathbb{T}^2} b(x) \nu_t(dx) \right] dt.$$

Since this holds for all ψ and b , we conclude that

$$(\pi_x)_* \nu_t^B = \nu_t \quad \text{for almost every } t \in \mathbb{R}. \quad (3.28)$$

□

Remark 3.5. *In contrast with the h -oscillation assumptions commonly used in long-time semi-classical analysis, we impose here the stronger spectral localization condition*

$$\lim_{h \rightarrow 0^+} \left\| \mathbb{1}_{[E_1, E_2]}(\widehat{H}_h) u_h^{(0)} - u_h^{(0)} \right\|_{L^2(\mathbb{T}^2, L)} = 0.$$

Under our assumptions on the energy window $[E_1, E_2]$, this guarantees that any semiclassical measure associated with the initial data is supported in

$$\Omega_{E_1, E_2} = H^{-1}([E_1, E_2]).$$

Moreover, the energy window $[E_1, E_2]$ is chosen so that no critical point of H lies in Ω_{E_1, E_2} . Therefore, the measure ν_t^B does not charge the critical set $\{\nabla H = 0\}$. This restriction is convenient for our analysis, since it excludes a possible component of the limiting measures concentrated on $\{\nabla H = 0\}$, for which no general description would be directly available through the upcoming arguments if one assumes only h -oscillation. In this sense, the spectral localization, together with the assumption $\nabla H \neq 0$ allows us to restrict the analysis from the beginning to the regular part of the energy surface.

In order to formulate the propagation properties of ν_t^B in terms of an initial phase-space measure, we also introduce the auxiliary space-time distribution associated with the initial data: for $a \in \mathcal{C}_c^\infty(T^*\mathbb{T}^2)$,

$$\langle W_{h,0}^B, a \rangle := \langle \text{Op}_h^B(a) u_h^{(0)}, u_h^{(0)} \rangle_{L^2(\mathbb{T}^2, L)}. \quad (3.29)$$

Possibly after extracting a further subsequence and using the same arguments as in Lemmas 3.2 and 3.4, $(W_{h,0}^B)_{h \rightarrow 0+}$ converges to a probability measure ν_0^B on $\mathbb{T}^2 \times \Omega_{E_1, E_2}$ satisfying

$$(\pi_x)_* \nu_0^B = \nu_0. \quad (3.30)$$

Before turning to the propagation properties of the family $(\nu_t^B)_{t \in \mathbb{R}}$, we record a complementary rigidity property of its ξ -marginal. Since the principal Hamiltonian H depends only on ξ , the ξ -distribution can only be affected by the subprincipal perturbation hR_h , which becomes visible at the scale $1/h$.

Lemma 3.6 (Description of the ξ -marginal). *Let $\pi_\xi : (x, \xi) \in T^*\mathbb{T}^2 \mapsto \xi$ be the canonical projection onto the frequency variable. For almost every $t \in \mathbb{R}$, define*

$$\lambda_t := (\pi_\xi)_* \nu_t^B \quad \text{and} \quad \lambda_0 := (\pi_\xi)_* \nu_0^B. \quad (3.31)$$

Then for almost every $t \in \mathbb{R}$, the following holds:

1. There exists a λ_t -measurable family $(\nu_{t,\xi}^B)_\xi$ of probability measures on $\mathbb{T}^2$ such that

$$\nu_t^B(dx, d\xi) = \nu_{t,\xi}^B(dx) \otimes \lambda_t(d\xi). \quad (3.32)$$

2. There exist a probability measure σ^B on $[E_1, E_2]$ and a σ^B -measurable family $(\lambda_{t,E})_{E \in [E_1, E_2]}$ of probability measures supported on $\mathbb{S}_E$, such that

$$\lambda_t(d\xi) = \int_{E_1}^{E_2} \lambda_{t,E}(d\xi) \sigma^B(dE). \quad (3.33)$$

Moreover, the measure σ^B is given by

$$\sigma^B = H_* \lambda_t = H_* \lambda_0 \quad \text{for a.e. } t \in \mathbb{R}. \quad (3.34)$$

3. If the time scale satisfies $\tau_h \ll h^{-1}$, then one has

$$\lambda_t = \lambda_0 := (\pi_\xi)_* \nu_0^B \quad \text{for a.e. } t \in \mathbb{R}. \quad (3.35)$$

Remark 3.7. Equivalently, combining items 1 and 2 for $a \in \mathcal{C}_c^\infty(T^*\mathbb{T}^2)$, leads, for almost every $t \in \mathbb{R}$, to

$$\int_{T^*\mathbb{T}^2} a(x, \xi) \nu_t^B(dx, d\xi) = \int_{E_1}^{E_2} \int_{\mathbb{S}_E} \int_{\mathbb{T}^2} a(x, \xi) \nu_{t, \xi}^B(dx) \lambda_{t, E}(d\xi) \sigma^B(dE).$$

The remainder of the analysis will consist in identifying these conditional measures on sufficiently long time scales: first $\nu_{t, \xi}^B$ in the configuration variable, and then $\lambda_{t, E}$ along the energy levels.

Proof. Since ν_t^B is a probability measure on $T^*\mathbb{T}^2$, one can apply the disintegration Theorem A.5, with respect to λ_t . This gives a λ_t -measurable family $(\nu_{t, \xi}^B)_\xi$ of probability measures on $\mathbb{T}^2$ such that

$$\nu_t^B(dx, d\xi) = \nu_{t, \xi}^B(dx) \otimes \lambda_t(d\xi). \quad (3.36)$$

We now prove the second item. Let $\psi, f \in \mathcal{C}_c^\infty(\mathbb{R})$. By the semiclassical functional calculus, as in the proof of Lemma 3.4, one has

$$\int_{\mathbb{R}} \psi(t) \left\langle \text{Op}_h^B(f \circ H) u_h(t\tau_h), u_h(t\tau_h) \right\rangle_{L^2} dt = \int_{\mathbb{R}} \psi(t) \left\langle f(\widehat{H}_h) u_h(t\tau_h), u_h(t\tau_h) \right\rangle_{L^2} dt + \mathcal{O}(h). \quad (3.37)$$

Using the unitarity of the propagator and the fact that $f(\widehat{H}_h)$ commutes with the propagator, one can write (3.37) as follows

$$\int_{\mathbb{R}} \psi(t) \left\langle \text{Op}_h^B(f \circ H) u_h(t\tau_h), u_h(t\tau_h) \right\rangle_{L^2} dt = \left(\int_{\mathbb{R}} \psi(t) dt \right) \left\langle \text{Op}_h^B(f \circ H) u_h^{(0)}, u_h^{(0)} \right\rangle_{L^2} + \mathcal{O}(h).$$

Passing to the limit along the subsequence defining λ_t and λ_0 , one gets

$$\int_{\mathbb{R}} \psi(t) \left(\int_{\mathbb{R}^2} f(H(\xi)) \lambda_t(d\xi) \right) dt = \left(\int_{\mathbb{R}} \psi(t) dt \right) \int_{\mathbb{R}^2} f(H(\xi)) \lambda_0(d\xi). \quad (3.38)$$

For almost every $t \in \mathbb{R}$, set

$$\sigma_t^B := H_* \lambda_t, \quad \sigma_0^B := H_* \lambda_0.$$

Since λ_t and λ_0 are probability measures supported in Ω_{E_1, E_2} , the measures σ_t^B and σ_0^B are probability measures supported in $[E_1, E_2]$. Hence, (3.38) becomes

$$\int_{\mathbb{R}} \psi(t) \left(\int_{E_1}^{E_2} f(E) \sigma_t^B(dE) \right) dt = \left(\int_{\mathbb{R}} \psi(t) dt \right) \int_{E_1}^{E_2} f(E) \sigma_0^B(dE).$$

Hence, for every fixed $f \in \mathcal{C}_c^\infty(\mathbb{R})$, one has

$$\int_{E_1}^{E_2} f(E) \sigma_t^B(dE) = \int_{E_1}^{E_2} f(E) \sigma_0^B(dE)$$

for almost every $t \in \mathbb{R}$. Since this holds for every $f \in \mathcal{C}_c^\infty(\mathbb{R})$, it follows that

$$\sigma_t^B = \sigma_0^B \quad \text{for almost every } t \in \mathbb{R}.$$

Setting

$$\sigma^B := \sigma_0^B = H_* \lambda_0,$$

we have proved that

$$H_* \lambda_t = H_* \lambda_0 = \sigma^B \quad \text{for almost every } t \in \mathbb{R}. \quad (3.39)$$

For such a t , apply the disintegration Theorem A.5 to the probability measure λ_t and to the map

$$H : \Omega_{E_1, E_2} \longrightarrow [E_1, E_2].$$

Since $H_* \lambda_t = \sigma^B$, there exists a σ^B -measurable family $(\lambda_{t,E})_{E \in [E_1, E_2]}$ of probability measures supported on $\mathbb{S}_E$ such that (3.33) holds.

To prove the third item, fix $a \in \mathcal{C}_c^\infty(\mathbb{R}^2)$ and set

$$F_h(t) := \langle \text{Op}_h^B(a) u_h(t\tau_h), u_h(t\tau_h) \rangle.$$

Using the fact that $(u_h)_{h \rightarrow 0+}$ is a solution to (3.1), one obtains

$$\frac{d}{dt} F_h(t) = -\tau_h \left\langle \frac{i}{h} [\text{Op}_h^B(a), \widehat{H}_h] u_h(t\tau_h), u_h(t\tau_h) \right\rangle_{L^2}. \quad (3.40)$$

Using the commutator expansion (2.15), the Calderón–Vaillancourt Theorem 2.2 and the fact that a and H depend only on the ξ -variable, one has

$$\frac{i}{h} [\text{Op}_h^B(a), \widehat{H}_h] = h \text{Op}_h^B(\{a, R_h\}) + \mathcal{O}_{L^2 \rightarrow L^2}(h). \quad (3.41)$$

Since a is compactly supported, the symbol $\{a, R_h\}$ belongs to $\mathcal{S}^0$ uniformly in h . Hence, combining (3.40) and (3.41) with the Calderón–Vaillancourt Theorem and the unitarity of the propagator, one gets

$$\left| \frac{d}{dt} F_h(t) \right| \leq C_a h \tau_h, \quad (3.42)$$

uniformly in t . Integrating (3.42) between 0 and t yields

$$|F_h(t) - F_h(0)| \leq C_a |t| h \tau_h. \quad (3.43)$$

Let now $\psi \in \mathcal{C}_c^\infty(\mathbb{R})$. It follows from (3.43) that

$$\left| \int_{\mathbb{R}} \psi(t) (F_h(t) - F_h(0)) dt \right| \leq C_a h \tau_h \int_{\mathbb{R}} |t \psi(t)| dt.$$

Since $\tau_h \ll h^{-1}$, the right-hand side tends to 0 as $h \rightarrow 0^+$. Moreover, by the definition of the initial momentum marginal λ_0 ,

$$F_h(0) = \langle \text{Op}_h^B(a) u_h^{(0)}, u_h^{(0)} \rangle_{L^2} \longrightarrow \int_{\mathbb{R}^2} a(\xi) \lambda_0(d\xi).$$

Consequently, one has

$$\int_{\mathbb{R}} \psi(t) F_h(t) dt \longrightarrow \left(\int_{\mathbb{R}} \psi(t) dt \right) \left(\int_{\mathbb{R}^2} a(\xi) \lambda_0(d\xi) \right).$$

On the other hand, by the definition of the time-dependent magnetic semiclassical measure, one has

$$\int_{\mathbb{R}} \psi(t) F_h(t) dt \longrightarrow \int_{\mathbb{R}} \psi(t) \left(\int_{\mathbb{R}^2} a(\xi) \lambda_t(d\xi) \right) dt.$$

Hence, one obtains

$$\int_{\mathbb{R}} \psi(t) \left(\int_{\mathbb{R}^2} a(\xi) \lambda_t(d\xi) - \int_{\mathbb{R}^2} a(\xi) \lambda_0(d\xi) \right) dt = 0.$$

Since this holds for every $\psi \in \mathcal{C}_c^\infty(\mathbb{R})$ and for every $a \in \mathcal{C}_c^\infty(\mathbb{R}^2)$, it follows that

$$\lambda_t = \lambda_0 \quad \text{for almost every } t \in \mathbb{R},$$

which proves (3.35). $\square$

The previous lemma shows that the energy marginal of the limiting measure is preserved along the evolution and below the scale h^{-1} , the ξ -marginal of the lifted measure is entirely determined by the initial data. We also proved that the disintegration of ν_t^B with respect to the energy variable involves a measure on $\mathbb{R}$ which is completely determined by the initial data.

In order to describe $\mathcal{N}(\widehat{H}_h, \tau_h)$, it is sufficient to describe the x -marginal of ν_t^B which is easier to work with. Having identified the lifted family of phase-space measures $(\nu_t^B)_{t \in \mathbb{R}}$ and its link with the configuration-space limits $(\nu_t)_{t \in \mathbb{R}}$, we now describe the dynamical constraints satisfied by ν_t^B . More precisely, we study how these constraints depend on the observation scale τ_h . This will clarify in which regimes the limiting measures are transported by the Hamiltonian flow of the principal symbol H , and in which regimes they satisfy an invariance property under this flow. It will also illustrate the kind of arguments that will have to be refined later on.

Lemma 3.8 (Propagation and invariance). *Let $(\nu_t^B)_{t \in \mathbb{R}}$ be an accumulation point of $(W_h^B(\tau_h))_{h \rightarrow 0+}$ in the sense of (3.27) and let φ_H^t be the Hamiltonian flow of $H(\xi)$ on $T^*\mathbb{T}^2$, namely*

$$\varphi_H^t(x, \xi) = (x + t\nabla H(\xi), \xi).$$

Then, the following holds:

1. *If $\tau_h \ll 1$, then ν_t^B is independent of t , i.e., $\nu_t^B = \nu_0^B$.*
2. *If $\tau_h = 1$, then ν_t^B is transported by the Hamiltonian flow of H , i.e. $\nu_t^B = (\varphi_H^t)_* \nu_0^B$.*
3. *If $\tau_h \gg 1$, then for a.e. t , the measure ν_t^B is invariant under the flow of H , i.e. for all $s \in \mathbb{R}$,*

$$(\varphi_H^s)_* \nu_t^B = \nu_t^B.$$

Remark 3.9. *Note that the relations given in the first and second point of Lemma 3.8 can be written in terms of Fourier coefficients for the measure ν_t^B . In the sense of distributions, one can decompose ν_t^B as a Fourier series:*

$$\nu_t^B(x, \xi) = \sum_{k \in \mathbb{Z}^2} \widehat{\nu}_t^B(k, \xi) e^{2i\pi k \cdot x}. \quad (3.44)$$

Hence, the first relation of the previous lemma gives us that for almost every $t \in \mathbb{R}$, one can identify the k -th Fourier coefficient of ν_t^B with its initial datum ν_0^B . The second one leads us to the following equality, for almost every $t \in \mathbb{R}$, one has

$$\widehat{\nu}_t^B(k, \xi) = e^{-2i\pi tk \cdot \nabla H(\xi)} \widehat{\nu}_0^B(k, \xi),$$

in particular, (3.44) becomes

$$\nu_t^B(x, \xi) = \sum_{k \in \mathbb{Z}^2} e^{-2i\pi tk \cdot \nabla H(\xi)} \widehat{\nu}_0^B(k, \xi) e^{2i\pi k \cdot x}.$$

Proof. Fix $\psi \in \mathcal{C}_c^\infty(\mathbb{R})$ and $b \in \mathcal{C}_c^\infty(T^*\mathbb{T}^2)$. Differentiate $t \mapsto \langle \text{Op}_h^B(b) u_h(t\tau_h), u_h(t\tau_h) \rangle$ using the fact that $(u_h)_{h \rightarrow 0^+}$ is a solution to (3.1) and integrate by parts:

$$\int_{\mathbb{R}} \psi'(t) \langle \text{Op}_h^B(b) u_h(t\tau_h), u_h(t\tau_h) \rangle dt = \tau_h \int_{\mathbb{R}} \psi(t) \left\langle \frac{i}{h} [\text{Op}_h^B(b), \widehat{H}_h] u_h(t\tau_h), u_h(t\tau_h) \right\rangle dt.$$

Arguing as in (3.41), one has

$$\begin{aligned} \int_{\mathbb{R}} \psi'(t) \langle \text{Op}_h^B(b) u_h(t\tau_h), u_h(t\tau_h) \rangle dt &= \tau_h \int_{\mathbb{R}} \psi(t) \langle \text{Op}_h^B(\{b, H\}) u_h(t\tau_h), u_h(t\tau_h) \rangle dt \\ &\quad + h\tau_h \int_{\mathbb{R}} \psi(t) \langle \text{Op}_h^B(\{b, R_h\}) u_h(t\tau_h), u_h(t\tau_h) \rangle dt + \mathcal{O}(h\tau_h). \end{aligned} \quad (3.45)$$

Since $b \in \mathcal{C}_c^\infty(\mathbb{T}^*\mathbb{T}^2)$, the symbols $\{b, H\}$ and $\{b, R_h\}$ are both in $\mathcal{S}^0(\mathbb{T}^2 \times \mathbb{R}^2)$ so that one can use the Calderón–Vaillancourt Theorem 2.2. Then, if $\tau_h \rightarrow 0$ we obtain $\int \psi'(t) \langle \nu_t^B, b \rangle dt = 0$ for all ψ , hence ν_t^B is constant in t . If $\tau_h = 1$, we obtain the weak transport equation $\partial_t \nu_t^B = \{\nu_t^B, H\}$, which means precisely that the family $(\nu_t^B)_{t \in \mathbb{R}}$ is the image of the initial measure ν_0^B under the Hamiltonian flow, namely $\nu_t^B = (\varphi_H^t)_* \nu_0^B$. Finally, if $\tau_h \rightarrow +\infty$, one can divide by τ_h in (3.45) and let h go to 0 to obtain $\{\nu_t^B, H\} = 0$ for a.e. t i.e., invariance under the Hamiltonian flow φ_H^s . $\square$

3.3 Decomposition of invariant measures

The long-time regime $\tau_h \rightarrow +\infty$ is the main focus of this paper: in that case, ν_t^B is invariant under φ_H^s and can be analyzed further by decomposing phase space according to resonant directions of the integrable flow. This is the starting point of the next section. This invariant setting provides a natural and robust framework for further analysis. In the next part, we follow the strategy developed by Macià, Anantharaman-Macià and Anantharaman-Fermanian-Macià in [37, 4, 5, 2] to refine the description of the invariant measures ν_t^B , by introducing finer time scales and studying how additional asymptotic regimes influence their structure.

On $\mathbb{T}^2 \times \mathbb{R}^2$, the trajectories of the Hamiltonian flow

$$\varphi_H^s(x, \xi) = (x + s\nabla H(\xi), \xi)$$

are either dense, periodic or fixed depending on arithmetic properties of the direction $\nabla H(\xi)$. Following the approach of [38, 2], we decompose the invariant measure according to these dynamical regimes.

We denote by $\mathcal{L}_1$ the collection of primitive rank-one submodules $\Lambda \subset \mathbb{Z}^2$, i.e. such that $\dim \langle \Lambda \rangle = 1$ and $\langle \Lambda \rangle \cap \mathbb{Z}^2 = \Lambda$. For every $\Lambda \in \mathcal{L}_1$, we choose $\mathbf{e}_\Lambda \in \mathbb{Z}^2 \setminus \{0\}$ such that $\Lambda = \mathbb{Z}\mathbf{e}_\Lambda$, we set $L_\Lambda := |\mathbf{e}_\Lambda|$, and we denote by $\Lambda^\perp := \mathbb{R}\mathbf{e}_\Lambda^\perp$ the orthogonal line (where $\mathbf{e}_\Lambda^\perp$ is a fixed orthogonal vector of length L_Λ). We introduce the resonant sets

$$E_{\Lambda^\perp \setminus \{0\}} := \{\xi \in \mathbb{R}^2 : \nabla H(\xi) \in \Lambda^\perp \setminus \{0\}\}, \quad (3.46)$$

and the non-resonant set

$$\Omega_{\text{nr}} := \left\{ \xi \in \mathbb{R}^2 : k \cdot \nabla H(\xi) \neq 0 \text{ for all } k \in \mathbb{Z}^2 \setminus \{0\} \right\}. \quad (3.47)$$

For $\xi \in \Omega_{\text{nr}}$, the orbit $x + s\nabla H(\xi)$ is dense in $\mathbb{T}^2$, whereas $\xi \in E_{\Lambda^\perp}$ imposes a rational constraint on the direction of motion and leads to periodic (linear) trajectories.

To make this dichotomy quantitative, let $a \in \mathcal{C}_c^\infty(T^*\mathbb{T}^2)$ and denote by

$$\widehat{a}_k(\xi) := \int_{\mathbb{T}^2} a(x, \xi) e^{-2\pi i k \cdot x} dx, \quad k \in \mathbb{Z}^2,$$

its Fourier coefficients in the x -variable. For $\Lambda \in \mathcal{L}_1$, we define the Λ -averaging operator by

$$\mathcal{I}_\Lambda(a)(x, \xi) := \sum_{k \in \Lambda} \widehat{a}_k(\xi) e^{2\pi i k \cdot x}. \quad (3.48)$$

Equivalently, $\mathcal{I}_\Lambda(a)$ keeps only the Fourier modes belonging to the submodule Λ , and it may be interpreted as the time average of a along the linear flow when $\xi \in \Lambda^\perp$. More precisely, on the one hand, one has

$$\forall (x, \xi) \in \mathbb{T}^2 \times E_{\Lambda^\perp \setminus \{0\}}, \quad \lim_{T \rightarrow +\infty} \frac{1}{T} \int_0^T a(x + t\nabla H(\xi), \xi) dt = \mathcal{I}_\Lambda(a)(x, \xi), \quad (3.49)$$

and on the other hand, one finds

$$\forall (x, \xi) \in \mathbb{T}^2 \times \Omega_{\text{nr}}, \quad \lim_{T \rightarrow +\infty} \frac{1}{T} \int_0^T a(x + t\nabla H(\xi), \xi) dt = \int_{\mathbb{T}^2} a(y, \xi) dy. \quad (3.50)$$

Remark 3.10. For $\tau_h \gg 1$, Lemma 3.8 tells us that the semiclassical measure ν_t^B is invariant under the geodesic flow φ_H^s . Hence, the measure ν_t^B is a probability measure on $T^*\mathbb{T}^2$ that can be decomposed as follows:

$$\nu_t^B = \nu_t^B|_{\mathbb{T}^2 \times \Omega_{\text{nr}}} + \sum_{\Lambda \in \mathcal{L}_1} \nu_t^B|_{\mathbb{T}^2 \times E_{\Lambda^\perp \setminus \{0\}}} + \nu_t^B|_{\mathbb{T}^2 \times \{\nabla H(\xi)=0\}}. \quad (3.51)$$

Since we made the hypothesis (1.6) and the spectral localization (3.8), one knows from Lemma 3.4 that the measure ν_t^B carries no mass on the set of the critical points of the Hamiltonian H and therefore ν_t^B can be decomposed as follows

$$\nu_t^B = \nu_t^B|_{\mathbb{T}^2 \times \Omega_{\text{nr}}} + \sum_{\Lambda \in \mathcal{L}_1} \nu_t^B|_{\mathbb{T}^2 \times E_{\Lambda^\perp \setminus \{0\}}}. \quad (3.52)$$

In what follows, we will mainly focus on the analysis of the regularity of each measure $\nu_t^B|_{\mathbb{T}^2 \times E_{\Lambda^\perp \setminus \{0\}}}$. Our assumptions on the energy layers $H \in [E_1, E_2]$ are precisely made to avoid the fixed points of the Hamiltonian that would require a finer analysis.

The next lemma summarizes the dynamical decomposition of φ_H^s -invariant measures on $T^*\mathbb{T}^2$ (see for instance [2, §2]).

Lemma 3.11. *Let μ be a probability measure on $T^*\mathbb{T}^2$ that is invariant under the Hamiltonian flow φ_H^t . Then, the following holds:*

1. *For any Λ in $\mathcal{L}_1$, $\mathcal{I}_\Lambda(\mu)$ is a finite nonnegative Radon measure on $T^*\mathbb{T}^2$.*
2. *For any Λ in $\mathcal{L}_1$, we have $\mu|_{\mathbb{T}^2 \times E_{\Lambda^\perp} \setminus \{0\}} = \mathcal{I}_\Lambda(\mu)|_{\mathbb{T}^2 \times E_{\Lambda^\perp} \setminus \{0\}}$.*
3. *$\hat{\mu}_0$ is a finite nonnegative Radon measure on $T^*\mathbb{T}^2$ and $\mu|_{\mathbb{T}^2 \times \Omega_{\text{nr}}} = \hat{\mu}_0|_{\mathbb{T}^2 \times \Omega_{\text{nr}}}$.*

Since $\tau_h \gg 1$, one can apply this lemma to the semiclassical measures ν_t^B given by (3.52) and decompose them accordingly:

$$\nu_t^B(x, \xi) = \hat{\nu}_t^B(0, \xi)|_{\mathbb{T}^2 \times \Omega_{\text{nr}}} + \sum_{\Lambda \in \mathcal{L}_1} \mathcal{I}_\Lambda(\nu_t^B)|_{\mathbb{T}^2 \times E_{\Lambda^\perp} \setminus \{0\}}. \quad (3.53)$$

Therefore, to derive regularity properties for ν_t along the x variable, it is enough to control how ν_t^B behaves in the x variable on each resonant layer, i.e. for each $\Lambda \in \mathcal{L}_1$, to analyze $\mathcal{I}_\Lambda(\nu_t^B)|_{\mathbb{T}^2 \times E_{\Lambda^\perp} \setminus \{0\}}$. This will be the purpose of the next section.

4 A two-microlocal procedure to analyze periodic orbits

Throughout the section, we work under the spectral localization assumption (3.8) and assume that the conditions (2.20) and (2.21) are satisfied. Let W^B be an accumulation point of the space-time magnetic Wigner distributions $W_h^B(\tau_h)$ introduced in (3.7). In this section, we analyze the information carried by W^B (more precisely ν^B) along a fixed rank-one primitive sublattice $\Lambda \in \mathcal{L}_1$, namely along the periodic directions of the Hamiltonian flow. More precisely, we aim at describing the component $\mathcal{I}_\Lambda(\nu_t^B)|_{\mathbb{T}^2 \times E_{\Lambda^\perp} \setminus \{0\}}$ in (3.53). We first perform a microlocal splitting between a neighborhood of the resonant set $\{\nabla H(\xi) \in \Lambda^\perp \setminus \{0\}\}$ and its complement, and prove that both pieces define nonnegative Radon measures with absolutely continuous time marginals and the expected propagation or invariance properties (Lemmas 4.1, 4.3 and 4.4). We then study the weakly resonant component and show, through a commutator argument based on the symbol introduced in Lemma 4.6, that all its nonzero Λ -Fourier modes vanish (Lemma 4.9). Next, we turn to the strong resonant component, introduce a two-microlocal lift, and establish its compactness, positivity, projection and time-disintegration properties (Lemma 4.11), together with the invariance of the lifted measure under the lifted geodesic flow (Lemma 4.13). Finally, after fixing the zoom scaling in Definition 4.7, we identify the effective dynamics of the lifted measure in the three regimes $\tau_h \ll h^{-1/2}$, $\tau_h = h^{-1/2}$, and $\tau_h \gg h^{-1/2}$ (Lemma 4.14). These properties will then be used in Sections 5 and 6 to describe the measures appearing in the decomposition (3.53) of the time-dependent Wigner distribution W^B .

4.1 A microlocal splitting around $\Lambda^\perp$

Fix $\Lambda \in \mathcal{L}_1$ and let $\mathfrak{e}_\Lambda$ be a generator of Λ . Let $\chi \in \mathcal{C}_c^\infty(\mathbb{R})$ be such that $0 \leq \chi \leq 1$, $\chi \equiv 1$ on $[-1, 1]$ and $\chi \equiv 0$ outside $[-2, 2]$. We fix a sequence $R_m \rightarrow +\infty$ (we simply write $R \rightarrow +\infty$).

Let $(\gamma_h)_{h \rightarrow 0+}$ be a positive scale (to be chosen later depending on how fast τ_h grows) satisfying

$$\gamma_h \xrightarrow{h \rightarrow 0+} 0, \quad \text{and} \quad \gamma_h \geq h^{2-2\delta} \quad \text{for a fixed } \delta > 0. \quad (4.1)$$

In the applications considered below, we will in fact choose $\delta \geq \frac{1}{2}$. It represents the size of the zoom around $\Lambda^\perp$. Define the frequency cut-off

$$\chi_{R\sqrt{\gamma_h}, \Lambda}(\xi) := \chi\left(\frac{\nabla H(\xi) \cdot \mathbf{e}_\Lambda}{R\sqrt{\gamma_h} L_\Lambda}\right). \quad (4.2)$$

For $a \in \mathcal{C}_c^\infty(\mathbb{R} \times T^*\mathbb{T}^2)$, we split

$$\langle W_h^B(\tau_h), a \rangle = \langle W_h^B(\tau_h), a \chi_{R\sqrt{\gamma_h}, \Lambda} \rangle + \langle W_h^B(\tau_h), a(1 - \chi_{R\sqrt{\gamma_h}, \Lambda}) \rangle.$$

This motivates the definition of the following two distributions:

$$W_{h,R,\Lambda}^B : a \mapsto \langle W_h^B(\tau_h), a \chi_{R\sqrt{\gamma_h}, \Lambda} \rangle, \quad (4.3)$$

$$W_{h,R}^{B,\Lambda} : a \mapsto \langle W_h^B(\tau_h), a(1 - \chi_{R\sqrt{\gamma_h}, \Lambda}) \rangle, \quad (4.4)$$

for which the Calderón–Vaillancourt Theorem 2.2 ensures that these expressions are well defined as distributions. Whenever we write $(\cdot)_{h \rightarrow 0^+, R \rightarrow +\infty}$, we mean the *iterated limit* $h \rightarrow 0^+$ followed by $R \rightarrow +\infty$.

Lemma 4.1 (Boundedness, positivity, and decomposition). *Assume $(u_h)_{h \rightarrow 0^+}$ solves the magnetic Schrödinger equation (3.1) and admits a semiclassical measure W^B . Then the following holds:*

1. *The families $(W_{h,R,\Lambda}^B)_{h \rightarrow 0^+, R \rightarrow +\infty}$ and $(W_{h,R}^{B,\Lambda})_{h \rightarrow 0^+, R \rightarrow +\infty}$ are bounded in $\mathcal{D}'(\mathbb{R} \times T^*\mathbb{T}^2)$.*
2. *Any accumulation points W_Λ^B and $W^{B,\Lambda}$ of these families are nonnegative Radon measures on $\mathbb{R} \times \mathbb{T}^2 \times \Omega_{E_1, E_2}$.*
3. *Recalling that W^B denotes an accumulation point of the magnetic Wigner distributions $W_h^B(\tau_h)$ defined in (1.16), as in Lemma 3.4, one has the measure identity*

$$W^B = W_\Lambda^B + W^{B,\Lambda}. \quad (4.5)$$

Proof. For items 1 and 2, the techniques involved in the proof are the same for $(W_{h,R,\Lambda}^B)_{h \rightarrow 0^+, R \rightarrow +\infty}$ and $(W_{h,R}^{B,\Lambda})_{h \rightarrow 0^+, R \rightarrow +\infty}$. We give each time the proof only for one of them. Let $a \in \mathcal{C}_c^\infty(\mathbb{R} \times T^*\mathbb{T}^2)$. Using the rescaling formula (2.16), one can rewrite

$$\langle W_{h,R,\Lambda}^B, a \rangle = \int_{\mathbb{R}} \langle \text{Op}_{\frac{h}{\sqrt{\gamma_h}}}^B(a(t, x, \sqrt{\gamma_h}\xi)) \chi\left(\frac{\nabla H(\sqrt{\gamma_h}\xi) \cdot \mathbf{e}_\Lambda}{R\sqrt{\gamma_h} L_\Lambda}\right) u_h(t\tau_h), u_h(t\tau_h) \rangle dt.$$

For $\gamma_h \in (0, 1)$ satisfying (4.1) and $R \geq 1$, the symbols

$$a(t, x, \sqrt{\gamma_h}\xi) \chi\left(\frac{\nabla H(\sqrt{\gamma_h}\xi) \cdot \mathbf{e}_\Lambda}{R\sqrt{\gamma_h} L_\Lambda}\right) \quad \text{and} \quad a(t, x, \sqrt{\gamma_h}\xi) \left(1 - \chi\left(\frac{\nabla H(\sqrt{\gamma_h}\xi) \cdot \mathbf{e}_\Lambda}{R\sqrt{\gamma_h} L_\Lambda}\right)\right)$$

belong to $\mathcal{S}^0$ uniformly in h and R . Indeed, on the support of $a(t, x, \sqrt{\gamma_h}\xi)$, the rescaled variable $\sqrt{\gamma_h}\xi$ remains in a fixed compact set. Hence all derivatives of H are evaluated on a compact set, and the chain rule gives uniform bounds for all derivatives of $\chi_{R\sqrt{\gamma_h}, \Lambda}$. Hence, the Calderón–Vaillancourt Theorem yields uniform boundedness, proving item 1 of the lemma.

For item 2, we repeat the positivity argument used in the proof of Lemma 3.2. The assumption (4.1) allows us to apply the same argument in the rescaled calculus, since the corresponding semi-classical parameter $h/\sqrt{\gamma_h}$ tends to 0. Fix $a \geq 0$ in $C_c^\infty(\mathbb{R} \times T^*\mathbb{T}^2)$ and let $[t_1, t_2] \subset \mathbb{R}$ be a compact interval containing the support of a in the time variable. For $\varepsilon > 0$, define for each $t \in [t_1, t_2]$

$$a_{\varepsilon,h}(t, x, \xi) := \sqrt{a(t, x, \sqrt{\gamma_h}\xi) \chi_{R\sqrt{\gamma_h}, \Lambda}(\sqrt{\gamma_h}\xi) + \varepsilon}.$$

Then $(a_{\varepsilon,h}(t))_{t \in [t_1, t_2]}$ is a bounded family in $\mathcal{S}^0$. Using the symbolic composition (2.12) and adjoint formula (2.11) in the magnetic calculus, for each $t \in [t_1, t_2]$, one has

$$\text{Op}_{\frac{h}{\sqrt{\gamma_h}}}^B(a_{\varepsilon,h}(t))^* \text{Op}_{\frac{h}{\sqrt{\gamma_h}}}^B(a_{\varepsilon,h}(t)) = \text{Op}_{\frac{h}{\sqrt{\gamma_h}}}^B(a(t, x, \sqrt{\gamma_h}\xi) \chi_{R\sqrt{\gamma_h}, \Lambda}(\sqrt{\gamma_h}\xi) + \varepsilon) + \mathcal{O}_{L^2 \rightarrow L^2}\left(\frac{h}{\sqrt{\gamma_h}}\right),$$

which implies positivity of the limiting functional as $\varepsilon \rightarrow 0$. This shows that any accumulation point defines a nonnegative distribution, hence a finite nonnegative Radon measure on $\mathbb{R} \times T^*\mathbb{T}^2$.

Up to extracting a common subsequence, we may assume both components converge, and then the identity $W_h^B(\tau_h) = W_{h,R,\Lambda}^B + W_{h,R}^B$ passes to the limit.

The support property of item 2 follows by Lemma 3.3 together with the nonnegativity of W_Λ^B and $W^{B,\Lambda}$ and the identity (4.5). $\square$

Remark 4.2. *The proof of Lemma 4.1 uses the fact that $(u_h)_{h \rightarrow 0^+}$ solves the Schrödinger equation (3.1) only to obtain the support property. The boundedness, positivity, and algebraic decomposition arguments rely only on the definition of the distributions involved and on the uniform boundedness properties of the corresponding pseudodifferential operators.*

Now that the limiting distributions have been identified as measures, we may perform on W_Λ^B and $W^{B,\Lambda}$ the same time disintegration with respect to their time marginal as in (3.20). Hence, we can write

$$W_\Lambda^B(dt, dx, d\xi) = \nu_{t,\Lambda}^B(dx, d\xi) \otimes \omega_\Lambda^B(dt), \quad W^{B,\Lambda}(dt, dx, d\xi) = \nu_t^{B,\Lambda}(dx, d\xi) \otimes \omega^{B,\Lambda}(dt), \quad (4.6)$$

where $\nu_{t,\Lambda}^B$ and $\nu_t^{B,\Lambda}$ are measurable families of probability measures on $\mathbb{T}^2 \times \Omega_{E_1, E_2}$, and $\omega_\Lambda^B, \omega^{B,\Lambda}$ are nonnegative Radon measures on $\mathbb{R}$. Since one can decompose W^B as we did in Lemma 3.4 as $W^B(dt, dx, d\xi) = \nu_t^B(dx, d\xi) \otimes dt$ with ν_t^B a probability measure for a.e. t , we can identify the time marginals of ω_Λ^B and $\omega^{B,\Lambda}$.

Lemma 4.3 (Absolute continuity of the time marginals). *There exist $h_\Lambda, h^\Lambda \in L_{\text{loc}}^1(\mathbb{R})$ such that*

$$\omega_\Lambda^B(dt) = h_\Lambda(t) dt, \quad \omega^{B,\Lambda}(dt) = h^\Lambda(t) dt,$$

and

$$h_\Lambda(t) + h^\Lambda(t) = 1 \quad \text{for a.e. } t. \quad (4.7)$$

Proof. Applying the Radon–Nikodym decomposition to ω_Λ^B and $\omega^{B,\Lambda}$ with respect to dt leads to the existence of h_Λ and h^Λ both in $L_{\text{loc}}^1(\mathbb{R})$ such that

$$\omega_\Lambda^B(dt) = h_\Lambda(t)dt + \omega_{\Lambda,\perp}^B(dt), \quad \omega^{B,\Lambda}(dt) = h^\Lambda(t)dt + \omega_\perp^{B,\Lambda}(dt), \quad (4.8)$$

where $\omega_\perp^{B,\Lambda}$ and $\omega_{\Lambda,\perp}^B$ are singular with respect to the Lebesgue measure dt . One can combine the decomposition (4.6) and (4.8) with (4.5) to get

$$\begin{aligned} \nu_t^B(dx, d\xi) \otimes dt &= \left[\nu_{t,\Lambda}^B(dx, d\xi) h_\Lambda(t) + \nu_t^{B,\Lambda}(dx, d\xi) h^\Lambda(t) \right] \otimes dt \\ &\quad + \nu_{t,\Lambda}^B(dx, d\xi) \otimes \omega_{\Lambda,\perp}^B(dt) + \nu_t^{B,\Lambda}(dx, d\xi) \otimes \omega_\perp^{B,\Lambda}(dt). \end{aligned} \quad (4.9)$$

Since the left-hand side of (4.9) is absolutely continuous with respect to the Lebesgue measure dt , the singular parts must vanish. Hence (4.9) becomes

$$\nu_t^B(dx, d\xi) \otimes dt = \left[\nu_{t,\Lambda}^B(dx, d\xi) h_\Lambda(t) + \nu_t^{B,\Lambda}(dx, d\xi) h^\Lambda(t) \right] \otimes dt. \quad (4.10)$$

Taking total masses in (4.10) and using that ν_t^B , $\nu_{t,\Lambda}^B$ and $\nu_t^{B,\Lambda}$ are probability measures for almost every t , we obtain (4.7). $\square$

The dynamics of $\nu_{t,\Lambda}^B$ and $\nu_t^{B,\Lambda}$ are governed by the same time-scale trichotomy as in Lemma 3.8. We record the statement for later use.

Lemma 4.4 (Propagation and invariance for the split components). *Let $\varphi_H^s(x, \xi) = (x + s\nabla H(\xi), \xi)$ be the Hamiltonian flow of H .*

1. *If $\tau_h \rightarrow 0$, then $\nu_{t,\Lambda}^B$ and $\nu_t^{B,\Lambda}$ are time-independent.*
2. *If $\tau_h = 1$, then they are transported by φ_H^s .*
3. *If $\tau_h \rightarrow +\infty$, then for a.e. t they are invariant under φ_H^s for every $s \in \mathbb{R}$.*

Proof. We only treat the component W_Λ^B , the proof for $W^{B,\Lambda}$ is identical replacing χ by $1 - \chi$. Let $\psi \in C_c^\infty(\mathbb{R})$, let $a \in C_c^\infty(T^*\mathbb{T}^2)$, and fix $R > 1$. Define

$$a_{h,R}(x, \xi) := a(x, \xi) \chi\left(\frac{\nabla H(\xi) \cdot \mathbf{e}_\Lambda}{R\sqrt{\gamma_h}L_\Lambda}\right) \quad \text{and} \quad \tilde{a}_{h,R}(x, \xi) = a_{h,R}(x, \sqrt{\gamma_h}\xi).$$

By the rescaling identity (2.16), one has

$$\text{Op}_h^B(a_{h,R}) = \text{Op}_{\frac{B_h}{\sqrt{\gamma_h}}}^B(\tilde{a}_{h,R}) \quad \text{and} \quad \hat{H}_h = \text{Op}_{\frac{B_h}{\sqrt{\gamma_h}}}^B(\tilde{p}_h),$$

where $\tilde{p}_h(x, \xi) := H(\sqrt{\gamma_h}\xi) + hR_h(x, \sqrt{\gamma_h}\xi)$. Since $(u_h)_{h \rightarrow 0^+}$ solves (3.1), integration by parts in time yields

$$\frac{1}{\tau_h} \int_{\mathbb{R}} \psi'(t) \langle \text{Op}_{\frac{B_h}{\sqrt{\gamma_h}}}^B(\tilde{a}_{h,R}) u_h(t\tau_h), u_h(t\tau_h) \rangle dt = \int_{\mathbb{R}} \psi(t) \left\langle \frac{i}{h} [\text{Op}_{\frac{B_h}{\sqrt{\gamma_h}}}^B(\tilde{a}_{h,R}), \hat{H}_h] u_h(t\tau_h), u_h(t\tau_h) \right\rangle dt. \quad (4.11)$$

By the same argument as in Lemma 4.1, the family $(\tilde{a}_{h,R})_h$ is bounded in $\mathcal{S}^0(T^*\mathbb{T}^2)$, uniformly in h for fixed R . In particular, by Calderón–Vaillancourt,

$$\left\| \text{Op}_{\frac{B_h}{\sqrt{\gamma_h}}}^B(\tilde{a}_{h,R}) \right\|_{L^2 \rightarrow L^2} \leq C_{a,R}. \quad (4.12)$$

Moreover, since a is compactly supported in the ξ -variable, there exists $C_a > 0$ such that

$$\text{supp}_\xi(\tilde{a}_{h,R}) \subset \left\{ \xi \in \mathbb{R}^2 : |\xi| \leq C_a \gamma_h^{-1/2} \right\}.$$

Consequently, on the support of $\tilde{a}_{h,R}$, one has

$$\gamma_h^{m/2} \langle \xi \rangle^m \leq C_{a,m}.$$

Since all derivatives of $\tilde{a}_{h,R}$ satisfy the same support condition, the uniform $\mathcal{S}^0$ -estimates imply that for fixed R , $\gamma_h^{m/2} \tilde{a}_{h,R}$ is uniformly bounded in $\mathcal{S}^{-m}$. We now compute the commutator. Applying (2.13) at the semiclassical parameter $h/\sqrt{\gamma_h}$, we obtain

$$\left[\text{Op}_{\frac{h}{\sqrt{\gamma_h}}}^B(\gamma_h^{m/2} \tilde{a}_{h,R}), \text{Op}_{\frac{h}{\sqrt{\gamma_h}}}^B(\tilde{p}_h) \right] = \frac{h \gamma_h^{(m-1)/2}}{i} \text{Op}_{\frac{h}{\sqrt{\gamma_h}}}^B(\{\tilde{a}_{h,R}, \tilde{p}_h\}) + \text{Op}_{\frac{h}{\sqrt{\gamma_h}}}^B(\tilde{r}_{h,R}). \quad (4.13)$$

We claim that

$$\tilde{r}_{h,R} = \mathcal{O}_{\mathcal{S}^0} \left(h^2 \gamma_h^{(m-1)/2} \right). \quad (4.14)$$

To prove (4.14), we use the precise structure of the composition formula (2.13). Consider first a term of order $k \geq 2$ involving $H(\sqrt{\gamma_h} \xi)$. Since H is independent of x , every derivative falling on this factor is a ξ -derivative. Therefore, one has

$$\partial_\xi^\beta H(\sqrt{\gamma_h} \xi) = \gamma_h^{|\beta|/2} (\partial^\beta H)(\sqrt{\gamma_h} \xi), \quad |\beta| = k.$$

Since the derivatives of $\gamma_h^{m/2} \tilde{a}_{h,R}$ are $\mathcal{O}_{\mathcal{S}^0}(\gamma_h^{m/2})$, every such contribution is bounded by

$$\gamma_h^{m/2} \left(\frac{h}{\sqrt{\gamma_h}} \right)^k \gamma_h^{k/2} = \gamma_h^{m/2} h^k = \mathcal{O}_{\mathcal{S}^0} \left(h^2 \gamma_h^{(m-1)/2} \right),$$

since $k \geq 2$.

For the terms involving $h R_h(x, \sqrt{\gamma_h} \xi)$, some derivatives may fall on the x -variable and therefore produce no factor $\sqrt{\gamma_h}$. Nevertheless, these terms always carry the additional prefactor h . The worst contribution is therefore

$$\gamma_h^{m/2} h \left(\frac{h}{\sqrt{\gamma_h}} \right)^2 = h^3 \gamma_h^{(m-2)/2} = h^2 \gamma_h^{(m-1)/2} \frac{h}{\sqrt{\gamma_h}} = o \left(h^2 \gamma_h^{(m-1)/2} \right),$$

because $h/\sqrt{\gamma_h} \rightarrow 0$.

It remains to control the final remainder in (2.13). Since $\gamma_h^{m/2} \tilde{a}_{h,R}$ and $\tilde{p}_h$ are uniformly bounded in $\mathcal{S}^{-m}$ and $\mathcal{S}^m$, respectively, the remainder of order N is bounded in $\mathcal{S}^0$ by

$$C_{N,R} \left(\frac{h}{\sqrt{\gamma_h}} \right)^N.$$

The assumption (4.1) implies

$$\frac{h}{\sqrt{\gamma_h}} \leq h^\delta.$$

Choosing N sufficiently large shows that this remainder is also of size

$$\mathcal{O}_{\mathcal{S}^0} \left(h^2 \gamma_h^{(m-1)/2} \right).$$

This proves (4.14). Hence, dividing (4.13) by $\gamma_h^{m/2}$ yields

$$[\text{Op}_{\frac{h}{\sqrt{\gamma_h}}}^B(\tilde{a}_{h,R}), \text{Op}_{\frac{h}{\sqrt{\gamma_h}}}^B(\tilde{p}_h)] = \frac{h}{i\sqrt{\gamma_h}} \text{Op}_{\frac{h}{\sqrt{\gamma_h}}}^B(\{\tilde{a}_{h,R}, \tilde{p}_h\}) + \text{Op}_{\frac{h}{\sqrt{\gamma_h}}}^B(r_{h,R}), \quad (4.15)$$

where

$$r_{h,R} := \gamma_h^{-m/2} \tilde{r}_{h,R} = \mathcal{O}_{\mathcal{S}^0} \left(\frac{h^2}{\sqrt{\gamma_h}} \right). \quad (4.16)$$

By the Calderón–Vaillancourt Theorem 2.2, one obtains

$$\left\| \text{Op}_{\frac{h}{\sqrt{\gamma_h}}}^B(r_{h,R}) \right\|_{L^2 \rightarrow L^2} \leq C_R \frac{h^2}{\sqrt{\gamma_h}}.$$

Consequently,

$$\frac{i}{h} [\text{Op}_{\frac{h}{\sqrt{\gamma_h}}}^B(\tilde{a}_{h,R}), \hat{H}_h] = \text{Op}_{\frac{h}{\sqrt{\gamma_h}}}^B \left(\frac{1}{\sqrt{\gamma_h}} \{\tilde{a}_{h,R}, \tilde{p}_h\} \right) + \mathcal{O}_{L^2 \rightarrow L^2} \left(\frac{h}{\sqrt{\gamma_h}} \right). \quad (4.17)$$

We next split the Poisson bracket:

$$\frac{1}{\sqrt{\gamma_h}} \{\tilde{a}_{h,R}, \tilde{p}_h\} = \frac{1}{\sqrt{\gamma_h}} \{\tilde{a}_{h,R}, H(\sqrt{\gamma_h} \xi)\} + \frac{h}{\sqrt{\gamma_h}} \{\tilde{a}_{h,R}, R_h(x, \sqrt{\gamma_h} \xi)\}.$$

For the second term, note that $(\tilde{a}_{h,R})_h$ is uniformly bounded in $\mathcal{S}^0$, while every positive-order ξ -derivative of $R_h(x, \sqrt{\gamma_h} \xi)$ is $\mathcal{O}(\sqrt{\gamma_h})$ on the support of $\tilde{a}_{h,R}$. It follows that

$$\frac{h}{\sqrt{\gamma_h}} \{\tilde{a}_{h,R}, R_h(x, \sqrt{\gamma_h} \xi)\} = \mathcal{O}_{\mathcal{S}^0} \left(\frac{h}{\sqrt{\gamma_h}} \right).$$

Hence, thanks to the Calderón–Vaillancourt Theorem (4.17) becomes

$$\frac{i}{h} [\text{Op}_{\frac{h}{\sqrt{\gamma_h}}}^B(\tilde{a}_{h,R}), \hat{H}_h] = \text{Op}_{\frac{h}{\sqrt{\gamma_h}}}^B \left(\frac{1}{\sqrt{\gamma_h}} \{\tilde{a}_{h,R}, H(\sqrt{\gamma_h} \xi)\} \right) + \mathcal{O}_{L^2 \rightarrow L^2} \left(\frac{h}{\sqrt{\gamma_h}} \right). \quad (4.18)$$

Computing explicitly the Poisson bracket, one gets

$$\frac{1}{\sqrt{\gamma_h}} \{\tilde{a}_{h,R}, H(\sqrt{\gamma_h} \xi)\} = -\nabla H(\sqrt{\gamma_h} \xi) \cdot \partial_x a(x, \sqrt{\gamma_h} \xi) \chi \left(\frac{\nabla H(\sqrt{\gamma_h} \xi) \cdot \mathbf{e}_\Lambda}{R\sqrt{\gamma_h} L_\Lambda} \right). \quad (4.19)$$

Finally, inserting (4.19) in (4.11) leads to

$$\begin{aligned} & \frac{1}{\tau_h} \int_{\mathbb{R}} \psi'(t) \langle \text{Op}_{\frac{h}{\sqrt{\gamma_h}}}^B(\tilde{a}_{h,R}) u_h(t\tau_h), u_h(t\tau_h) \rangle dt \\ &= - \int_{\mathbb{R}} \psi(t) \left\langle \text{Op}_{\frac{h}{\sqrt{\gamma_h}}}^B \left(\nabla H(\sqrt{\gamma_h} \xi) \cdot \partial_x a(x, \sqrt{\gamma_h} \xi) \chi \left(\frac{\nabla H(\sqrt{\gamma_h} \xi) \cdot \mathbf{e}_\Lambda}{R\sqrt{\gamma_h} L_\Lambda} \right) \right) u_h(t\tau_h), u_h(t\tau_h) \right\rangle dt \\ &+ \mathcal{O} \left(\frac{h}{\sqrt{\gamma_h}} \right). \end{aligned} \quad (4.20)$$

From (4.1), one can infer that $h/\sqrt{\gamma_h} \rightarrow 0$ as $h \rightarrow 0^+$, and from (4.12) the left-hand side of (4.20) passes to the limit. We now distinguish the three regimes.

If $\tau_h \rightarrow 0$, multiplying (4.20) by τ_h and passing to the limit $h \rightarrow 0^+$, then $R \rightarrow +\infty$, yields

$$\int_{\mathbb{R}} \psi'(t) \langle a, \nu_{t,\Lambda}^B \rangle h_\Lambda(t) dt = 0.$$

Hence $\nu_{t,\Lambda}^B$ is independent of t for $h_\Lambda(t) dt$ -almost every t .

If $\tau_h = 1$, passing to the limit in (4.20) and using Theorem 2.2, together with $h/\sqrt{\gamma_h} \rightarrow 0$, we obtain

$$\int_{\mathbb{R}} \psi'(t) \langle a, \nu_{t,\Lambda}^B \rangle h_\Lambda(t) dt = - \int_{\mathbb{R}} \psi(t) \langle \nabla H(\xi) \cdot \partial_x a, \nu_{t,\Lambda}^B \rangle h_\Lambda(t) dt.$$

This is the weak form of the transport equation

$$\partial_t \nu_{t,\Lambda}^B + \nabla H(\xi) \cdot \partial_x \nu_{t,\Lambda}^B = 0.$$

Thus $\nu_{t,\Lambda}^B$ is transported by φ_H^s , in the weak sense, for $h_\Lambda(t) dt$ -almost every t .

If $\tau_h \rightarrow +\infty$, the left-hand side of (4.20) tends to 0, and the same limit procedure yields

$$\int_{\mathbb{R}} \psi(t) \langle \nabla H(\xi) \cdot \partial_x a, \nu_{t,\Lambda}^B \rangle h_\Lambda(t) dt = 0.$$

Therefore, for $h_\Lambda(t) dt$ -almost every t ,

$$\langle \nabla H(\xi) \cdot \partial_x a, \nu_{t,\Lambda}^B \rangle = 0,$$

which is exactly the infinitesimal formulation of the invariance of $\nu_{t,\Lambda}^B$ under φ_H^s . $\square$

We shall use this splitting together with the standard Fourier structure of invariant measures under the completely integrable flow generated by H . Indeed, in the long-time regime $\tau_h \rightarrow +\infty$, Lemma 4.4 implies that both split components are invariant under φ_H^s , for almost every time with respect to their time marginals. Applying Lemma 3.11 to each component gives the following consequence.

Corollary 4.5. *Assume that $\tau_h \rightarrow +\infty$. Then, for every $\Lambda \in \mathcal{L}_1$, the measures $\mathcal{I}_\Lambda(\nu_{t,\Lambda}^B)$ and $\mathcal{I}_\Lambda(\nu_t^{B,\Lambda})$ are finite nonnegative Radon measures such that*

$$\nu_{t,\Lambda}^B|_{\mathbb{T}^2 \times E_{\Lambda^\perp} \setminus \{0\}} = \mathcal{I}_\Lambda(\nu_{t,\Lambda}^B)|_{\mathbb{T}^2 \times E_{\Lambda^\perp} \setminus \{0\}}, \quad \nu_t^{B,\Lambda}|_{\mathbb{T}^2 \times E_{\Lambda^\perp} \setminus \{0\}} = \mathcal{I}_\Lambda(\nu_t^{B,\Lambda})|_{\mathbb{T}^2 \times E_{\Lambda^\perp} \setminus \{0\}}, \quad (4.21)$$

and

$$\nu_{t,\Lambda}^B(x, \xi)|_{\mathbb{T}^2 \times \Omega_{\text{nr}}} = \widehat{\nu}_{t,\Lambda}^B(0, \xi)|_{\mathbb{T}^2 \times \Omega_{\text{nr}}}, \quad \nu_t^{B,\Lambda}(x, \xi)|_{\mathbb{T}^2 \times \Omega_{\text{nr}}} = \widehat{\nu}_t^{B,\Lambda}(0, \xi)|_{\mathbb{T}^2 \times \Omega_{\text{nr}}}. \quad (4.22)$$

Proof. By Lemma 4.4, for ω_Λ^B -almost every t , the measure $\nu_{t,\Lambda}^B$ is invariant under φ_H^s , and for $\omega^{B,\Lambda}$ -almost every t , the measure $\nu_t^{B,\Lambda}$ is invariant under φ_H^s . Since both measures are probability measures, Lemma 3.11 can be applied to each of them. $\square$

4.2 The weakly-resonant part: vanishing of Λ -modes

From now on, we assume that $\tau_h \rightarrow +\infty$, so that the decomposition of Corollary 4.5 holds. We now turn to the weakly resonant component $\nu_t^{B,\Lambda}|_{\mathbb{T}^2 \times E_{\Lambda^\perp} \setminus \{0\}}$ and prove that its nonzero Λ -Fourier modes vanish. Let $k \in \Lambda \setminus \{0\}$ and $b \in \mathcal{C}_c^\infty(\mathbb{R}^2)$. The idea is to test the k -th Fourier coefficient away from the resonant set associated with Λ , namely the set where

$$\nabla H(\xi) \cdot \mathbf{e}_\Lambda = 0.$$

For this purpose, we introduce a cutoff which removes a neighborhood of this set at scale $R\sqrt{\gamma_h}$. We define, for $\gamma_h > 0$ and $R > 0$,

$$s_{\gamma_h,R}(x, \xi) = e^{2i\pi k \cdot x} \frac{b(\sqrt{\gamma_h}\xi)}{\nabla H(\sqrt{\gamma_h}\xi) \cdot \mathbf{e}_\Lambda} \left(1 - \chi \left(\frac{\nabla H(\sqrt{\gamma_h}\xi) \cdot \mathbf{e}_\Lambda}{RL_\Lambda \sqrt{\gamma_h}} \right) \right). \quad (4.23)$$

The role of this symbol is to make the desired Fourier mode appear as the principal symbol in the commutator with the Hamiltonian part. More precisely, when taking the commutator between the symbol $s_{\gamma_h,R}$ and the Hamiltonian $H(\sqrt{\gamma_h}\xi)$, the first-order term in the symbolic expansion (2.13) is given by the Poisson bracket

$$\{s_{\gamma_h,R}, H(\sqrt{\gamma_h}\xi)\}.$$

By construction of $s_{\gamma_h,R}$, this Poisson bracket is, up to an explicit nonzero constant and a power of γ_h , precisely the truncated Fourier mode. More precisely,

$$\{s_{\gamma_h,R}, H(\sqrt{\gamma_h}\xi)\} = -2i\pi \frac{k \cdot \mathbf{e}_\Lambda}{L_\Lambda^2} \sqrt{\gamma_h} e^{2i\pi k \cdot x} b(\sqrt{\gamma_h}\xi) \left(1 - \chi \left(\frac{\nabla H(\sqrt{\gamma_h}\xi) \cdot \mathbf{e}_\Lambda}{RL_\Lambda \sqrt{\gamma_h}} \right) \right). \quad (4.24)$$

Thus, after the appropriate normalization, the commutator recovers the k -th Fourier coefficient that we want to prove vanishes.

The next lemma gives the symbol estimates needed to use $s_{\gamma_h,R}$ in the commutator argument. In particular, $\sqrt{\gamma_h} s_{\gamma_h,R}$ is an admissible test symbol in the $h/\sqrt{\gamma_h}$ -semiclassical calculus, uniformly with respect to h for fixed R .

Lemma 4.6. *Let $k \in \Lambda \setminus \{0\}$ and $b \in \mathcal{C}_c^\infty(\mathbb{R}^2)$. For $(\gamma_h)_{h \rightarrow 0^+}$ satisfying (4.1) and $R > 0$, the following holds:*

1. $\sqrt{\gamma_h} s_{\gamma_h,R} \in \mathcal{S}^0(\mathbb{T}^2 \times \mathbb{R}^2)$, uniformly in h .
2. For every $m \in \mathbb{N}^*$, $\gamma_h^{m/2} s_{\gamma_h,R} \in \mathcal{S}^{1-m}(\mathbb{T}^2 \times \mathbb{R}^2)$, uniformly in h .

Moreover, for $R \geq 1$, the following estimates hold uniformly with respect to h :

$$\sqrt{\gamma_h} s_{\gamma_h,R} = \mathcal{O}_{\mathcal{S}^0} \left(\frac{1}{R} \right). \quad (4.25)$$

For every $j \in \{1, 2\}$ and every $m \in \mathbb{N}^*$, one also has

$$\partial_{x_j} \left(\gamma_h^{m/2} s_{\gamma_h,R} \right) = \mathcal{O}_{\mathcal{S}^0} \left(\frac{\gamma_h^{(m-1)/2}}{R} \right) = \mathcal{O}_{\mathcal{S}^{1-m}} \left(\frac{1}{R} \right), \quad (4.26)$$

and

$$\partial_{\xi_j} \left(\gamma_h^{m/2} s_{\gamma_h,R} \right) = \mathcal{O}_{\mathcal{S}^0} \left(\frac{\gamma_h^{m/2}}{R} \right) + \mathcal{O}_{\mathcal{S}^0} \left(\frac{\gamma_h^{(m-1)/2}}{R^2} \right) = \mathcal{O}_{\mathcal{S}^{1-m}} \left(\frac{1}{R} \right). \quad (4.27)$$

Proof. Fix $R > 0$. Write

$$\sqrt{\gamma_h} s_{\gamma_h, R}(x, \xi) = e^{2i\pi k \cdot x} b(\sqrt{\gamma_h} \xi) F_R(G_{\gamma_h}(\xi)), \quad G_{\gamma_h}(\xi) := \frac{\nabla H(\sqrt{\gamma_h} \xi) \cdot \mathbf{e}_\Lambda}{\sqrt{\gamma_h}},$$

with

$$F_R(\eta) := \frac{1 - \chi(\eta/(RL_\Lambda))}{\eta}.$$

Since $1 - \chi$ vanishes near 0 and is identically equal to 1 near infinity, $F_R \in \mathcal{C}^\infty(\mathbb{R})$ and one has for every $N \in \mathbb{N}$,

$$\sup_{\eta \in \mathbb{R}} |F_R^{(N)}(\eta)| \leq \frac{C_N}{R^{N+1}}. \quad (4.28)$$

Moreover, for any multi-index $\alpha \neq 0$,

$$\partial_\xi^\alpha G_{\gamma_h}(\xi) = \gamma_h^{(|\alpha|-1)/2} (\partial^\alpha (\nabla H \cdot \mathbf{e}_\Lambda))(\sqrt{\gamma_h} \xi), \quad (4.29)$$

hence $|\partial_\xi^\alpha G_{\gamma_h}(\xi)| \leq C_\alpha \gamma_h^{(|\alpha|-1)/2}$ on $\text{supp}(b(\sqrt{\gamma_h} \cdot))$ as b is compactly supported. Using Leibniz' rule and repeated chain rule, (4.28)–(4.29) yield, for all α, β ,

$$\sup_{x, \xi} |\partial_x^\alpha \partial_\xi^\beta (\sqrt{\gamma_h} s_{\gamma_h, R})(x, \xi)| \leq \frac{C_{\alpha, \beta}}{R},$$

which proves both item 1 and (4.25). Item 2 follows from item 1. Indeed, for $m \geq 1$, one has

$$\gamma_h^{m/2} s_{\gamma_h, R} = \gamma_h^{(m-1)/2} \sqrt{\gamma_h} s_{\gamma_h, R}.$$

By item 1, the family $\sqrt{\gamma_h} s_{\gamma_h, R}$ is bounded in $\mathcal{S}^0$, uniformly in h (for each fixed $R > 0$). Moreover, since b is compactly supported, there exists $M > 0$ such that

$$|\xi| \leq M \gamma_h^{-1/2} \quad \text{on } \text{supp}(s_{\gamma_h, R}).$$

It follows that, on this support,

$$\gamma_h^{(m-1)/2} \leq C_m \langle \xi \rangle^{1-m}. \quad (4.30)$$

Therefore, for all multi-indices α, β ,

$$|\partial_x^\alpha \partial_\xi^\beta (\gamma_h^{m/2} s_{\gamma_h, R})(x, \xi)| \leq C_{\alpha, \beta, R} \gamma_h^{(m-1)/2} \leq C_{\alpha, \beta, m, R} \langle \xi \rangle^{1-m},$$

which proves that $\gamma_h^{m/2} s_{\gamma_h, R} \in \mathcal{S}^{1-m}$ uniformly in h . It remains to prove the refined estimates (4.26) and (4.27). For the x -derivatives, the only dependence on x is the factor $e^{2i\pi k \cdot x}$. Hence, for $j \in \{1, 2\}$ one has

$$\partial_{x_j} \left(\gamma_h^{\frac{m}{2}} s_{\gamma_h, R} \right) = 2i\pi k_j \gamma_h^{\frac{m-1}{2}} (\sqrt{\gamma_h} s_{\gamma_h, R}).$$

By (4.25), this gives

$$\partial_{x_j} \left(\gamma_h^{m/2} s_{\gamma_h, R} \right) = \mathcal{O}_{\mathcal{S}^0} \left(\frac{\gamma_h^{(m-1)/2}}{R} \right).$$

Using again the fact that (4.30) holds on $\text{supp}(s_{\gamma_h, R})$, we obtain

$$\partial_{x_j} \left(\gamma_h^{m/2} s_{\gamma_h, R} \right) = \mathcal{O}_{\mathcal{S}^0} \left(\frac{\gamma_h^{(m-1)/2}}{R} \right) = \mathcal{O}_{\mathcal{S}^{1-m}} \left(\frac{1}{R} \right).$$

Then, (4.26) follows from (4.25). We finally estimate the ξ -derivatives. Differentiating

$$\sqrt{\gamma_h} s_{\gamma_h, R} = e^{2i\pi k \cdot x} b(\sqrt{\gamma_h} \xi) F_R(G_{\gamma_h}(\xi))$$

with respect to ξ_j , we get

$$\partial_{\xi_j} (\sqrt{\gamma_h} s_{\gamma_h, R}) = e^{2i\pi k \cdot x} \sqrt{\gamma_h} (\partial_{\xi_j} b)(\sqrt{\gamma_h} \xi) F_R(G_{\gamma_h}(\xi)) + e^{2i\pi k \cdot x} b(\sqrt{\gamma_h} \xi) F'_R(G_{\gamma_h}(\xi)) \partial_{\xi_j} G_{\gamma_h}(\xi). \quad (4.31)$$

The first term in (4.31) is

$$\mathcal{O}_{\mathcal{S}^0} \left(\frac{\sqrt{\gamma_h}}{R} \right).$$

Indeed, derivatives falling on $b(\sqrt{\gamma_h} \xi)$ only produce nonnegative powers of $\sqrt{\gamma_h}$, while (4.28), (4.29), and the repeated chain rule give

$$F_R(G_{\gamma_h}) = \mathcal{O}_{\mathcal{S}^0} \left(\frac{1}{R} \right).$$

Similarly, using again (4.28), (4.29), Leibniz' rule, and the repeated chain rule, one obtains

$$F'_R(G_{\gamma_h}) \partial_{\xi_j} G_{\gamma_h} = \mathcal{O}_{\mathcal{S}^0} \left(\frac{1}{R^2} \right).$$

Therefore

$$\partial_{\xi_j} (\sqrt{\gamma_h} s_{\gamma_h, R}) = \mathcal{O}_{\mathcal{S}^0} \left(\frac{\sqrt{\gamma_h}}{R} \right) + \mathcal{O}_{\mathcal{S}^0} \left(\frac{1}{R^2} \right)$$

which by multiplying by $\gamma_h^{(m-1)/2}$, leads to

$$\partial_{\xi_j} \left(\gamma_h^{m/2} s_{\gamma_h, R} \right) = \mathcal{O}_{\mathcal{S}^0} \left(\frac{\gamma_h^{m/2}}{R} \right) + \mathcal{O}_{\mathcal{S}^0} \left(\frac{\gamma_h^{(m-1)/2}}{R^2} \right).$$

Since $0 < \gamma_h \leq 1$ for h small enough and $R \geq 1$, we have

$$\frac{\gamma_h^{m/2}}{R} \leq \frac{\gamma_h^{(m-1)/2}}{R}, \quad \frac{\gamma_h^{(m-1)/2}}{R^2} \leq \frac{\gamma_h^{(m-1)/2}}{R}. \quad (4.32)$$

Consequently, combining (4.32) with (4.30), we finally obtain

$$\partial_{\xi_j} \left(\gamma_h^{m/2} s_{\gamma_h, R} \right) = \mathcal{O}_{\mathcal{S}^0} \left(\frac{\gamma_h^{(m-1)/2}}{R} \right) = \mathcal{O}_{\mathcal{S}^{1-m}} \left(\frac{1}{R} \right),$$

which proves (4.27). □

So far, the scaling parameter $(\gamma_h)_{h \rightarrow 0^+}$ was only assumed to satisfy (4.1). We now make a definite choice for $(\gamma_h)_{h \rightarrow 0^+}$, which will be used throughout the remainder of this article.

Definition 4.7. *The scaling parameter $(\gamma_h)_{h \rightarrow 0^+}$ is defined by*

$$\gamma_h = \begin{cases} \frac{1}{\tau_h^2}, & \text{if } \tau_h \ll \frac{1}{\sqrt{h}}, \\ h, & \text{if } \tau_h \gtrsim \frac{1}{\sqrt{h}}. \end{cases} \quad (4.33)$$

Remark 4.8. *This choice may seem somewhat ad hoc at first. However, the proof of the next lemma shows that this dichotomy is naturally dictated by the asymptotic behavior of the time scale $(\tau_h)_{h \rightarrow 0^+}$, and the corresponding computations justify the definition above.*

Lemma 4.9 (Vanishing of Λ -Fourier modes). *Let $k \in \Lambda \setminus \{0\}$, $(\gamma_h)_{h \rightarrow 0^+}$ satisfying (4.33) and let $(\psi, b) \in \mathcal{C}_c^\infty(\mathbb{R}) \times \mathcal{C}_c^\infty(\mathbb{R}^2)$. Then, one has*

$$\int_{\mathbb{R}} \psi(t) \left[\int_{T^*\mathbb{T}^2} e^{2i\pi k \cdot x} b(\xi) \nu_t^{B,\Lambda}(dx, d\xi) \right] h^\Lambda(t) dt = 0. \quad (4.34)$$

In particular, $\mathcal{I}_\Lambda(\nu_t^{B,\Lambda})$ is independent of x (for a.e. t).

Proof. The goal is to prove that the following limit vanishes:

$$\lim_{R \rightarrow \infty} \lim_{h \rightarrow 0^+} \int_{\mathbb{R}} \psi(t) \left\langle \text{Op}_h^B \left(e^{2i\pi k \cdot x} b(\xi) \left(1 - \chi \left(\frac{\nabla H(\xi) \cdot \mathbf{e}_\Lambda}{RL_\Lambda \sqrt{\gamma_h}} \right) \right) \right) u_h(t\tau_h), u_h(t\tau_h) \right\rangle dt. \quad (4.35)$$

To do so, we have already seen that $\sqrt{\gamma_h} s_{\gamma_h, R} \in \mathcal{S}^0$ and, more generally, $\gamma_h^{(m+1)/2} s_{\gamma_h, R} \in \mathcal{S}^{-m}$ by Lemma 4.6. This symbol was introduced to analyze the following commutator expression:

$$\int_{\mathbb{R}} \psi(t) \left\langle \left[\text{Op}_{\frac{h}{\sqrt{\gamma_h}}}^B \left(\gamma_h^{\frac{m+1}{2}} s_{\gamma_h, R} \right), \text{Op}_h^B (H + hR_h) \right] u_h(t\tau_h), u_h(t\tau_h) \right\rangle dt. \quad (4.36)$$

Using that $(u_h)_{h \rightarrow 0^+}$ solves (3.1), we obtain that (4.36) is equal to

$$-\frac{ih\gamma_h^{\frac{m}{2}}}{\tau_h} \int_{\mathbb{R}} \psi'(t) \left\langle \text{Op}_{\frac{h}{\sqrt{\gamma_h}}}^B (\sqrt{\gamma_h} s_{\gamma_h, R}) u_h(t\tau_h), u_h(t\tau_h) \right\rangle dt. \quad (4.37)$$

By Lemma 4.6 and the Calderón–Vaillancourt theorem, the quantity in (4.37) is $\mathcal{O}\left(\frac{h\gamma_h^{m/2}}{\tau_h R}\right)$. As we shall see below, the leading Hamiltonian contribution to the commutator is of order $h\gamma_h^{(m+1)/2}$ and produces precisely the quantity appearing in (4.35). Therefore, after normalization by $h\gamma_h^{(m+1)/2}$, the contribution of (4.37) is

$$\mathcal{O}\left(\frac{1}{R\tau_h\sqrt{\gamma_h}}\right) = \mathcal{O}\left(\frac{1}{R}\right),$$

where the last estimate follows from (4.33), since $\tau_h\sqrt{\gamma_h} \geq 1$. Thus, this term vanishes as $R \rightarrow +\infty$. On the other hand, using the rescaling formula (2.16) in (4.36) to write $\widehat{H}_h$ in the $\frac{h}{\sqrt{\gamma_h}}$ -quantization,

we are left with the following commutator, which we split into its Hamiltonian and perturbative parts:

$$\left[\text{Op}_{\frac{h}{\sqrt{\gamma_h}}}^B \left(\gamma_h^{\frac{m+1}{2}} s_{\gamma_h, R} \right), \text{Op}_{\frac{h}{\sqrt{\gamma_h}}}^B (H(\sqrt{\gamma_h} \xi) + h R_h(x, \sqrt{\gamma_h} \xi)) \right] = \mathcal{C}_h^{\text{Ham}} + h \mathcal{C}_h^{\text{pert}}. \quad (4.38)$$

Since $\gamma_h^{\frac{m+1}{2}} s_{\gamma_h, R}$ is in the class $\mathcal{S}^{-m}$ (uniformly in h) and since the Hamiltonian H belongs to $\mathcal{S}^m$ (and the perturbation $R_h \in \mathcal{S}^m$ with $m > 1$), all symbols arising in the commutator expansion are uniformly bounded in $\mathcal{S}^0$.

1. Hamiltonian contribution. We first analyze the Hamiltonian contribution using the formula (2.15) to see that the principal symbol arising from the Poisson bracket is exactly the one in (4.35). We now compute the commutator

$$\mathcal{C}_h^{\text{Ham}} = \left[\text{Op}_{\frac{h}{\sqrt{\gamma_h}}}^B \left(\gamma_h^{\frac{m+1}{2}} s_{\gamma_h, R} \right), \text{Op}_{\frac{h}{\sqrt{\gamma_h}}}^B (H(\sqrt{\gamma_h} \xi)) \right]. \quad (4.39)$$

Applying the composition formula (2.13) in the rescaled $h/\sqrt{\gamma_h}$ quantization, we obtain for every integer $N \geq 4$

$$\left[\text{Op}_{\frac{h}{\sqrt{\gamma_h}}}^B \left(\gamma_h^{\frac{m+1}{2}} s_{\gamma_h, R} \right), \text{Op}_{\frac{h}{\sqrt{\gamma_h}}}^B (H(\sqrt{\gamma_h} \xi)) \right] = \text{Op}_{\frac{h}{\sqrt{\gamma_h}}}^B (\mathcal{H}_{h, N}) + \text{Op}_{\frac{h}{\sqrt{\gamma_h}}}^B (\rho_{N, h, R}), \quad (4.40)$$

with

$$\mathcal{H}_{h, N} = \sum_{p=0}^{\lfloor (N-2)/2 \rfloor} \frac{2}{(2p+1)!} \left(\frac{ih}{2\sqrt{\gamma_h}} \right)^{2p+1} \text{L}_h^{2p+1} \left[\gamma_h^{\frac{m+1}{2}} s_{\gamma_h, R}(x, \xi) H(\sqrt{\gamma_h} \xi') \right] \Big|_{(x, \xi) = (x', \xi')}, \quad (4.41)$$

where

$$\text{L}_h = \Omega_{\frac{h}{\sqrt{\gamma_h}} B}(D_x, D_\xi, D_{x'}, D_{\xi'})$$

and, for each fixed $R > 0$,

$$\rho_{N, h, R} = \mathcal{O}_{\mathcal{S}^0, R} \left(\left(\frac{h}{\sqrt{\gamma_h}} \right)^N \right) = \mathcal{O}_{\mathcal{S}^0, R} (h^{N/2}), \quad (4.42)$$

because Definition 4.7 implies $\gamma_h \geq h$. We now isolate the contribution $p = 0$ from the remaining odd terms. Making the term $p = 0$ explicit gives

$$\begin{aligned} & \left[\text{Op}_{\frac{h}{\sqrt{\gamma_h}}}^B \left(\gamma_h^{\frac{m+1}{2}} s_{\gamma_h, R} \right), \text{Op}_{\frac{h}{\sqrt{\gamma_h}}}^B (H(\sqrt{\gamma_h} \xi)) \right] \\ &= \frac{h}{i\sqrt{\gamma_h}} \text{Op}_{\frac{h}{\sqrt{\gamma_h}}}^B \left(\left\{ \gamma_h^{\frac{m+1}{2}} s_{\gamma_h, R}, H(\sqrt{\gamma_h} \xi) \right\} \right) \end{aligned} \quad (4.43)$$

$$+ \frac{\widehat{B}_0 h^2}{i\gamma_h} \text{Op}_{\frac{h}{\sqrt{\gamma_h}}}^B \left(\partial_{\xi_2} \left(\gamma_h^{\frac{m+1}{2}} s_{\gamma_h, R} \right) \partial_{\xi_1} (H(\sqrt{\gamma_h} \xi)) - \partial_{\xi_1} \left(\gamma_h^{\frac{m+1}{2}} s_{\gamma_h, R} \right) \partial_{\xi_2} (H(\sqrt{\gamma_h} \xi)) \right) \quad (4.44)$$

$$+ \text{Op}_{\frac{h}{\sqrt{\gamma_h}}}^B (\widetilde{\mathcal{H}}_{N, h, R}), \quad (4.45)$$

where $\tilde{\mathcal{H}}_{N,h,R}$ collects the terms with $p \geq 1$ in (4.41) together with the symbolic remainder $\rho_{N,h,R}$. Using (4.24), and since $\gamma_h^{\frac{m+1}{2}}$ is independent of (x, ξ) , one directly obtains

$$\left\{ \gamma_h^{\frac{m+1}{2}} s_{\gamma_h,R}, H(\sqrt{\gamma_h} \xi) \right\} = -2i\pi \frac{k \cdot \epsilon_\Lambda}{L_\Lambda^2} \gamma_h^{\frac{m+2}{2}} e^{2i\pi k \cdot x} b(\sqrt{\gamma_h} \xi) \left(1 - \chi \left(\frac{\nabla H(\sqrt{\gamma_h} \xi) \cdot \epsilon_\Lambda}{RL_\Lambda \sqrt{\gamma_h}} \right) \right). \quad (4.46)$$

Up to the factor $\gamma_h^{\frac{m+2}{2}}$, this is exactly the quantity in which we are interested in to compute the k -Fourier coefficient of our measure in (4.35).

We next turn to the magnetic contribution of order h^2/γ_h coming from the symbol in (4.44). By the estimate (4.27) in Lemma 4.6, for $j = 1, 2$, one has

$$\partial_{\xi_j} \left(\gamma_h^{\frac{m+1}{2}} s_{\gamma_h,R} \right) = \mathcal{O}_{\mathcal{S}^0} \left(\frac{\gamma_h^{\frac{m+1}{2}}}{R} \right) + \mathcal{O}_{\mathcal{S}^0} \left(\frac{\gamma_h^{m/2}}{R^2} \right). \quad (4.47)$$

Inserting (4.47) into (4.44), and using that $\partial_{\xi_j}(H(\sqrt{\gamma_h} \xi)) = \mathcal{O}_{\mathcal{S}^0}(\sqrt{\gamma_h})$ on the support of $b(\sqrt{\gamma_h} \xi)$, yields

$$\partial_{\xi_2} \left(\gamma_h^{\frac{m+1}{2}} s_{\gamma_h,R} \right) \partial_{\xi_1} (H(\sqrt{\gamma_h} \xi)) - \partial_{\xi_1} \left(\gamma_h^{\frac{m+1}{2}} s_{\gamma_h,R} \right) \partial_{\xi_2} (H(\sqrt{\gamma_h} \xi)) = \mathcal{O}_{\mathcal{S}^0} \left(\frac{\gamma_h^{\frac{m+2}{2}}}{R} \right) + \mathcal{O}_{\mathcal{S}^0} \left(\frac{\gamma_h^{\frac{m+1}{2}}}{R^2} \right). \quad (4.48)$$

Using (4.48) and the Calderón–Vaillancourt theorem, it follows that

$$\begin{aligned} & \frac{\hat{B}_0 h^2}{i\gamma_h} \text{Op}_{\frac{B}{\sqrt{\gamma_h}}} \left(\partial_{\xi_2} \left(\gamma_h^{\frac{m+1}{2}} s_{\gamma_h,R} \right) \partial_{\xi_1} (H(\sqrt{\gamma_h} \xi)) - \partial_{\xi_1} \left(\gamma_h^{\frac{m+1}{2}} s_{\gamma_h,R} \right) \partial_{\xi_2} (H(\sqrt{\gamma_h} \xi)) \right) \\ &= \mathcal{O}_{L^2 \rightarrow L^2} \left(\frac{h^2 \gamma_h^{\frac{m}{2}}}{R} \right) + \mathcal{O}_{L^2 \rightarrow L^2} \left(\frac{h^2 \gamma_h^{\frac{m-1}{2}}}{R^2} \right). \end{aligned} \quad (4.49)$$

It remains to control the higher-order odd terms in (4.45). Each term with $p \geq 1$ carries at least the factor $(h/\sqrt{\gamma_h})^3$, since $2p+1 \geq 3$. Moreover, $\Omega_{\frac{h}{\sqrt{\gamma_h}} B}^{2p+1}$ involves only finitely many derivatives of the two symbols. The derivatives of $H(\sqrt{\gamma_h} \xi)$ are uniformly bounded on the support of $b(\sqrt{\gamma_h} \xi)$. The estimate (4.25) of Lemma 4.6 implies that, for every pair of multi-indices α, β , there exists a constant $C_{\alpha,\beta} > 0$, independent of h and R , such that

$$|\partial_x^\alpha \partial_\xi^\beta \gamma_h^{\frac{m+1}{2}} s_{\gamma_h,R}(x, \xi)| = \gamma_h^{\frac{m}{2}} |\partial_x^\alpha \partial_\xi^\beta \sqrt{\gamma_h} s_{\gamma_h,R}(x, \xi)| \leq \frac{\gamma_h^{\frac{m}{2}} C_{\alpha,\beta}}{R},$$

where $C_{\alpha,\beta}$ depends on finitely many derivatives of H , b , and χ . Therefore, for each fixed N , all the terms with $p \geq 1$ in (4.40), together with the estimate (4.42) of the symbolic remainder $\rho_{N,h,R}$, can be absorbed into

$$\tilde{\mathcal{H}}_{N,h,R} = \gamma_h^{\frac{m+1}{2}} \mathcal{O}_{\mathcal{S}^0,R} \left(\frac{h^3}{\sqrt{\gamma_h}} \right) + \mathcal{O}_{\mathcal{S}^0,R} \left(h^{N/2} \right) = \mathcal{O}_{\mathcal{S}^0,R} \left(h^3 \gamma_h^{\frac{m}{2}} \right) + \mathcal{O}_{\mathcal{S}^0,R} \left(h^{N/2} \right), \quad (4.50)$$

where the constant entering into the remainder depends on finitely many derivatives of H , b , and χ , and has at most polynomial growth in R^{-1} . Since $\gamma_h \geq h$, choosing N large enough allows us to

absorb the second term into the first one. Hence, one has

$$\tilde{\mathcal{H}}_{N,h,R} = \mathcal{O}_{\mathcal{S}^0,R} \left(h^3 \gamma_h^{\frac{m}{2}} \right).$$

By the Calderón–Vaillancourt theorem, it follows that

$$\mathrm{Op}_{\frac{h}{\sqrt{\gamma_h}}}^B(\tilde{\mathcal{H}}_{N,h,R}) = \mathcal{O}_{L^2 \rightarrow L^2,R} \left(h^3 \gamma_h^{\frac{m}{2}} \right). \quad (4.51)$$

Combining (4.46), (4.49), and (4.51), we finally obtain the decomposition

$$\begin{aligned} & \left[\mathrm{Op}_{\frac{h}{\sqrt{\gamma_h}}}^B \left(\gamma_h^{\frac{m+1}{2}} s_{\gamma_h,R} \right), \mathrm{Op}_{\frac{h}{\sqrt{\gamma_h}}}^B (H(\sqrt{\gamma_h} \xi)) \right] \\ &= -\frac{2\pi(k \cdot \mathbf{e}_\Lambda)}{L_\Lambda^2} h \mathrm{Op}_{\frac{h}{\sqrt{\gamma_h}}}^B \left(\gamma_h^{\frac{m+1}{2}} e^{2i\pi k \cdot x} b(\sqrt{\gamma_h} \xi) \left(1 - \chi \left(\frac{\nabla H(\sqrt{\gamma_h} \xi) \cdot \mathbf{e}_\Lambda}{RL_\Lambda \sqrt{\gamma_h}} \right) \right) \right) \\ &+ \mathcal{O}_{L^2 \rightarrow L^2} \left(\frac{h^2 \gamma_h^{\frac{m}{2}}}{R} \right) + \mathcal{O}_{L^2 \rightarrow L^2} \left(\frac{h^2 \gamma_h^{\frac{m-1}{2}}}{R^2} \right) + \mathcal{O}_{L^2 \rightarrow L^2,R} \left(h^3 \gamma_h^{\frac{m}{2}} \right). \end{aligned} \quad (4.52)$$

2. Perturbative contribution. Now that we have analyzed the Hamiltonian contribution, we show that the perturbative contribution is also negligible in the limit as for (4.50). The key point is that the perturbation carries an additional factor h , which makes the whole contribution negligible in the limit. Applying (2.15) in the $h/\sqrt{\gamma_h}$ -quantization, we obtain

$$\mathcal{C}_h^{\mathrm{pert}} = \left[\mathrm{Op}_{\frac{h}{\sqrt{\gamma_h}}}^B \left(\gamma_h^{\frac{m+1}{2}} s_{\gamma_h,R} \right), \mathrm{Op}_{\frac{h}{\sqrt{\gamma_h}}}^B (R_h(x, \sqrt{\gamma_h} \xi)) \right] \quad (4.53)$$

$$= \frac{h}{i\sqrt{\gamma_h}} \mathrm{Op}_{\frac{h}{\sqrt{\gamma_h}}}^B \left(\left\{ \gamma_h^{\frac{m+1}{2}} s_{\gamma_h,R}, R_h(x, \sqrt{\gamma_h} \xi) \right\} \right) \quad (4.54)$$

$$+ \frac{\hat{B}_0 h^2}{i\gamma_h} \mathrm{Op}_{\frac{h}{\sqrt{\gamma_h}}}^B \left(\partial_{\xi_2} \left(\gamma_h^{\frac{m+1}{2}} s_{\gamma_h,R} \right) \partial_{\xi_1} (R_h(x, \sqrt{\gamma_h} \xi)) - \partial_{\xi_1} \left(\gamma_h^{\frac{m+1}{2}} s_{\gamma_h,R} \right) \partial_{\xi_2} (R_h(x, \sqrt{\gamma_h} \xi)) \right) \quad (4.55)$$

$$+ \mathrm{Op}_{\frac{h}{\sqrt{\gamma_h}}}^B(\tilde{\mathcal{R}}_{N,h,R}),$$

where $\tilde{\mathcal{R}}_{N,h,R}$ denotes the remainder obtained by collecting all the terms of order $2p+1 \geq 3$ in the odd symbolic expansion of the commutator, together with the symbolic remainder, exactly as in the Hamiltonian contribution.

On the support of $b(\sqrt{\gamma_h} \xi)$, one has for $j = 1, 2$

$$\partial_{x_j} (R_h(x, \sqrt{\gamma_h} \xi)) = \mathcal{O}_{\mathcal{S}^0}(1), \quad \partial_{\xi_j} (R_h(x, \sqrt{\gamma_h} \xi)) = \mathcal{O}_{\mathcal{S}^0}(\sqrt{\gamma_h}).$$

Moreover, from Lemma 4.6, one can use (4.26) to have $\partial_{x_j} (\gamma_h^{\frac{m+1}{2}} s_{\gamma_h,R}) = \mathcal{O}_{\mathcal{S}^0} \left(\frac{\gamma_h^{\frac{m}{2}}}{R} \right)$. Hence, using (4.47) one gets

$$\left\{ \gamma_h^{\frac{m+1}{2}} s_{\gamma_h,R}, R_h(x, \sqrt{\gamma_h} \xi) \right\} = \mathcal{O}_{\mathcal{S}^0} \left(\frac{\gamma_h^{\frac{m+1}{2}}}{R} \right) + \mathcal{O}_{\mathcal{S}^0} \left(\frac{\gamma_h^{m/2}}{R^2} \right). \quad (4.56)$$

Similarly, using again (4.47), we obtain

$$\begin{aligned} \partial_{\xi_2} \left(\gamma_h^{\frac{m+1}{2}} s_{\gamma_h, R} \right) \partial_{\xi_1} (R_h(x, \sqrt{\gamma_h} \xi)) - \partial_{\xi_1} \left(\gamma_h^{\frac{m+1}{2}} s_{\gamma_h, R} \right) \partial_{\xi_2} (R_h(x, \sqrt{\gamma_h} \xi)) \\ = \mathcal{O}_{\mathcal{S}^0} \left(\frac{\gamma_h^{\frac{m+2}{2}}}{R} \right) + \mathcal{O}_{\mathcal{S}^0} \left(\frac{\gamma_h^{\frac{m+1}{2}}}{R^2} \right). \end{aligned} \quad (4.57)$$

Finally, the higher odd terms in the full symbolic expansion are treated as in the Hamiltonian contribution. Using the estimates above, together with the uniform symbol bounds on $R_h(x, \sqrt{\gamma_h} \xi)$, the terms of order $2p+1 \geq 3$ are bounded, for fixed $R > 0$, by

$$\mathcal{O}_{\mathcal{S}^0, R} \left(h^3 \gamma_h^{\frac{m}{2}} \right).$$

It remains to control the symbolic remainder in the composition formula. Since $\gamma_h \geq h$, the same argument as in (4.42) gives

$$\rho_{N, h, R}^{\text{pert}} = \mathcal{O}_{\mathcal{S}^0, R} \left(\left(\frac{h}{\sqrt{\gamma_h}} \right)^N \right) = \mathcal{O}_{\mathcal{S}^0, R} \left(h^{N/2} \right).$$

Thus the full perturbative remainder satisfies

$$\tilde{\mathcal{R}}_{N, h, R} = \mathcal{O}_{\mathcal{S}^0, R} \left(h^3 \gamma_h^{\frac{m}{2}} \right) + \mathcal{O}_{\mathcal{S}^0, R} \left(h^{N/2} \right).$$

Choosing N large enough allows us to absorb the second term into the first one. Hence, one has

$$\tilde{\mathcal{R}}_{N, h, R} = \mathcal{O}_{\mathcal{S}^0, R} \left(h^3 \gamma_h^{\frac{m}{2}} \right). \quad (4.58)$$

The constants implicit in these estimates depend on finitely many seminorms of R_h , uniformly with respect to h , and on finitely many derivatives of H , b , and χ . They may depend on R , but are independent of h . Combining (4.56), (4.57) and (4.58) and using the Calderón–Vaillancourt theorem, we obtain

$$\begin{aligned} h \mathcal{C}_h^{\text{pert}} = \text{Op}_{\frac{h}{\sqrt{\gamma_h}}}^B \left(\mathcal{O}_{\mathcal{S}^0} \left(\frac{h^2 \gamma_h^{m/2}}{R} \right) + \mathcal{O}_{\mathcal{S}^0} \left(\frac{h^2 \gamma_h^{\frac{m-1}{2}}}{R^2} \right) + \mathcal{O}_{\mathcal{S}^0} \left(\frac{h^3 \gamma_h^{m/2}}{R} \right) + \mathcal{O}_{\mathcal{S}^0} \left(\frac{h^3 \gamma_h^{\frac{m-1}{2}}}{R^2} \right) \right) \\ + \mathcal{O}_{L^2 \rightarrow L^2, R} \left(h^4 \gamma_h^{\frac{m}{2}} \right). \end{aligned} \quad (4.59)$$

Finally, combining (4.52) with (4.59) in (4.38), and applying the Calderón–Vaillancourt theorem to the symbolic remainders, we obtain

$$\begin{aligned} \mathcal{C}_h^{\text{Ham}} + h \mathcal{C}_h^{\text{pert}} = - \frac{2\pi(k \cdot \mathbf{e}_\Lambda)}{L_\Lambda^2} h \gamma_h^{\frac{m+1}{2}} \text{Op}_{\frac{h}{\sqrt{\gamma_h}}}^B \left(e^{2i\pi k \cdot x} b(\sqrt{\gamma_h} \xi) \left(1 - \chi \left(\frac{\nabla H(\sqrt{\gamma_h} \xi) \cdot \mathbf{e}_\Lambda}{R L_\Lambda \sqrt{\gamma_h}} \right) \right) \right) \\ + \mathcal{O}_{L^2 \rightarrow L^2, R} \left(\frac{h^2 \gamma_h^{m/2}}{R} + \frac{h^2 \gamma_h^{\frac{m-1}{2}}}{R^2} + h^3 \gamma_h^{\frac{m}{2}} \right). \end{aligned} \quad (4.60)$$

We now conclude the proof. Inserting the decomposition (4.60) into the commutator expression (4.36), and comparing the resulting identity with (4.37), we obtain, after using the rescaling formula (2.16), that for every fixed $R > 1$,

$$\begin{aligned} & 2\pi \frac{k \cdot \mathbf{e}_\Lambda}{L_\Lambda^2} \int_{\mathbb{R}} \psi(t) \left\langle \text{Op}_h^B \left(e^{2i\pi k \cdot x} b(\xi) \left(1 - \chi \left(\frac{\nabla H(\xi) \cdot \mathbf{e}_\Lambda}{RL_\Lambda \sqrt{\gamma_h}} \right) \right) \right) u_h(t\tau_h), u_h(t\tau_h) \right\rangle dt \\ &= \frac{i}{\tau_h \sqrt{\gamma_h}} \int_{\mathbb{R}} \psi'(t) \left\langle \text{Op}_{\frac{h}{\sqrt{\gamma_h}}}^B (\sqrt{\gamma_h} s_{\gamma_h, R}) u_h(t\tau_h), u_h(t\tau_h) \right\rangle dt \\ &+ \mathcal{O}_R \left(\frac{h}{R\sqrt{\gamma_h}} + \frac{h}{\gamma_h R^2} + \frac{h^2}{\sqrt{\gamma_h}} \right). \end{aligned} \quad (4.61)$$

Here the last term follows from (4.60), after pairing the operator remainder with $u_h(t\tau_h)$, integrating against $\psi(t)$, and using the conservation of the L^2 -norm. Moreover, by the estimate (4.25) of Lemma 4.6, one has

$$\sqrt{\gamma_h} s_{\gamma_h, R} = \mathcal{O}_{\mathcal{S}^0} \left(\frac{1}{R} \right),$$

uniformly in h . Therefore, by the Calderón–Vaillancourt theorem and the conservation of the L^2 -norm, one has

$$\int_{\mathbb{R}} \psi'(t) \left\langle \text{Op}_{\frac{h}{\sqrt{\gamma_h}}}^B (\sqrt{\gamma_h} s_{\gamma_h, R}) u_h(t\tau_h), u_h(t\tau_h) \right\rangle dt = \mathcal{O} \left(\frac{1}{R} \right),$$

uniformly in h . Hence (4.61) yields

$$\begin{aligned} & 2\pi \frac{k \cdot \mathbf{e}_\Lambda}{L_\Lambda^2} \int_{\mathbb{R}} \psi(t) \left\langle \text{Op}_h^B \left(e^{2i\pi k \cdot x} b(\xi) \left(1 - \chi \left(\frac{\nabla H(\xi) \cdot \mathbf{e}_\Lambda}{RL_\Lambda \sqrt{\gamma_h}} \right) \right) \right) u_h(t\tau_h), u_h(t\tau_h) \right\rangle dt \\ &= \mathcal{O}_R \left(\frac{1}{\tau_h \sqrt{\gamma_h} R} + \frac{h}{R\sqrt{\gamma_h}} + \frac{h}{\gamma_h R^2} + \frac{h^2}{\sqrt{\gamma_h}} \right). \end{aligned} \quad (4.62)$$

By the property of $(\gamma_h)_{h \rightarrow 0^+}$ in Definition 4.7, one gets

$$\frac{1}{\tau_h \sqrt{\gamma_h}} = \mathcal{O}_{h \rightarrow 0^+}(1), \quad \frac{h}{\sqrt{\gamma_h}} \rightarrow 0, \quad \frac{h}{\gamma_h} = \mathcal{O}_{h \rightarrow 0^+}(1), \quad \frac{h^2}{\sqrt{\gamma_h}} \rightarrow 0.$$

Therefore, for every fixed $R > 1$, one has

$$\limsup_{h \rightarrow 0^+} \left| \int_{\mathbb{R}} \psi(t) \left\langle \text{Op}_h^B \left(e^{2i\pi k \cdot x} b(\xi) \left(1 - \chi \left(\frac{\nabla H(\xi) \cdot \mathbf{e}_\Lambda}{RL_\Lambda \sqrt{\gamma_h}} \right) \right) \right) u_h(t\tau_h), u_h(t\tau_h) \right\rangle dt \right| = \mathcal{O} \left(\frac{1}{R} + \frac{1}{R^2} \right).$$

Letting $R \rightarrow +\infty$ proves that the limit in (4.35) vanishes. $\square$

4.3 The resonant part: a two-microlocal lift

To capture the fine structure near $\nabla H(\xi) \cdot \mathbf{e}_\Lambda = 0$, we introduce a two-microlocal variable

$$\eta_{\gamma_h}(\xi) := \frac{\nabla H(\xi) \cdot \mathbf{e}_\Lambda}{L_\Lambda \sqrt{\gamma_h}}, \quad (4.63)$$

where $(\gamma_h)_{h \rightarrow 0^+}$ is defined as in (4.33). The cutoff $\chi \left(\frac{\nabla H(\xi) \cdot \mathbf{e}_\Lambda}{RL_\Lambda \sqrt{\gamma_h}} \right)$ localizes to a $R\sqrt{\gamma_h}$ -neighborhood of the resonant set $\{\nabla H(\xi) \cdot \mathbf{e}_\Lambda = 0\}$, and the variable η remains of order R in this region.

Definition 4.10 (Two-microlocal distribution near Λ). *For $a \in \mathcal{C}_c^\infty(\mathbb{R} \times T^*\mathbb{T}^2 \times \mathbb{R})$, set*

$$\langle \widetilde{W}_{h,R,\Lambda}^B, a \rangle := \int_{\mathbb{R}} \left\langle \text{Op}_h^B \left(a(t, x, \xi, \eta_{\gamma_h}(\xi)) \chi \left(\frac{\eta_{\gamma_h}(\xi)}{R} \right) \right) u_h(t\tau_h), u_h(t\tau_h) \right\rangle dt. \quad (4.64)$$

This two-microlocal object is a refined version of the previous Wigner distributions, with test symbols depending on the additional variable η . We first establish its compactness, positivity, projection and time-disintegration properties in Lemma 4.11. We then prove its invariance under the lifted geodesic flow in Lemma 4.13, before deriving the effective dynamics of its accumulation points in Lemma 4.14.

Lemma 4.11 (Compactness and projection). *Let $(u_h)_{h \rightarrow 0^+}$ solve (3.1) and be spectrally localized in the sense of (3.8). The two-microlocal distribution introduced above satisfies the following:*

1. $(\widetilde{W}_{h,R,\Lambda}^B)_{h \rightarrow 0^+, R \rightarrow +\infty}$ is bounded in $\mathcal{D}'(\mathbb{R} \times T^*\mathbb{T}^2 \times \mathbb{R})$.
2. Any accumulation point $\widetilde{W}_\Lambda^B$ is a nonnegative Radon measure supported on $\mathbb{R} \times \mathbb{T}^2 \times \Omega_{E_1, E_2} \times \mathbb{R}$.
3. The marginal in η recovers the resonant part:

$$W_\Lambda^B(t, x, \xi) = \int_{\mathbb{R}} \widetilde{W}_\Lambda^B(t, x, \xi, d\eta). \quad (4.65)$$

4. There exists a measurable family $(\widetilde{\nu}_{t,\Lambda}^B)$ of probability measures on $\mathbb{T}^2 \times \Omega_{E_1, E_2} \times \mathbb{R}$ such that

$$\widetilde{W}_\Lambda^B(dt, dx, d\xi, d\eta) = \widetilde{\nu}_{t,\Lambda}^B(dx, d\xi, d\eta) \otimes h_\Lambda(t) dt, \quad (4.66)$$

where $h_\Lambda \in L^1_{\text{loc}}(\mathbb{R})$ is given by Lemma 4.3. Moreover, for a.e $t \in \mathbb{R}$, one has

$$\nu_{t,\Lambda}^B(dx, d\xi) = \int_{\mathbb{R}} \widetilde{\nu}_{t,\Lambda}^B(dx, d\xi, d\eta). \quad (4.67)$$

Equivalently,

$$(\pi_{x,\xi})_* \widetilde{\nu}_{t,\Lambda}^B = \nu_{t,\Lambda}^B,$$

where $\pi_{x,\xi}$ denotes the canonical projection defined by $\pi_{x,\xi} : (x, \xi, \eta) \in \mathbb{T}^2 \times \mathbb{R}^2 \times \mathbb{R} \mapsto (x, \xi)$.

Proof. Using the same argument as in Lemma 3.2, after the rescaling $\xi \mapsto \sqrt{\gamma_h} \xi$ the symbol is uniformly bounded in $\mathcal{S}^0$ uniformly in time, so Theorem 2.2 gives the boundedness of $(\widetilde{W}_{h,R,\Lambda}^B)$ in $\mathcal{D}'(\mathbb{R} \times T^*\mathbb{T}^2 \times \mathbb{R})$. Positivity follows by the usual regularization argument: for $a \geq 0$, set $a_\varepsilon = \sqrt{a + \varepsilon}$ and use the magnetic adjoint (2.11) and composition formulas (2.12) exactly as in the proof of Lemma 3.2. Hence any accumulation point $\widetilde{W}_\Lambda^B$ is a nonnegative Radon measure on $\mathbb{R} \times T^*\mathbb{T}^2 \times \mathbb{R}$. The support property in the ξ variable is proved exactly as for W^B ; see Lemma 3.3. The marginal identity is obtained by testing against $a(t, x, \xi) \chi(\eta/R_1)$, then letting $R_1 \rightarrow \infty$ and comparing with the definition of W_Λ^B . More precisely, one has

$$\begin{aligned} & \int_{\mathbb{R} \times T^*\mathbb{T}^2 \times \mathbb{R}} a(t, x, \xi) \chi \left(\frac{\eta}{R_1} \right) \widetilde{W}_\Lambda^B(dt, dx, d\xi, d\eta) \\ &= \lim_{R \rightarrow \infty} \lim_{h \rightarrow 0^+} \int_{\mathbb{R}} \left\langle \text{Op}_h^B \left[a(t, x, \xi) \chi \left(\frac{\nabla H(\xi) \cdot \mathbf{e}_\Lambda}{R_1 L_\Lambda \sqrt{\gamma_h}} \right) \chi \left(\frac{\nabla H(\xi) \cdot \mathbf{e}_\Lambda}{R L_\Lambda \sqrt{\gamma_h}} \right) \right] u_h(t\tau_h), u_h(t\tau_h) \right\rangle dt. \end{aligned}$$

Since $\chi \equiv 1$ near 0, for every fixed $R_1 > 0$ there exists $C_{R_1} > 0$ such that, for every $R \geq C_{R_1}$,

$$\chi \left(\frac{\nabla H(\xi) \cdot \mathbf{e}_\Lambda}{R L_\Lambda \sqrt{\gamma_h}} \right) \chi \left(\frac{\nabla H(\xi) \cdot \mathbf{e}_\Lambda}{R_1 L_\Lambda \sqrt{\gamma_h}} \right) = \chi \left(\frac{\nabla H(\xi) \cdot \mathbf{e}_\Lambda}{R_1 L_\Lambda \sqrt{\gamma_h}} \right).$$

Hence, for every fixed $R_1 > 0$, one gets

$$\begin{aligned} \int_{\mathbb{R} \times T^* \mathbb{T}^2 \times \mathbb{R}} a(t, x, \xi) \chi \left(\frac{\eta}{R_1} \right) \widetilde{W}_\Lambda^B(dt, dx, d\xi, d\eta) \\ = \lim_{h \rightarrow 0^+} \int_{\mathbb{R}} \left\langle \text{Op}_h^B \left[a(t, x, \xi) \chi \left(\frac{\nabla H(\xi) \cdot \mathbf{e}_\Lambda}{R_1 L_\Lambda \sqrt{\gamma_h}} \right) \right] u_h(t\tau_h), u_h(t\tau_h) \right\rangle dt. \end{aligned}$$

Letting $R_1 \rightarrow \infty$ and using the definition of W_Λ^B , we obtain

$$\lim_{R_1 \rightarrow \infty} \int_{\mathbb{R} \times T^* \mathbb{T}^2 \times \mathbb{R}} a(t, x, \xi) \chi \left(\frac{\eta}{R_1} \right) \widetilde{W}_\Lambda^B(dt, dx, d\xi, d\eta) = \int_{\mathbb{R} \times T^* \mathbb{T}^2} a(t, x, \xi) W_\Lambda^B(dt, dx, d\xi).$$

This proves that the marginal of $\widetilde{W}_\Lambda^B$ in the η variable is W_Λ^B , namely

$$W_\Lambda^B(t, x, \xi) = \int_{\mathbb{R}} \widetilde{W}_\Lambda^B(t, x, \xi, d\eta).$$

In order to prove the time disintegration of $\widetilde{W}_\Lambda^B$, define

$$\begin{aligned} \pi_{(t,x,\xi)} : \mathbb{R} \times \mathbb{T}^2 \times \mathbb{R}^2 \times \mathbb{R} &\longrightarrow \mathbb{R} \times \mathbb{T}^2 \times \mathbb{R}^2, & (t, x, \xi, \eta) &\longmapsto (t, x, \xi), \\ \pi_t : \mathbb{R} \times \mathbb{T}^2 \times \mathbb{R}^2 &\longrightarrow \mathbb{R}, & (t, x, \xi) &\longmapsto t, \\ \widetilde{\pi}_t &:= \pi_t \circ \pi_{(t,x,\xi)}. \end{aligned} \tag{4.68}$$

Then the marginal identity (4.65) can be rewritten as

$$(\pi_{(t,x,\xi)})_* \widetilde{W}_\Lambda^B = W_\Lambda^B. \tag{4.69}$$

Applying the push-forward by π_t to both sides of (4.69), we obtain

$$(\pi_t)_* [(\pi_{(t,x,\xi)})_* \widetilde{W}_\Lambda^B] = (\pi_t)_* W_\Lambda^B. \tag{4.70}$$

By functoriality of push-forwards, the left-hand side is

$$(\pi_t \circ \pi_{(t,x,\xi)})_* \widetilde{W}_\Lambda^B = (\widetilde{\pi}_t)_* \widetilde{W}_\Lambda^B.$$

On the other hand, by (4.6) and Lemma 4.3, one has

$$(\pi_t)_* W_\Lambda^B = \omega_\Lambda^B(dt) = h_\Lambda(t) dt.$$

Therefore, one gets

$$(\widetilde{\pi}_t)_* \widetilde{W}_\Lambda^B = h_\Lambda(t) dt.$$

Since $h_\Lambda \in L^1_{\text{loc}}(\mathbb{R})$, this measure is σ -finite. The disintegration Theorem A.5 may therefore be applied to $\widetilde{W}_\Lambda^B$ with respect to the time marginal, yielding a measurable family $(\widetilde{\nu}_{t,\Lambda}^B)_{t \in \mathbb{R}}$ of probability measures such that

$$\widetilde{W}_\Lambda^B(dt, dx, d\xi, d\eta) = \widetilde{\nu}_{t,\Lambda}^B(dx, d\xi, d\eta) \otimes h_\Lambda(t) dt. \tag{4.71}$$

The identity $(\pi_{x,\xi})_* \tilde{\nu}_{t,\Lambda}^B = \nu_{t,\Lambda}^B$ is then a direct consequence of the uniqueness of disintegration. Indeed, since

$$(\pi_{t,x,\xi})_* \widetilde{W}_\Lambda^B = W_\Lambda^B,$$

pushing forward the above disintegration (4.71) gives

$$W_\Lambda^B = ((\pi_{x,\xi})_* \tilde{\nu}_{t,\Lambda}^B) \otimes h_\Lambda(t) dt.$$

Comparing with the disintegration $W_\Lambda^B = \nu_{t,\Lambda}^B \otimes h_\Lambda(t) dt$ yields

$$(\pi_{x,\xi})_* \tilde{\nu}_{t,\Lambda}^B = \nu_{t,\Lambda}^B$$

for $h_\Lambda(t) dt$ -almost every t . $\square$

In order to analyze the properties of the measure $\tilde{\nu}_{t,\Lambda}^B$, we first introduce a rescaled symbol adapted to the $h/\sqrt{\gamma_h}$ -calculus. The purpose of the next lemma is to ensure that these symbols belong to the appropriate symbol classes uniformly in h , so that the same commutator strategy as in Lemma 4.4 can be applied.

Lemma 4.12. *Let $a \in \mathcal{C}_c^\infty(T^*\mathbb{T}^2 \times \mathbb{R})$, $\chi \in \mathcal{C}_c^\infty(\mathbb{R})$ and $(\gamma_h)_{h \rightarrow 0+}$ satisfying (4.1). The symbol $\tilde{s}_{\gamma_h,R}$ defined by*

$$\tilde{s}_{\gamma_h,R} : (x, \xi) \in T^*\mathbb{T}^2 \mapsto a(x, \sqrt{\gamma_h}\xi, \eta_{\gamma_h}(\sqrt{\gamma_h}\xi)) \chi\left(\frac{\eta_{\gamma_h}(\sqrt{\gamma_h}\xi)}{R}\right), \quad (4.72)$$

belongs to the class $\mathcal{S}^0(T^\mathbb{T}^2)$ uniformly in h (with seminorms depending on R). Moreover, one has $\gamma_h^{m/2} \tilde{s}_{\gamma_h,R} \in \mathcal{S}^{-m}(T^*\mathbb{T}^2)$ uniformly in h for any $m \geq 0$. Finally, for every $j \in \{1, 2\}$, one has in $\mathcal{S}^0(T^*\mathbb{T}^2)$, uniformly with respect to h ,*

$$\begin{aligned} \partial_{\xi_j} \tilde{s}_{\gamma_h,R} &= \left(\text{Hess}(H)(\sqrt{\gamma_h}\xi) \frac{\mathbf{e}_\Lambda}{L_\Lambda} \cdot e_j \right) \partial_{\eta} a(x, \sqrt{\gamma_h}\xi, \eta_{\gamma_h}(\sqrt{\gamma_h}\xi)) \chi\left(\frac{\eta_{\gamma_h}(\sqrt{\gamma_h}\xi)}{R}\right) \\ &\quad + \mathcal{O}_{\mathcal{S}^0,R}(\sqrt{\gamma_h}) + \mathcal{O}_{\mathcal{S}^0,R}\left(\frac{1}{R}\right), \end{aligned} \quad (4.73)$$

where (e_1, e_2) denotes the canonical basis of $\mathbb{R}^2$.

Proof. Fix $R > 1$. From the definition (4.63), one can write

$$\tilde{s}_{\gamma_h,R}(x, \xi) = a(x, \sqrt{\gamma_h}\xi, \eta_{\gamma_h}(\sqrt{\gamma_h}\xi)) \chi\left(\frac{\eta_{\gamma_h}(\sqrt{\gamma_h}\xi)}{R}\right).$$

Since $a \in \mathcal{C}_c^\infty(T^*\mathbb{T}^2 \times \mathbb{R})$, there exists a compact set $K \subset \mathbb{R}^2 \times \mathbb{R}$ such that $\text{supp } a \subset \mathbb{T}^2 \times K$. Let $K_1 \subset \mathbb{R}^2$ be the projection of K onto the $\mathbb{R}^2$ variable. Then, on $\text{supp}(\tilde{s}_{\gamma_h,R})$, one necessarily has $\sqrt{\gamma_h}\xi$ bounded. Moreover, for every $|\beta| \geq 2$, one has $\partial_\xi^\beta(\sqrt{\gamma_h}\xi) = 0$. In particular, all derivatives of $\sqrt{\gamma_h}\xi$ are uniformly bounded. One also has that, for every $|\beta| \geq 1$,

$$\partial_\xi^\beta(\eta_{\gamma_h}(\sqrt{\gamma_h}\xi)) = \frac{\gamma_h^{(|\beta|-1)/2}}{L_\Lambda} (\partial^\beta(\nabla H \cdot \mathbf{e}_\Lambda))(\sqrt{\gamma_h}\xi).$$

Since $(\gamma_h)_{h \rightarrow 0^+}$ satisfies (4.1), in particular $\gamma_h \leq 1$ for h small, and since $\sqrt{\gamma_h}\xi$ stays in the fixed compact set K_1 on the support of $\tilde{s}_{\gamma_h, R}$, one has uniformly in h ,

$$|\partial_\xi^\alpha(\sqrt{\gamma_h}\xi)| \leq C_\alpha, \quad |\partial_\xi^\alpha \eta_{\gamma_h}(\sqrt{\gamma_h}\xi)| \leq C_\alpha \quad \text{for } |\alpha| \geq 1.$$

We do not need any uniform bound on $\eta_{\gamma_h}(\sqrt{\gamma_h}\xi)$ itself due to the fact that when differentiating the composition above, the chain rule only involves derivatives of $\eta_{\gamma_h}(\sqrt{\gamma_h}\xi)$ of order at least one, while the factors involving a , χ and their derivatives are uniformly bounded. Since a and χ are smooth with compact support, all their derivatives are bounded. Therefore, Leibniz' rule and repeated chain rule imply that for every multi-index α, β ,

$$\sup_{x, \xi} |\partial_x^\alpha \partial_\xi^\beta \tilde{s}_{\gamma_h, R}(x, \xi)| \leq C_{\alpha, \beta, R},$$

which proves that $\tilde{s}_{\gamma_h, R} \in \mathcal{S}^0(T^*\mathbb{T}^2)$ uniformly in h .

Moreover, since a is compactly supported in the ξ variable, there exists $M > 0$ such that

$$|\xi| \leq M\gamma_h^{-1/2} \quad \text{on } \text{supp } \tilde{s}_{\gamma_h, R}.$$

Therefore, on this support, one has

$$\gamma_h^{m/2} \leq C_m \langle \xi \rangle^{-m}.$$

Since $\tilde{s}_{\gamma_h, R}$ is uniformly bounded in $\mathcal{S}^0$, we obtain for all multi-indices α, β ,

$$|\partial_x^\alpha \partial_\xi^\beta (\gamma_h^{m/2} \tilde{s}_{\gamma_h, R})(x, \xi)| \leq C_{\alpha, \beta, R} \gamma_h^{m/2} \leq C_{\alpha, \beta, m, R} \langle \xi \rangle^{-m},$$

which proves the claim. In order to prove the estimate (4.73), one has by differentiating $\tilde{s}_{\gamma_h, R}$ with respect to ξ_j

$$\begin{aligned} \partial_{\xi_j} \tilde{s}_{\gamma_h, R} &= \sqrt{\gamma_h} \partial_{\xi_j} a(x, \sqrt{\gamma_h}\xi, \eta_{\gamma_h}(\sqrt{\gamma_h}\xi)) \chi \left(\frac{\nabla H(\sqrt{\gamma_h}\xi) \cdot \mathbf{e}_\Lambda}{RL_\Lambda \sqrt{\gamma_h}} \right) \\ &\quad + \partial_\eta a(x, \sqrt{\gamma_h}\xi, \eta_{\gamma_h}(\sqrt{\gamma_h}\xi)) \partial_{\xi_j} (\eta_{\gamma_h}(\sqrt{\gamma_h}\xi)) \chi \left(\frac{\nabla H(\sqrt{\gamma_h}\xi) \cdot \mathbf{e}_\Lambda}{RL_\Lambda \sqrt{\gamma_h}} \right) \\ &\quad + \frac{1}{R} a(x, \sqrt{\gamma_h}\xi, \eta_{\gamma_h}(\sqrt{\gamma_h}\xi)) \chi' \left(\frac{\nabla H(\sqrt{\gamma_h}\xi) \cdot \mathbf{e}_\Lambda}{RL_\Lambda \sqrt{\gamma_h}} \right) \partial_{\xi_j} (\eta_{\gamma_h}(\sqrt{\gamma_h}\xi)). \end{aligned} \quad (4.74)$$

The first term in (4.74) is $\mathcal{O}_{\mathcal{S}^0}(\sqrt{\gamma_h})$. Indeed, further derivatives only fall on smooth compactly supported factors composed with $\sqrt{\gamma_h}\xi$ and $\eta_{\gamma_h}(\sqrt{\gamma_h}\xi)$ and derivatives of $\eta_{\gamma_h}(\sqrt{\gamma_h}\xi)$ of order at least one are uniformly bounded while the prefactor $\sqrt{\gamma_h}$ remains. For the second term, using

$$\partial_{\xi_j} \eta_{\gamma_h}(\sqrt{\gamma_h}\xi) = \text{Hess}(H)(\sqrt{\gamma_h}\xi) \frac{\mathbf{e}_\Lambda}{L_\Lambda} \cdot e_j,$$

we obtain the leading contribution. Finally, the third term is $\mathcal{O}_{\mathcal{S}^0}(1/R)$. The factor $1/R$ is explicit, and further derivatives produce uniformly bounded factors, possibly with additional powers of $1/R$ from derivatives of the cutoff. Therefore, one has

$$\begin{aligned} \partial_{\xi_j} \tilde{s}_{\gamma_h, R} &= \text{Hess}(H)(\sqrt{\gamma_h}\xi) \frac{\mathbf{e}_\Lambda}{L_\Lambda} \cdot e_j \partial_\eta a(x, \sqrt{\gamma_h}\xi, \eta_{\gamma_h}(\sqrt{\gamma_h}\xi)) \chi \left(\frac{\eta_{\gamma_h}(\sqrt{\gamma_h}\xi)}{R} \right) \\ &\quad + \mathcal{O}_{\mathcal{S}^0, R}(\sqrt{\gamma_h}) + \mathcal{O}_{\mathcal{S}^0, R} \left(\frac{1}{R} \right). \end{aligned} \quad (4.75)$$

□

Lemma 4.13. *The measure $\tilde{\nu}_{t,\Lambda}^B$ is invariant under the lifted geodesic flow*

$$\tilde{\varphi}_H^s(x, \xi, \eta) := (x + s\nabla H(\xi), \xi, \eta).$$

Proof. Let $\psi \in \mathcal{C}_c^\infty(\mathbb{R})$ and let $a \in \mathcal{C}_c^\infty(T^*\mathbb{T}^2 \times \mathbb{R})$. For $R > 1$, define

$$\mathfrak{s}_{h,R}(x, \xi) := a\left(x, \xi, \frac{\nabla H(\xi) \cdot \mathbf{e}_\Lambda}{L_\Lambda \sqrt{\gamma_h}}\right) \chi\left(\frac{\nabla H(\xi) \cdot \mathbf{e}_\Lambda}{RL_\Lambda \sqrt{\gamma_h}}\right).$$

By the rescaling identity (2.16), in the $h/\sqrt{\gamma_h}$ quantization one has

$$\mathrm{Op}_h^B(\mathfrak{s}_{h,R}) = \mathrm{Op}_{\frac{h}{\sqrt{\gamma_h}}}^B(\tilde{s}_{\gamma_h,R}),$$

where $\tilde{s}_{\gamma_h,R}$ is defined as in (4.72) and also

$$\hat{H}_h = \mathrm{Op}_{\frac{h}{\sqrt{\gamma_h}}}^B(\tilde{p}_h), \quad \tilde{p}_h(x, \xi) := H(\sqrt{\gamma_h}\xi) + h R_h(x, \sqrt{\gamma_h}\xi). \quad (4.76)$$

Since $(u_h)_{h \rightarrow 0+}$ solves (3.1), an integration by parts in time yields

$$\begin{aligned} & \frac{\gamma_h^{m/2}}{\tau_h} \int_{\mathbb{R}} \psi'(t) \left\langle \mathrm{Op}_{\frac{h}{\sqrt{\gamma_h}}}^B(\tilde{s}_{\gamma_h,R}) u_h(t\tau_h), u_h(t\tau_h) \right\rangle dt \\ &= \int_{\mathbb{R}} \psi(t) \left\langle \frac{i}{h} [\mathrm{Op}_{\frac{h}{\sqrt{\gamma_h}}}^B(\gamma_h^{m/2} \tilde{s}_{\gamma_h,R}), \hat{H}_h] u_h(t\tau_h), u_h(t\tau_h) \right\rangle dt. \end{aligned} \quad (4.77)$$

By Lemma 4.12, the family $\tilde{s}_{\gamma_h,R}$ is bounded in $\mathcal{S}^0(T^*\mathbb{T}^2)$, uniformly in h (for fixed R). Hence, by the Calderón–Vaillancourt theorem 2.2 and the unitarity of the propagator, the left-hand side of (4.77) is

$$\mathcal{O}_R\left(\frac{\gamma_h^{m/2}}{\tau_h}\right) = o_R\left(\gamma_h^{m/2}\right), \quad (4.78)$$

since $\tau_h \rightarrow +\infty$.

We now compute the commutator, using the fact that $\tilde{p}_h \in \mathcal{S}^m$ and $\gamma_h^{m/2} \tilde{s}_{\gamma_h,R} \in \mathcal{S}^{-m}$ by Lemma 4.12. Applying the commutator formula (2.15) with semiclassical parameter $h/\sqrt{\gamma_h}$, one gets

$$[\mathrm{Op}_{\frac{h}{\sqrt{\gamma_h}}}^B(\gamma_h^{m/2} \tilde{s}_{\gamma_h,R}), \mathrm{Op}_{\frac{h}{\sqrt{\gamma_h}}}^B(\tilde{p}_h)] = \frac{h}{i\sqrt{\gamma_h}} \mathrm{Op}_{\frac{h}{\sqrt{\gamma_h}}}^B\left(\{\gamma_h^{m/2} \tilde{s}_{\gamma_h,R}, \tilde{p}_h\}\right) + \mathrm{Op}_{\frac{h}{\sqrt{\gamma_h}}}^B(r_{h,R}), \quad (4.79)$$

where the remainder symbol $r_{h,R}$ satisfies

$$r_{h,R} = \mathcal{O}_{\mathcal{S}^0}\left(h^2 \gamma_h^{(m-1)/2}\right). \quad (4.80)$$

Using the commutator formula (2.15) in the rescaled setting, a direct application first gives that $r_{h,R}$ is $\mathcal{O}_{\mathcal{S}^0}(h^2 \gamma_h^{(m-2)/2})$, but this estimate can be improved. Indeed, the symbolic expansion consists of a finite sum of explicit terms, together with a remainder controlled by the same symbol seminorms. The explicit terms are products of derivatives of $\gamma_h^{m/2} \tilde{s}_{\gamma_h,R}$ and derivatives of $\tilde{p}_h$,

multiplied by powers of $h/\sqrt{\gamma_h}$ of order at least 2. By Lemma 4.12, all derivatives of $\tilde{s}_{\gamma_h, R}$ are uniformly bounded, and therefore all derivatives of $\gamma_h^{m/2} \tilde{s}_{\gamma_h, R}$ are bounded by $\mathcal{O}_{\mathcal{S}^0}(\gamma_h^{m/2})$.

Moreover, for all multi-indices α and β with $|\alpha| \geq 1$ and $|\beta| \geq 1$ one has

$$\partial_\xi^\beta H(\sqrt{\gamma_h} \xi) = \gamma_h^{|\beta|/2} (\partial^\beta H)(\sqrt{\gamma_h} \xi), \quad \text{and} \quad \partial_x^\alpha \partial_\xi^\beta (h R_h(x, \sqrt{\gamma_h} \xi)) = h \gamma_h^{|\beta|/2} (\partial_x^\alpha \partial_\xi^\beta R_h)(x, \sqrt{\gamma_h} \xi).$$

On the support of $\tilde{s}_{\gamma_h, R}$ and of all its derivatives, the variable $\sqrt{\gamma_h} \xi$ stays in a fixed compact set, since a is compactly supported in the ξ -variable. Hence, on that support, every ξ -derivative of $H(\sqrt{\gamma_h} \xi)$ is $\mathcal{O}(\sqrt{\gamma_h})$, and every ξ -derivative of $h R_h(x, \sqrt{\gamma_h} \xi)$ is $\mathcal{O}(h \sqrt{\gamma_h})$. The derivatives in x of the perturbative term do not give a factor $\sqrt{\gamma_h}$, but they still carry the factor h , which is enough since $h/\sqrt{\gamma_h} \rightarrow 0$. This yields (4.80). By another use of Calderón–Vaillancourt,

$$\left\| \text{Op}_{\frac{B_h}{\sqrt{\gamma_h}}}^B(r_{h, R}) \right\|_{L^2 \rightarrow L^2} \leq C_{a, R} h^2 \gamma_h^{(m-1)/2}.$$

Hence, one has

$$\frac{i}{h} [\text{Op}_{\frac{B_h}{\sqrt{\gamma_h}}}^B(\gamma_h^{m/2} \tilde{s}_{\gamma_h, R}), \hat{H}_h] = \text{Op}_{\frac{B_h}{\sqrt{\gamma_h}}}^B \left(\gamma_h^{(m-1)/2} \{ \tilde{s}_{\gamma_h, R}, \tilde{p}_h \} \right) + \mathcal{O}_{L^2 \rightarrow L^2} \left(h \gamma_h^{(m-1)/2} \right). \quad (4.81)$$

We next split the Poisson bracket:

$$\gamma_h^{(m-1)/2} \{ \tilde{s}_{\gamma_h, R}, \tilde{p}_h \} = \gamma_h^{(m-1)/2} \{ \tilde{s}_{\gamma_h, R}, H(\sqrt{\gamma_h} \xi) \} + h \gamma_h^{(m-1)/2} \{ \tilde{s}_{\gamma_h, R}, R_h(x, \sqrt{\gamma_h} \xi) \}.$$

For the principal term, since $H(\sqrt{\gamma_h} \xi)$ depends only on ξ , one has

$$\gamma_h^{(m-1)/2} \{ \tilde{s}_{\gamma_h, R}, H(\sqrt{\gamma_h} \xi) \} = -\gamma_h^{m/2} \nabla H(\sqrt{\gamma_h} \xi) \cdot \partial_x (\tilde{s}_{\gamma_h, R}). \quad (4.82)$$

Note that the rescaled quantization of the subprincipal term can be absorbed into the remainder $\mathcal{O}_{L^2 \rightarrow L^2} \left(h \gamma_h^{(m-1)/2} \right)$ of (4.81) from the fact that $\tilde{s}_{\gamma_h, R}$ is bounded in $\mathcal{S}^0$ and the fact that the derivatives of $R_h(x, \sqrt{\gamma_h} \xi)$ are defined on the support of a which is compact, hence they are bounded. Combining (4.81) and (4.82), we obtain

$$\frac{i}{h} [\text{Op}_{\frac{B_h}{\sqrt{\gamma_h}}}^B(\gamma_h^{m/2} \tilde{s}_{\gamma_h, R}), \hat{H}_h] = -\gamma_h^{m/2} \text{Op}_{\frac{B_h}{\sqrt{\gamma_h}}}^B(\nabla H(\sqrt{\gamma_h} \xi) \cdot \partial_x (\tilde{s}_{\gamma_h, R})) + \mathcal{O}_{L^2 \rightarrow L^2} \left(h \gamma_h^{(m-1)/2} \right). \quad (4.83)$$

Dividing (4.77) by $\gamma_h^{m/2}$, the estimate (4.78) shows that the left-hand side of (4.77) is

$$\mathcal{O}_R \left(\frac{1}{\tau_h} \right) = o_R(1),$$

since $\tau_h \rightarrow +\infty$. Moreover, by (4.1), one has $h/\sqrt{\gamma_h} \rightarrow 0$. Hence, letting $h \rightarrow 0^+$ first and then $R \rightarrow \infty$ in (4.77), we obtain

$$\int_{\mathbb{R}} \psi(t) \left\langle -\nabla H(\xi) \cdot \partial_x a(x, \xi, \eta), \tilde{\nu}_{t, \Lambda}^B \right\rangle h_\Lambda(t) dt = 0. \quad (4.84)$$

Since (4.84) holds for all $\psi \in \mathcal{C}_c^\infty(\mathbb{R})$, we infer the infinitesimal form of the invariance of $\tilde{\nu}_{t, \Lambda}^B$ under the lifted flow $\tilde{\varphi}_H^s$. $\square$

By construction, the measure $\tilde{\nu}_{t,\Lambda}^B$ is supported, in the ξ variable, on $E_{\Lambda^\perp \setminus \{0\}} \cap \Omega_{E_1, E_2}$. Together with the invariance under the lifted geodesic flow and Lemma 3.11, this implies that, for every $k \notin \Lambda$, $\psi \in C_c^\infty(\mathbb{R})$ and $b \in C_c^\infty(\mathbb{R}^2 \times \mathbb{R})$,

$$\int_{\mathbb{R}} \psi(t) \langle e^{2i\pi k \cdot x} b(\xi, \eta), \tilde{\nu}_{t,\Lambda}^B \rangle h_\Lambda(t) dt = 0,$$

which implies that

$$\tilde{\nu}_{t,\Lambda}^B = \mathcal{I}_\Lambda(\tilde{\nu}_{t,\Lambda}^B). \quad (4.85)$$

4.4 Effective dynamics for the two-microlocal lift

To formulate the extra propagation properties of the measure $\tilde{\nu}_{t,\Lambda}^B$ in terms of an initial two-microlocal measure, we also introduce the auxiliary space-time distribution associated with the initial data. For $a \in C_c^\infty(T^*\mathbb{T}^2 \times \mathbb{R})$, we define

$$\langle \tilde{\nu}_{0,h,R,\Lambda}^B, a \rangle := \left\langle \text{Op}_h^B \left(a(x, \xi, \eta_{\gamma_h}(\xi)) \chi \left(\frac{\nabla H(\xi) \cdot \mathbf{e}_\Lambda}{RL_\Lambda \sqrt{\gamma_h}} \right) \right) u_h^{(0)}, u_h^{(0)} \right\rangle_{L^2(\mathbb{T}^2, L)}. \quad (4.86)$$

Possibly after extracting a further subsequence and arguing exactly as in the proof of the compactness results for the two-microlocal distributions, the family $(\tilde{\nu}_{0,h,R,\Lambda}^B)_{h \rightarrow 0^+, R \rightarrow +\infty}$ converges to a nonnegative Radon measure $\tilde{\nu}_{0,\Lambda}^B$ on $T^*\mathbb{T}^2 \times \mathbb{R}$. The two-microlocal measure satisfies additional transport/invariance constraints depending on the size of $\tau_h \sqrt{h}$.

Lemma 4.14. *Let X_Λ be the vector field*

$$X_\Lambda := \eta \frac{\mathbf{e}_\Lambda}{L_\Lambda} \partial_x - G_{\widehat{B}_0, \Lambda}(x, \xi) \partial_\eta, \quad (4.87)$$

where

$$G_{\widehat{B}_0, \Lambda}(x, \xi) := \text{Hess}(H)(\xi) \frac{\mathbf{e}_\Lambda}{L_\Lambda} \cdot (\nabla H(\xi)^\perp \widehat{B}_0 + \partial_x \mathcal{I}_\Lambda(\mathcal{R})(x, \xi)), \quad (4.88)$$

with the following convention $\begin{pmatrix} u_1 \\ u_2 \end{pmatrix}^\perp = \begin{pmatrix} -u_2 \\ u_1 \end{pmatrix}$. We denote by $\phi_{X_\Lambda}^t$ the flow on (x, ξ, η) generated by X_Λ , that is,

$$\dot{x}(t) = \eta(t) \frac{\mathbf{e}_\Lambda}{L_\Lambda}, \quad \dot{\xi}(t) = 0, \quad \dot{\eta}(t) = -G_{\widehat{B}_0, \Lambda}(x(t), \xi(t)). \quad (4.89)$$

Then the following holds for $\tilde{\nu}_{t,\Lambda}^B$:

1. If $\tau_h \ll h^{-1/2}$, then $\tilde{\nu}_{t,\Lambda}^B$ is transported by the free flow

$$\phi_0^t(x, \xi, \eta) := \left(x + t \eta \frac{\mathbf{e}_\Lambda}{L_\Lambda}, \xi, \eta \right), \quad \text{i.e.} \quad \tilde{\nu}_{t,\Lambda}^B = (\phi_0^t)_* \mathcal{I}_\Lambda(\tilde{\nu}_{0,\Lambda}^B). \quad (4.90)$$

2. If $\tau_h = h^{-1/2}$, then $\tilde{\nu}_{t,\Lambda}^B$ is transported by the flow $\phi_{X_\Lambda}^t$:

$$\tilde{\nu}_{t,\Lambda}^B = (\phi_{X_\Lambda}^t)_* \mathcal{I}_\Lambda(\tilde{\nu}_{0,\Lambda}^B). \quad (4.91)$$

3. If $\tau_h \gg h^{-1/2}$, then $\tilde{\nu}_{t,\Lambda}^B$ is invariant under the flow $\phi_{X_\Lambda}^s$ generated by X_Λ , namely

$$(\phi_{X_\Lambda}^s)_* \tilde{\nu}_{t,\Lambda}^B = \tilde{\nu}_{t,\Lambda}^B \quad \forall s \in \mathbb{R}. \quad (4.92)$$

This lemma shows that the evolution of $\tilde{\nu}_{t,\Lambda}^B$ is explicit in the subcritical and critical regimes $\tau_h \ll h^{-1/2}$ and $\tau_h = h^{-1/2}$, since in both cases the measure is completely determined by its initial data through the relevant transport flow. As a consequence, its projection $\nu_{t,\Lambda}^B := (\pi_{x,\xi})_* \tilde{\nu}_{t,\Lambda}^B$ onto $T^*\mathbb{T}^2$ given by (4.67) is also entirely determined by the initial two-microlocal measure $\tilde{\nu}_{0,\Lambda}^B$.

The proof of the lemma follows the same strategy as in Lemma 4.9: we test the invariance identity through a commutator argument. The Schrödinger equation (3.1) gives the integration by parts in time, while the magnetic symbolic commutator formula identifies the first-order term, namely the sum of the classical Poisson bracket and the magnetic correction.

As a consequence of Lemma 4.12, $\text{Op}_{\frac{h}{\sqrt{\gamma_h}}}^B(\tilde{s}_{\gamma_h,R})$ is an admissible pseudodifferential test operator, and we can repeat the commutator argument used in Lemma 4.9. We emphasize that the zoom scaling γ_h has been fixed (depending on the time regime of τ_h) in Definition 4.7.

Proof. Fix $\psi \in C_c^\infty(\mathbb{R})$ and $a \in C_c^\infty(T^*\mathbb{T}^2 \times \mathbb{R})$. By (4.85), it is enough to assume that $\mathcal{I}_\Lambda(a) = a$. As in the proof of Lemma 4.9, we consider the commutator

$$\int_{\mathbb{R}} \psi(t) \left\langle \left[\text{Op}_{\frac{h}{\sqrt{\gamma_h}}}^B \left(\gamma_h^{m/2} \tilde{s}_{\gamma_h,R} \right), \text{Op}_h^B(H(\xi) + hR_h(x, \xi)) \right] u_h(t\tau_h), u_h(t\tau_h) \right\rangle dt. \quad (4.93)$$

The symbols entering this commutator are uniformly bounded in the expected classes: by Lemma 4.12, the family $\gamma_h^{m/2} \tilde{s}_{\gamma_h,R}$ is uniformly bounded in $\mathcal{S}^{-m}$, whereas $H(\sqrt{\gamma_h}\xi) + hR_h(x, \sqrt{\gamma_h}\xi)$ is uniformly bounded in $\mathcal{S}^m$. Since $(u_h)_{h \rightarrow 0+}$ solves (3.1), an integration by parts in time shows that (4.93) is equal to

$$-i \frac{h\gamma_h^{m/2}}{\tau_h} \int_{\mathbb{R}} \psi'(t) \left\langle \text{Op}_{\frac{h}{\sqrt{\gamma_h}}}^B(\tilde{s}_{\gamma_h,R}) u_h(t\tau_h), u_h(t\tau_h) \right\rangle dt. \quad (4.94)$$

On the other hand, using the rescaling formula (2.16), we rewrite the commutator in the $\frac{h}{\sqrt{\gamma_h}}$ -quantization and split the commutator into its Hamiltonian and perturbative contributions:

$$\left[\text{Op}_{\frac{h}{\sqrt{\gamma_h}}}^B \left(\gamma_h^{m/2} \tilde{s}_{\gamma_h,R} \right), \text{Op}_{\frac{h}{\sqrt{\gamma_h}}}^B(H(\sqrt{\gamma_h}\xi) + hR_h(x, \sqrt{\gamma_h}\xi)) \right] = \mathcal{C}_h^{\text{Ham}} + h \mathcal{C}_h^{\text{pert}}, \quad (4.95)$$

where

$$\mathcal{C}_h^{\text{Ham}} := \left[\text{Op}_{\frac{h}{\sqrt{\gamma_h}}}^B \left(\gamma_h^{m/2} \tilde{s}_{\gamma_h,R} \right), \text{Op}_{\frac{h}{\sqrt{\gamma_h}}}^B(H(\sqrt{\gamma_h}\xi)) \right],$$

and

$$\mathcal{C}_h^{\text{pert}} := \left[\text{Op}_{\frac{h}{\sqrt{\gamma_h}}}^B \left(\gamma_h^{m/2} \tilde{s}_{\gamma_h,R} \right), \text{Op}_{\frac{h}{\sqrt{\gamma_h}}}^B(R_h(x, \sqrt{\gamma_h}\xi)) \right].$$

Since $\gamma_h^{m/2} \tilde{s}_{\gamma_h,R} \in \mathcal{S}^{-m}$ uniformly and $H(\sqrt{\gamma_h}\xi) + hR_h(x, \sqrt{\gamma_h}\xi)$ is uniformly bounded in $\mathcal{S}^m$,

the magnetic symbolic calculus in the $\frac{h}{\sqrt{\gamma_h}}$ -quantization gives, for every integer N large enough,

$$\mathcal{C}_h^{\text{Ham}} = \frac{h}{i\sqrt{\gamma_h}} \text{Op}_{\frac{h}{\sqrt{\gamma_h}}}^B \left(\left\{ \gamma_h^{m/2} \tilde{s}_{\gamma_h, R}, H(\sqrt{\gamma_h} \xi) \right\} \right) \quad (4.96)$$

$$+ \frac{\widehat{B}_0 h^2}{i\gamma_h} \text{Op}_{\frac{h}{\sqrt{\gamma_h}}}^B \left(\partial_{\xi_2} \left(\gamma_h^{m/2} \tilde{s}_{\gamma_h, R} \right) \partial_{\xi_1} (H(\sqrt{\gamma_h} \xi)) - \partial_{\xi_1} \left(\gamma_h^{m/2} \tilde{s}_{\gamma_h, R} \right) \partial_{\xi_2} (H(\sqrt{\gamma_h} \xi)) \right) \quad (4.97)$$

$$+ \text{Op}_{\frac{h}{\sqrt{\gamma_h}}}^B (\mathfrak{H}_{N, h, R}) + \text{Op}_{\frac{h}{\sqrt{\gamma_h}}}^B (\tilde{\rho}_{N, h, R}), \quad (4.98)$$

where the symbol $\mathfrak{H}_{N, h, R}$ collects the explicit odd terms of order ≥ 3 in the symbolic expansion up to order N . Here, $\tilde{\rho}_{N, h, R}$ denotes the symbolic remainder exactly as in the proof of Lemma 4.9 and satisfies for each fixed $R > 0$,

$$\tilde{\rho}_{N, h, R} = \mathcal{O}_{\mathcal{S}^0, R} \left(\left(\frac{h}{\sqrt{\gamma_h}} \right)^N \right) = \mathcal{O}_{\mathcal{S}^0, R} (h^{N/2}), \quad (4.99)$$

because Definition 4.7 implies $\gamma_h \geq h$.

1. Hamiltonian contribution. We begin with the Poisson bracket. Since $H(\sqrt{\gamma_h} \xi)$ is independent of x and since $\mathcal{I}_\Lambda(a) = a$, all the x -Fourier modes of a are parallel to $\mathbf{e}_\Lambda$. Computing the Poisson bracket explicitly, we obtain

$$\left\{ \gamma_h^{m/2} \tilde{s}_{\gamma_h, R}, H(\sqrt{\gamma_h} \xi) \right\} = -\gamma_h^{\frac{m}{2}+1} \eta_{\gamma_h}(\sqrt{\gamma_h} \xi) \frac{\mathbf{e}_\Lambda}{L_\Lambda} \cdot \partial_x(a)(x, \sqrt{\gamma_h} \xi, \eta_{\gamma_h}(\sqrt{\gamma_h} \xi)) \chi_{R\sqrt{\gamma_h}, \Lambda}(\sqrt{\gamma_h} \xi), \quad (4.100)$$

where we recall that the notation $\chi_{R\sqrt{\gamma_h}, \Lambda}$ was introduced in (4.2).

We next compute the magnetic correction in (4.97). This is where the Hessian of H appears. Using the estimate (4.73) of Lemma 4.12, one has

$$\begin{aligned} \partial_{\xi_j} \tilde{s}_{\gamma_h, R} &= \text{Hess}(H)(\sqrt{\gamma_h} \xi) \frac{\mathbf{e}_\Lambda}{L_\Lambda} \cdot e_j \partial_\eta(a)(x, \sqrt{\gamma_h} \xi, \eta_{\gamma_h}(\sqrt{\gamma_h} \xi)) \chi_{R\sqrt{\gamma_h}, \Lambda}(\sqrt{\gamma_h} \xi) \\ &\quad + \mathcal{O}_{\mathcal{S}^0, R}(\sqrt{\gamma_h}) + \mathcal{O}_{\mathcal{S}^0, R} \left(\frac{1}{R} \right), \end{aligned} \quad (4.101)$$

where (e_1, e_2) denotes the canonical basis of $\mathbb{R}^2$. Since $\partial_{\xi_j} (H(\sqrt{\gamma_h} \xi)) = \sqrt{\gamma_h} \partial_{\xi_j} H(\sqrt{\gamma_h} \xi)$, and since $\sqrt{\gamma_h} \xi$ remains in a fixed compact set on the support of $\tilde{s}_{\gamma_h, R}$, we infer from (4.101) that

$$\begin{aligned} &\partial_{\xi_2} \left(\gamma_h^{m/2} \tilde{s}_{\gamma_h, R} \right) \partial_{\xi_1} (H(\sqrt{\gamma_h} \xi)) - \partial_{\xi_1} \left(\gamma_h^{m/2} \tilde{s}_{\gamma_h, R} \right) \partial_{\xi_2} (H(\sqrt{\gamma_h} \xi)) \\ &= \gamma_h^{\frac{m+1}{2}} \left[\text{Hess}(H)(\sqrt{\gamma_h} \xi) \frac{\mathbf{e}_\Lambda}{L_\Lambda} \cdot \nabla H(\sqrt{\gamma_h} \xi)^\perp \right] \partial_\eta a(x, \sqrt{\gamma_h} \xi, \eta_{\gamma_h}(\sqrt{\gamma_h} \xi)) \chi_{R\sqrt{\gamma_h}, \Lambda}(\sqrt{\gamma_h} \xi) \\ &\quad + \mathcal{O}_{\mathcal{S}^0, R} \left(\gamma_h^{\frac{m+2}{2}} \right) + \mathcal{O}_{\mathcal{S}^0, R} \left(\frac{\gamma_h^{\frac{m+1}{2}}}{R} \right). \end{aligned} \quad (4.102)$$

It remains to control the remainder symbol $\tilde{\mathfrak{H}}_{N, h, R}$. By definition, it collects all odd terms of order $2p+1 \geq 3$ in the symbolic expansion of the commutator together with the symbolic remainder. Since $H(\sqrt{\gamma_h} \xi')$ is independent of x' , every surviving monomial in

$$\text{Op}_{\frac{h}{\sqrt{\gamma_h}}}^B (D_x, D_\xi, D_{x'}, D_{\xi'})^{2p+1} \left[\gamma_h^{m/2} \tilde{s}_{\gamma_h, R}(x, \xi) H(\sqrt{\gamma_h} \xi') \right]$$

contains at least $2p + 1$ derivatives in the ξ' variable falling on $H(\sqrt{\gamma_h}\xi')$. Hence it gains a factor $\gamma_h^{(2p+1)/2}$. Since $\tilde{s}_{\gamma_h,R}$ and all its derivatives are uniformly bounded in $\mathcal{S}^0$ for each fixed $R > 1$, we infer that, for every $p \geq 1$,

$$\left(\frac{h}{\sqrt{\gamma_h}}\right)^{2p+1} \Omega_{\frac{h}{\sqrt{\gamma_h}}B}(D_x, D_\xi, D_{x'}, D_{\xi'})^{2p+1} \left[\gamma_h^{m/2} \tilde{s}_{\gamma_h,R}(x, \xi) H(\sqrt{\gamma_h}\xi') \right] = \mathcal{O}_{\mathcal{S}^0,R} \left(h^{2p+1} \gamma_h^{m/2} \right),$$

where the constant involved depends on finitely many seminorms of H , a , and χ . In particular, using (4.99), all odd terms of order ≥ 3 are absorbed into

$$\tilde{\mathfrak{H}}_{N,h,R} := \mathfrak{H}_{N,h,R} + \tilde{\rho}_{N,h,R} = \mathcal{O}_{\mathcal{S}^0,R} \left(h^3 \gamma_h^{m/2} \right) + \mathcal{O}_{\mathcal{S}^0,R} \left(h^{N/2} \right). \quad (4.103)$$

Again, taking N large enough leads to

$$\tilde{\mathfrak{H}}_{N,h,R} = \mathcal{O}_{\mathcal{S}^0,R} \left(h^3 \gamma_h^{m/2} \right). \quad (4.104)$$

Hence, using the Calderón–Vaillancourt Theorem and combining (4.100), (4.102) and (4.104) all together in the Hamiltonian commutator, one gets

$$\begin{aligned} \mathcal{C}_h^{\text{Ham}} &= -\gamma_h^{\frac{m+1}{2}} \frac{h}{i} \text{Op}_{\frac{h}{\sqrt{\gamma_h}}}^B \left(\eta_{\gamma_h}(\sqrt{\gamma_h}\xi) \frac{\mathbf{e}_\Lambda}{L_\Lambda} \cdot \partial_x(a)(x, \sqrt{\gamma_h}\xi, \eta_{\gamma_h}(\sqrt{\gamma_h}\xi)) \chi_{R\sqrt{\gamma_h},\Lambda}(\sqrt{\gamma_h}\xi) \right) \\ &\quad + \gamma_h^{\frac{m-1}{2}} \frac{\widehat{B}_0 h^2}{i} \text{Op}_{\frac{h}{\sqrt{\gamma_h}}}^B \left(\left[\text{Hess}(H)(\sqrt{\gamma_h}\xi) \frac{\mathbf{e}_\Lambda}{L_\Lambda} \cdot \nabla H(\sqrt{\gamma_h}\xi)^\perp \right] \partial_\eta a(x, \sqrt{\gamma_h}\xi, \eta_{\gamma_h}(\sqrt{\gamma_h}\xi)) \chi_{R\sqrt{\gamma_h},\Lambda}(\sqrt{\gamma_h}\xi) \right) \\ &\quad + \mathcal{O}_{L^2 \rightarrow L^2} \left(h^2 \gamma_h^{\frac{m}{2}} \right) + \mathcal{O}_{L^2 \rightarrow L^2} \left(h^2 \frac{\gamma_h^{\frac{m-1}{2}}}{R} \right) + \mathcal{O}_{L^2 \rightarrow L^2,R} \left(h^3 \gamma_h^{m/2} \right). \end{aligned} \quad (4.105)$$

2. Perturbative contribution. We now argue similarly to treat the perturbative contribution, which is given by

$$\mathcal{C}_h^{\text{pert}} = \frac{h}{i\sqrt{\gamma_h}} \text{Op}_{\frac{h}{\sqrt{\gamma_h}}}^B \left(\left\{ \gamma_h^{m/2} \tilde{s}_{\gamma_h,R}, R_h(x, \sqrt{\gamma_h}\xi) \right\} \right) \quad (4.106)$$

$$+ \frac{\widehat{B}_0 h^2}{i\gamma_h} \text{Op}_{\frac{h}{\sqrt{\gamma_h}}}^B \left(\partial_{\xi_2} \left(\gamma_h^{m/2} \tilde{s}_{\gamma_h,R} \right) \partial_{\xi_1} (R_h(x, \sqrt{\gamma_h}\xi)) - \partial_{\xi_1} \left(\gamma_h^{m/2} \tilde{s}_{\gamma_h,R} \right) \partial_{\xi_2} (R_h(x, \sqrt{\gamma_h}\xi)) \right) \quad (4.107)$$

$$+ \text{Op}_{\frac{h}{\sqrt{\gamma_h}}}^B (\tilde{\mathfrak{R}}_{N,h,R}), \quad (4.108)$$

where $\tilde{\mathfrak{R}}_{N,h,R}$ collects the odd terms of order at least 3 together with the symbolic remainder involved in the composition formula (2.12). We first discuss the Poisson bracket in (4.106) which will give the main contribution. One has

$$\{\tilde{s}_{\gamma_h,R}, R_h(x, \sqrt{\gamma_h}\xi)\} = \partial_\xi \tilde{s}_{\gamma_h,R} \cdot \partial_x R_h(x, \sqrt{\gamma_h}\xi) - \partial_x \tilde{s}_{\gamma_h,R} \cdot \partial_\xi (R_h(x, \sqrt{\gamma_h}\xi)).$$

Using (4.75) and $\partial_\xi(R_h(x, \sqrt{\gamma_h}\xi)) = \sqrt{\gamma_h} \partial_\xi R_h(x, \sqrt{\gamma_h}\xi)$, we obtain

$$\begin{aligned} & \left\{ \gamma_h^{\frac{m}{2}} \tilde{s}_{\gamma_h, R}, R_h(x, \sqrt{\gamma_h}\xi) \right\} \\ &= \gamma_h^{\frac{m}{2}} \left[\text{Hess}(H)(\sqrt{\gamma_h}\xi) \frac{\mathbf{e}_\Lambda}{L_\Lambda} \cdot \partial_x R_h(x, \sqrt{\gamma_h}\xi) \right] \partial_\eta a(x, \sqrt{\gamma_h}\xi, \eta_{\gamma_h}(\sqrt{\gamma_h}\xi)) \chi_{R\sqrt{\gamma_h}, \Lambda}(\sqrt{\gamma_h}\xi) \\ &+ \mathcal{O}_{\mathcal{S}^0}(\gamma_h^{\frac{m+1}{2}}) + \mathcal{O}_{\mathcal{S}^0}\left(\frac{\gamma_h^{\frac{m}{2}}}{R}\right). \end{aligned} \quad (4.109)$$

We now turn to the magnetic correction in (4.107). Since R_h is uniformly bounded in $\mathcal{S}^m$ and since, on the support of $\tilde{s}_{\gamma_h, R}$ and of its derivatives, the variable $\sqrt{\gamma_h}\xi$ remains in a fixed compact set, one has

$$\partial_{\xi_j}(R_h(x, \sqrt{\gamma_h}\xi)) = \mathcal{O}_{\mathcal{S}^0, R}(\sqrt{\gamma_h}).$$

Moreover, Lemma 4.12 implies that

$$\partial_{\xi_j}(\gamma_h^{m/2} \tilde{s}_{\gamma_h, R}) = \gamma_h^{m/2} \mathcal{O}_{\mathcal{S}^0, R}(1).$$

Hence, one gets

$$\partial_{\xi_2}(\gamma_h^{m/2} \tilde{s}_{\gamma_h, R}) \partial_{\xi_1}(R_h(x, \sqrt{\gamma_h}\xi)) - \partial_{\xi_1}(\gamma_h^{m/2} \tilde{s}_{\gamma_h, R}) \partial_{\xi_2}(R_h(x, \sqrt{\gamma_h}\xi)) = \mathcal{O}_{\mathcal{S}^0, R}(\gamma_h^{\frac{m+1}{2}}). \quad (4.110)$$

Finally, the higher odd terms and the symbolic remainder collected by $\tilde{\mathfrak{R}}_{N, h, R}$ in the expansion of $\mathcal{C}_h^{\text{pert}}$ are handled exactly as for $\tilde{\mathcal{R}}_{N, h, R}$ in (4.58) and yield

$$\tilde{\mathfrak{R}}_{N, h, R} = \mathcal{O}_{\mathcal{S}^0, R}(h^3 \gamma_h^{\frac{m-3}{2}}). \quad (4.111)$$

Finally, combining (4.109), (4.110), (4.111) and using the Calderón–Vaillancourt theorem in the perturbative commutator, we obtain

$$\begin{aligned} \mathcal{C}_h^{\text{pert}} &= \frac{h\gamma_h^{\frac{m-1}{2}}}{i} \text{Op}_{\frac{B_h}{\sqrt{\gamma_h}}} \left(\text{Hess}(H)(\sqrt{\gamma_h}\xi) \frac{\mathbf{e}_\Lambda}{L_\Lambda} \cdot \partial_x R_h(x, \sqrt{\gamma_h}\xi) \partial_\eta a(x, \sqrt{\gamma_h}\xi, \eta_{\gamma_h}(\sqrt{\gamma_h}\xi)) \chi_{R\sqrt{\gamma_h}, \Lambda}(\sqrt{\gamma_h}\xi) \right) \\ &+ \mathcal{O}_{L^2 \rightarrow L^2}(h\gamma_h^{m/2}) + \mathcal{O}_{L^2 \rightarrow L^2}\left(h\frac{\gamma_h^{\frac{m-1}{2}}}{R}\right) + \mathcal{O}_{L^2 \rightarrow L^2}(h^2\gamma_h^{\frac{m-1}{2}}) + \mathcal{O}_{L^2 \rightarrow L^2, R}(h^3\gamma_h^{\frac{m-3}{2}}). \end{aligned} \quad (4.112)$$

3. Renormalized identity. Combining (4.105), (4.112) together in (4.93), linking it to (4.94),

and dividing by $\frac{h\gamma_h^{\frac{m+1}{2}}}{i}$, we obtain

$$\begin{aligned} & \frac{1}{\tau_h \sqrt{\gamma_h}} \int_{\mathbb{R}} \psi'(t) \left\langle \text{Op}_{\frac{h}{\sqrt{\gamma_h}}}^B(\tilde{s}_{\gamma_h, R}) u_h(t\tau_h), u_h(t\tau_h) \right\rangle dt \\ &= \int_{\mathbb{R}} \psi(t) \left\langle \text{Op}_{\frac{h}{\sqrt{\gamma_h}}}^B \left(-\eta_{\gamma_h}(\sqrt{\gamma_h}\xi) \frac{\mathbf{e}_\Lambda}{L_\Lambda} \cdot \partial_x a(x, \sqrt{\gamma_h}\xi, \eta_{\gamma_h}(\sqrt{\gamma_h}\xi)) \chi_{R\sqrt{\gamma_h}, \Lambda}(\sqrt{\gamma_h}\xi) \right) u_h(t\tau_h), u_h(t\tau_h) \right\rangle dt \\ &+ \frac{h}{\gamma_h} \int_{\mathbb{R}} \psi(t) \left\langle \text{Op}_{\frac{h}{\sqrt{\gamma_h}}}^B \left(\tilde{G}_h(x, \sqrt{\gamma_h}\xi) \partial_\eta a(x, \sqrt{\gamma_h}\xi, \eta_{\gamma_h}(\sqrt{\gamma_h}\xi)) \chi_{R\sqrt{\gamma_h}, \Lambda}(\sqrt{\gamma_h}\xi) \right) u_h(t\tau_h), u_h(t\tau_h) \right\rangle dt \\ &+ \mathcal{O}\left(\frac{h}{\sqrt{\gamma_h}}\right) + \mathcal{O}\left(\frac{h}{\gamma_h R}\right) + \mathcal{O}\left(\frac{h^2}{\gamma_h}\right) + \mathcal{O}_R\left(\frac{h}{\sqrt{\gamma_h}}\right) + \mathcal{O}_R\left(\frac{h^3}{\gamma_h^2}\right), \end{aligned} \quad (4.113)$$

where

$$\tilde{G}_h(x, \sqrt{\gamma_h}\xi) = \text{Hess}(H)(\sqrt{\gamma_h}\xi) \frac{\mathbf{e}_\Lambda}{L_\Lambda} \cdot \left[\widehat{B}_0 \nabla H(\sqrt{\gamma_h}\xi)^\perp + \partial_x R_h(x, \sqrt{\gamma_h}\xi) \right]. \quad (4.114)$$

From the definition (2.22) of R_h and using that $\sqrt{\gamma_h}\xi$ remains in a compact set on the support of a , one can write

$$\tilde{G}_h(x, \sqrt{\gamma_h}\xi) = \text{Hess}(H)(\sqrt{\gamma_h}\xi) \frac{\mathbf{e}_\Lambda}{L_\Lambda} \cdot \left[\widehat{B}_0 \nabla H(\sqrt{\gamma_h}\xi)^\perp + \partial_x(\mathcal{R})(x, \sqrt{\gamma_h}\xi) + \mathcal{O}_{\mathcal{S}^0}(h) \right]. \quad (4.115)$$

Hence, we finally write on the support of $a(x, \sqrt{\gamma_h}\xi)$

$$\tilde{G}_h(x, \sqrt{\gamma_h}\xi) = G_{\widehat{B}_0, \Lambda}(x, \sqrt{\gamma_h}\xi) + \mathcal{O}_{\mathcal{S}^0}(h), \quad (4.116)$$

where

$$G_{\widehat{B}_0, \Lambda}(x, \sqrt{\gamma_h}\xi) = \text{Hess}(H)(\sqrt{\gamma_h}\xi) \frac{\mathbf{e}_\Lambda}{L_\Lambda} \cdot \left[\widehat{B}_0 \nabla H(\sqrt{\gamma_h}\xi)^\perp + \partial_x(\mathcal{R})(x, \sqrt{\gamma_h}\xi) \right].$$

We may now let $h \rightarrow 0^+$ and then $R \rightarrow +\infty$, using the fact that $a = \mathcal{I}_\Lambda(a)$, Definition 4.7 of $(\gamma_h)_{h \rightarrow 0^+}$ and the definition of the two-microlocal measure $\tilde{\nu}_{t, \Lambda}^B$.

Case 1: $\tau_h \ll h^{-1/2}$. In this regime, $\gamma_h = \tau_h^{-2}$, so that

$$\frac{1}{\tau_h \sqrt{\gamma_h}} = 1, \quad \frac{h}{\gamma_h} = h\tau_h^2 \rightarrow 0.$$

Passing to the limit in (4.113), we obtain

$$\int_{\mathbb{R}} \psi'(t) \langle a, \tilde{\nu}_{t, \Lambda}^B \rangle h_\Lambda(t) dt = \int_{\mathbb{R}} \psi(t) \left\langle -\eta \frac{\mathbf{e}_\Lambda}{L_\Lambda} \cdot \partial_x(a), \tilde{\nu}_{t, \Lambda}^B \right\rangle h_\Lambda(t) dt.$$

This is the weak formulation of (4.90), hence

$$\tilde{\nu}_{t, \Lambda}^B = (\phi_0^t)_* \mathcal{I}_\Lambda(\tilde{\nu}_{0, \Lambda}^B).$$

Case 2: $\tau_h = h^{-1/2}$. Here $\gamma_h = h$, so that

$$\frac{h}{\gamma_h} = 1, \quad \tau_h \sqrt{\gamma_h} = \tau_h \sqrt{h} \rightarrow 1.$$

Passing to the limit in (4.113), we obtain

$$\int_{\mathbb{R}} \psi'(t) \langle a, \tilde{\nu}_{t,\Lambda}^B \rangle h_{\Lambda}(t) dt = \int_{\mathbb{R}} \psi(t) \left\langle -\eta \frac{\mathbf{e}_{\Lambda}}{L_{\Lambda}} \cdot \partial_x a + G_{\widehat{B}_0,\Lambda}(x, \xi) \partial_{\eta} a, \tilde{\nu}_{t,\Lambda}^B \right\rangle h_{\Lambda}(t) dt.$$

This is exactly the weak formulation of (4.91), namely

$$\tilde{\nu}_{t,\Lambda}^B = (\phi_{X_{\Lambda}}^t)_* \mathcal{I}_{\Lambda}(\tilde{\nu}_{0,\Lambda}^B).$$

Case 3: $\tau_h \gg h^{-1/2}$. Again $\gamma_h = h$, but now one has

$$\frac{1}{\tau_h \sqrt{\gamma_h}} = \frac{1}{\tau_h \sqrt{h}} \rightarrow 0, \quad \frac{h}{\gamma_h} = 1.$$

Passing to the limit in (4.113), the time derivative disappears and we are left with

$$\int_{\mathbb{R}} \psi(t) \left\langle -\eta \frac{\mathbf{e}_{\Lambda}}{L_{\Lambda}} \cdot \partial_x a + G_{\widehat{B}_0,\Lambda}(x, \xi) \partial_{\eta} a, \tilde{\nu}_{t,\Lambda}^B \right\rangle h_{\Lambda}(t) dt = 0.$$

Since ψ is arbitrary, this implies that for a.e. t ,

$$\left\langle -\eta \frac{\mathbf{e}_{\Lambda}}{L_{\Lambda}} \cdot \partial_x a + G_{\widehat{B}_0,\Lambda}(x, \xi) \partial_{\eta} a, \tilde{\nu}_{t,\Lambda}^B \right\rangle = 0 \quad \forall a \in \mathcal{C}_c^{\infty}(T^*\mathbb{T}^2 \times \mathbb{R}).$$

This is precisely the infinitesimal form of the invariance

$$(\phi_{X_{\Lambda}}^s)_* \tilde{\nu}_{t,\Lambda}^B = \tilde{\nu}_{t,\Lambda}^B, \quad \forall s \in \mathbb{R}.$$

This proves (4.92). □

5 Regularity of configuration-space quantum limits

The purpose of this section is to prove the configuration-space equidistribution part of Theorem 1.3. Combined with Lemma 3.6, this will also yield item (i) of that theorem. We shall further recover Theorem 1.1 and its specialization to the magnetic Laplacian by verifying the geometric non-vanishing condition along the periodic directions of the Hamiltonian flow.

Throughout this section, we assume that H and R_h satisfy the standing assumptions of Section 2.4, that the initial data satisfy the spectral localization condition (3.8), and that

$$\tau_h \gg h^{-1/2}.$$

Let W^B be a time-dependent magnetic semiclassical measure associated with the rescaled evolution $(u_h(t\tau_h))_{h \rightarrow 0^+}$. More precisely, W^B is a subsequential limit of the magnetic Wigner distributions $W_h^B(\tau_h)$ defined in (1.16); see also Section 3. By Lemma 3.4, it admits the time disintegration

$$W^B(dt, dx, d\xi) = \nu_t^B(dx, d\xi) \otimes dt,$$

where, for almost every $t \in \mathbb{R}$, ν_t^B is a probability measure supported in $\mathbb{T}^2 \times \Omega_{E_1, E_2}$. We denote its momentum marginal, introduced in Lemma 3.6, by

$$\lambda_t := (\pi_\xi)_* \nu_t^B.$$

Our goal is to prove the product decomposition (1.22), namely

$$\nu_t^B(dx, d\xi) = dx \otimes \lambda_t(d\xi) \quad \text{for almost every } t \in \mathbb{R}.$$

Equivalently, we shall prove that all nonzero Fourier coefficients of ν_t^B in the configuration variable vanish.

Remark 5.1. *The analysis of the compact two-microlocal measures carried out in Subsection 4.4, and summarized in Lemma 4.14, explains why $h^{-1/2}$ is the relevant threshold. Indeed, the first two assertions of Lemma 4.14 show that, in the regimes*

$$\tau_h \ll h^{-1/2} \quad \text{and} \quad \tau_h = h^{-1/2},$$

the compact two-microlocal measures are transported from their initial two-microlocal data. Consequently, there is in general no mechanism forcing their nonzero Fourier modes in the configuration variable to vanish. By contrast, when

$$\tau_h \gg h^{-1/2},$$

the third assertion of Lemma 4.14 shows that these measures become invariant under an effective flow. This is the supercritical regime considered throughout the present section.

The strategy is the following. Since $\tau_h \rightarrow +\infty$, Lemma 3.8 implies that, for almost every $t \in \mathbb{R}$, the measure ν_t^B is invariant under the Hamiltonian flow

$$\varphi_H^s(x, \xi) = (x + s\nabla H(\xi), \xi).$$

Using the resonant decomposition established in Section 3.3, and more precisely (3.53), this invariance already implies spatial equidistribution on the non-resonant region Ω_{nr} . Moreover, Lemma 4.9 shows that the non-compact resonant contributions carry only the zero Fourier mode in the configuration variable. Consequently, the only possible obstruction to spatial equidistribution is carried by the compact two-microlocal measures introduced in Lemma 4.11, associated with the resonant sets

$$E_{\Lambda^+ \setminus \{0\}}, \quad \Lambda \in \mathcal{L}_1.$$

We first make this reduction precise in Subsection 5.1. We then use, in Subsection 5.2, the invariance of the compact two-microlocal measures under the effective vector field X_Λ introduced in Lemma 4.14. Under the non-vanishing condition

$$G_{\widehat{B}_0, \Lambda}(x, \xi) \neq 0,$$

the corresponding dynamics escapes in the two-microlocal variable η , which rules out nonzero finite invariant compact measures. This yields the spatial equidistribution result of Theorem 5.3. We conclude the section by applying this criterion to the magnetic Laplacian.

5.1 Reduction to the compact two-microlocal contribution

We first prove that all nonzero Fourier modes in the configuration variable are carried by the compact two-microlocal contributions. Throughout this subsection, we only need to assume that $\tau_h \rightarrow +\infty$. By Lemma 3.8, for a.e. $t \in \mathbb{R}$, the measure ν_t^B is invariant under the Hamiltonian flow φ_H^s . We also work under the spectral localization assumption (3.8), and we assume that the corresponding energy window $[E_1, E_2]$ contains no critical value of H , see (2.20). Under these assumptions, we recall from (3.53) that, for a.e. $t \in \mathbb{R}$, one has the following identity of finite measures

$$\nu_t^B = \nu_t^B|_{\mathbb{T}^2 \times \Omega_{\text{nr}}} + \sum_{\Lambda \in \mathcal{L}_1} \mathcal{I}_\Lambda(\nu_t^B)|_{\mathbb{T}^2 \times E_{\Lambda^\perp \setminus \{0\}}}. \quad (5.1)$$

On the non-resonant region Ω_{nr} , the flow $x \mapsto x + s\nabla H(\xi)$ is dense in $\mathbb{T}^2$ for every $\xi \in \Omega_{\text{nr}}$. Hence the invariance of ν_t^B forces the conditional measure in the x -variable to be the Haar measure. Equivalently,

$$\nu_t^B|_{\mathbb{T}^2 \times \Omega_{\text{nr}}} = dx \otimes \lambda_t|_{\Omega_{\text{nr}}}, \quad (5.2)$$

where the measure λ_t is defined as $\lambda_t := (\pi_\xi)_* \nu_t^B$. Therefore, one gets

$$\nu_t^B = dx \otimes \lambda_t|_{\Omega_{\text{nr}}} + \sum_{\Lambda \in \mathcal{L}_1} \mathcal{I}_\Lambda(\nu_t^B)|_{\mathbb{T}^2 \times E_{\Lambda^\perp \setminus \{0\}}}. \quad (5.3)$$

Thus the non-resonant part is already spatially equidistributed. It remains to analyze the resonant contributions.

Lemma 5.2. *For a.e. $t \in \mathbb{R}$, the measure ν_t^B can be written in the form*

$$\nu_t^B = dx \otimes \left(\lambda_t|_{\Omega_{\text{nr}}} + \sum_{\Lambda \in \mathcal{L}_1} h^\Lambda(t) \beta_t^\Lambda \right) + \sum_{\Lambda \in \mathcal{L}_1} h_\Lambda(t) (\pi_{x,\xi})_* \tilde{\nu}_{t,\Lambda}^B|_{\mathbb{T}^2 \times E_{\Lambda^\perp \setminus \{0\}}}, \quad (5.4)$$

where, for each $\Lambda \in \mathcal{L}_1$, β_t^Λ is a finite nonnegative measure supported on $E_{\Lambda^\perp \setminus \{0\}}$ and where $\tilde{\nu}_{t,\Lambda}^B$ denotes the compact two-microlocal measure appearing in the time disintegration (4.66).

The preceding lemma shows that all terms in (5.4) are already independent of x , except possibly the weighted compact two-microlocal contributions

$$h_\Lambda(t) (\pi_{x,\xi})_* \tilde{\nu}_{t,\Lambda}^B.$$

Thus the question of spatial equidistribution is reduced to the analysis of these terms.

Proof. Fix $\Lambda \in \mathcal{L}_1$. Using the microlocal splitting of Section 4 and the time disintegrations of Lemma 4.3, one has

$$\nu_t^B dt = \nu_{t,\Lambda}^B h_\Lambda(t) dt + \nu_t^{B,\Lambda} h^\Lambda(t) dt.$$

Therefore, for a.e. t , as finite measures on $T^*\mathbb{T}^2$, one has

$$\nu_t^B = h_\Lambda(t) \nu_{t,\Lambda}^B + h^\Lambda(t) \nu_t^{B,\Lambda}.$$

Applying $\mathcal{I}_\Lambda$ and restricting to $\mathbb{T}^2 \times E_{\Lambda^\perp \setminus \{0\}}$, we obtain

$$\mathcal{I}_\Lambda(\nu_t^B)|_{\mathbb{T}^2 \times E_{\Lambda^\perp \setminus \{0\}}} = h^\Lambda(t) \mathcal{I}_\Lambda(\nu_t^{B,\Lambda})|_{\mathbb{T}^2 \times E_{\Lambda^\perp \setminus \{0\}}} + h_\Lambda(t) \mathcal{I}_\Lambda(\nu_{t,\Lambda}^B)|_{\mathbb{T}^2 \times E_{\Lambda^\perp \setminus \{0\}}}. \quad (5.5)$$

We first consider the non-compact term. By Lemma 4.9, all nonzero Λ -Fourier modes of $\nu_t^{B,\Lambda}$ vanish on $E_{\Lambda^\perp} \setminus \{0\}$. Hence, one has

$$\mathcal{I}_\Lambda(\nu_t^{B,\Lambda})|_{\mathbb{T}^2 \times E_{\Lambda^\perp} \setminus \{0\}} = \widehat{\nu}_t^{B,\Lambda}(0, d\xi)|_{\mathbb{T}^2 \times E_{\Lambda^\perp} \setminus \{0\}}.$$

Equivalently, there exists a finite nonnegative measure β_t^Λ on $E_{\Lambda^\perp} \setminus \{0\}$ such that

$$\mathcal{I}_\Lambda(\nu_t^{B,\Lambda})|_{\mathbb{T}^2 \times E_{\Lambda^\perp} \setminus \{0\}} = dx \otimes \beta_t^\Lambda. \quad (5.6)$$

We now consider the compact term. By Lemma 4.11, the compact component admits the two-microlocal lift $\widetilde{\nu}_{t,\Lambda}^B$, and by (4.67) its (x, ξ) -marginal is precisely the compact resonant component:

$$(\pi_{x,\xi})_* \widetilde{\nu}_{t,\Lambda}^B = \nu_{t,\Lambda}^B \quad \text{for } h_\Lambda(t) dt\text{-a.e. } t.$$

Moreover, by (4.85), this measure has only Λ -Fourier modes in the x -variable. Therefore, one has

$$h_\Lambda(t) \mathcal{I}_\Lambda(\nu_{t,\Lambda}^B) = h_\Lambda(t) (\pi_{x,\xi})_* \widetilde{\nu}_{t,\Lambda}^B. \quad (5.7)$$

Combining (5.5), (5.6) and (5.7), we get

$$\mathcal{I}_\Lambda(\nu_t^B)|_{\mathbb{T}^2 \times E_{\Lambda^\perp} \setminus \{0\}} = dx \otimes h^\Lambda(t) \beta_t^\Lambda + h_\Lambda(t) (\pi_{x,\xi})_* \widetilde{\nu}_{t,\Lambda}^B|_{\mathbb{T}^2 \times E_{\Lambda^\perp} \setminus \{0\}}. \quad (5.8)$$

Inserting this identity into (5.3) and summing over $\Lambda \in \mathcal{L}_1$ gives (5.4). $\square$

5.2 A geometric criterion for equidistribution

We now consider the regime

$$\tau_h \gg h^{-1/2}.$$

By Lemma 5.2, all nonzero Fourier modes of ν_t^B are carried by the weighted compact two-microlocal measures $h_\Lambda(t) \widetilde{\nu}_{t,\Lambda}^B$. Thus, to prove spatial equidistribution, it is enough to prove that these weighted measures vanish. In this regime, the third assertion of Lemma 4.14 implies that, for every $\Lambda \in \mathcal{L}_1$, the compact two-microlocal measure $\widetilde{\nu}_{t,\Lambda}^B$ is invariant, for $h_\Lambda(t) dt$ -a.e. t , under the flow generated by

$$X_\Lambda = \eta \frac{\mathbf{e}_\Lambda}{L_\Lambda} \cdot \partial_x - G_{\widehat{B}_0, \Lambda}(x, \xi) \partial_\eta, \quad (5.9)$$

where $G_{\widehat{B}_0, \Lambda}$ is defined in Lemma 4.14. We now give a criterion which prevents the existence of nonzero finite invariant measures for this flow. Recall that E_1 and E_2 were fixed so that

$$\Omega_{E_1, E_2} := \{\xi \in \mathbb{R}^2 : E_1 \leq H(\xi) \leq E_2\}$$

contains no critical point of H .

Theorem 5.3 (Equidistribution above the two-microlocal scale). *Let $(u_h)_{h \rightarrow 0+}$ be a family of solutions to (3.1). Assume that H and R_h satisfy the standing assumptions of Section 2.4, and that $(u_h^{(0)})_{h \rightarrow 0+}$ satisfies the spectral localization condition (3.8). Assume that*

$$\tau_h \gg h^{-1/2},$$

and that, for every $(\Lambda, x, \xi) \in \mathcal{L}_1 \times \mathbb{T}^2 \times (E_{\Lambda^\perp \setminus \{0\}} \cap \Omega_{E_1, E_2})$, one has

$$G_{\widehat{B}_0, \Lambda}(x, \xi) \neq 0. \quad (5.10)$$

Then every time-dependent magnetic semiclassical measure is independent of the configuration variable. More precisely, for a.e. $t \in \mathbb{R}$, one has

$$\nu_t^B(dx, d\xi) = dx \otimes \lambda_t(d\xi), \quad \lambda_t := (\pi_\xi)_* \nu_t^B. \quad (5.11)$$

Consequently, the associated configuration-space quantum limit is spatially equidistributed:

$$\nu_t = dx \quad \text{for a.e. } t.$$

If, in addition, $\tau_h \ll h^{-1}$, then one has

$$\lambda_t = \lambda_0 := (\pi_\xi)_* \nu_0^B \quad \text{for a.e. } t,$$

and therefore

$$\nu_t^B(dx, d\xi) = dx \otimes \lambda_0(d\xi).$$

This theorem proves item (i) of Theorem 1.3. The sufficient conditions stated in Theorem 1.1, as well as the specialization to the magnetic Laplacian, will be obtained below by verifying the geometric non-vanishing condition on $G_{\widehat{B}_0, \Lambda}$. We first record the elementary escape argument on which the proof of Theorem 5.3 relies.

Lemma 5.4. *Let $\Lambda \in \mathcal{L}_1$, and let μ be a finite nonnegative measure on*

$$\mathbb{T}^2 \times (E_{\Lambda^\perp \setminus \{0\}} \cap \Omega_{E_1, E_2}) \times \mathbb{R}.$$

Assume that μ is invariant under the flow generated by X_Λ , and that

$$\forall (x, \xi) \in \mathbb{T}^2 \times (E_{\Lambda^\perp \setminus \{0\}} \cap \Omega_{E_1, E_2}), \quad G_{\widehat{B}_0, \Lambda}(x, \xi) \neq 0. \quad (5.12)$$

Then, one has $\mu = 0$.

Proof. We argue by contradiction. Assume that $\mu \neq 0$. Let

$$s \mapsto (x(s), \xi(s), \eta(s))$$

be an integral curve of X_Λ . Then, one has

$$\dot{x}(s) = \eta(s) \frac{\mathbf{e}_\Lambda}{L_\Lambda}, \quad \dot{\xi}(s) = 0, \quad \dot{\eta}(s) = -G_{\widehat{B}_0, \Lambda}(x(s), \xi(s)).$$

Since $\mathbb{T}^2 \times (E_{\Lambda^\perp \setminus \{0\}} \cap \Omega_{E_1, E_2})$ is compact and $G_{\widehat{B}_0, \Lambda}$ is continuous, the assumption (5.12) implies that there exists $c_\Lambda > 0$ such that

$$|G_{\widehat{B}_0, \Lambda}(x, \xi)| \geq c_\Lambda$$

on this set. Moreover, $G_{\widehat{B}_0, \Lambda}$ has a constant sign on each connected component. We treat the restriction of the measure to a component on which $G_{\widehat{B}_0, \Lambda} < 0$; the argument for the case $G_{\widehat{B}_0, \Lambda} > 0$ is identical. On such a component, one has

$$G_{\widehat{B}_0, \Lambda}(x, \xi) \leq -c_\Lambda.$$

Therefore, along the flow of X_Λ , for every $s \geq 0$,

$$\eta(s) = \eta(0) - \int_0^s G_{\widehat{B}_0, \Lambda}(x(\sigma), \xi) d\sigma \geq \eta(0) + c_\Lambda s,$$

which implies that

$$\eta(s) \xrightarrow{s \rightarrow \infty} +\infty. \quad (5.13)$$

One can set $\mu^- := \mathbb{1}_{\{G_{\widehat{B}_0, \Lambda} < 0\}} \mu$ that is assumed to be a nonzero measure. Since μ is a nonzero finite positive measure and the set $\{G_{\widehat{B}_0, \Lambda} < 0\}$ is invariant under the flow, then the measure μ^- is an invariant finite positive measure. Hence, there exists $R_0 > 0$ such that $\mu^-(\mathbb{T}^2 \times \mathbb{R}^2 \times \{|\eta| \leq R_0\}) \geq \varepsilon > 0$. Moreover, as (5.13) holds, there exists $R_1 > 0$ such that, on $\text{supp } \mu^-$,

$$\phi_{X_\Lambda}^{s_1}(\{|\eta| \leq R_0\}) \subset \{2R_0 \leq |\eta| \leq R_1\} \text{ for } s_1 \text{ large enough.}$$

From the invariance of μ^- under $\phi_{X_\Lambda}^s$, one has,

$$\mu^-(\phi_{X_\Lambda}^{s_1}(\{|\eta| \leq R_0\})) = \mu^-(\{|\eta| \leq R_0\}) \geq \varepsilon > 0.$$

Hence, one has

$$\mu^-(\mathbb{T}^2 \times \mathbb{R}^2 \times \{|\eta| \leq R_0\} \sqcup \mathbb{T}^2 \times \mathbb{R}^2 \times \{2R_0 \leq |\eta| \leq R_1\}) \geq 2\varepsilon.$$

In that manner, one can construct a sequence (s_n, R_n) such that,

$$\mu^-\left(\mathbb{T}^2 \times \mathbb{R}^2 \times \{|\eta| \leq R_0\} \sqcup \bigsqcup_{k=1}^n \mathbb{T}^2 \times \mathbb{R}^2 \times \{2R_{k-1} \leq |\eta| \leq R_k\}\right) \geq (n+1)\varepsilon.$$

This contradicts the finiteness of μ^- on $\mathbb{T}^2 \times (E_{\Lambda^\perp \setminus \{0\}} \cap \Omega_{E_1, E_2}) \times \mathbb{R}$ and leads to $\mu^- = 0$. The same argument applied to $\mu^+ := \mathbb{1}_{\{G_{\widehat{B}_0, \Lambda} > 0\}} \mu$ gives $\mu^+ = 0$. Since $\mu = \mu^- + \mu^+$, we obtain $\mu = 0$. $\square$

We are now in position to prove Theorem 5.3.

Proof of Theorem 5.3. By Lemma 5.2, all terms in the decomposition of ν_t^B are already independent of x , except possibly the weighted compact two-microlocal contributions

$$h_\Lambda(t)(\pi_{x, \xi})_* \tilde{\nu}_{t, \Lambda}^B.$$

It is therefore enough to prove that, for every $\Lambda \in \mathcal{L}_1$,

$$h_\Lambda(t) \tilde{\nu}_{t, \Lambda}^B = 0 \quad \text{for a.e. } t.$$

Equivalently, for $h_\Lambda(t)dt$ -a.e. t , it is enough to show that $\tilde{\nu}_{t, \Lambda}^B = 0$.

Fix $\Lambda \in \mathcal{L}_1$. By Lemma 4.14, for $h_\Lambda(t)dt$ -a.e. t , the measure $\tilde{\nu}_{t, \Lambda}^B$ is invariant under the flow generated by X_Λ . Moreover, by the spectral localization assumption and by the definition of the compact two-microlocal component, it is supported in

$$\mathbb{T}^2 \times (E_{\Lambda^\perp \setminus \{0\}} \cap \Omega_{E_1, E_2}) \times \mathbb{R}.$$

By the assumption (5.10), the measure $\tilde{\nu}_{t, \Lambda}^B$ satisfies the hypotheses of Lemma 5.4 for $h_\Lambda(t)dt$ -a.e. t . Hence, one has

$$\tilde{\nu}_{t, \Lambda}^B = 0 \quad \text{for } h_\Lambda(t)dt\text{-a.e. } t.$$

Equivalently, one gets

$$h_\Lambda(t)\tilde{\nu}_{t,\Lambda}^B = 0 \quad \text{for dt-a.e. } t.$$

By Lemma 5.2, the only terms which could carry nonzero Fourier modes in x were precisely the weighted compact two-microlocal contributions. Since these terms vanish, all nonzero Fourier modes of ν_t^B vanish. Hence ν_t^B is independent of x . Since the ξ -marginal of ν_t^B is λ_t , we obtain

$$\nu_t^B(dx, d\xi) = dx \otimes \lambda_t(d\xi),$$

which proves (5.11). Projecting onto the configuration variable gives

$$\nu_t = dx \quad \text{for a.e. } t.$$

If $\tau_h \ll h^{-1}$, Lemma 3.6 gives

$$\lambda_t = \lambda_0 \quad \text{for a.e. } t,$$

and therefore

$$\nu_t^B(dx, d\xi) = dx \otimes \lambda_0(d\xi).$$

□

Corollary 5.5. *Let $(u_h)_{h \rightarrow 0^+}$ be a family of solutions to (3.1). Assume that H and R_h satisfy the standing assumptions of Section 2.4, and that the initial data $(u_h^{(0)})_{h \rightarrow 0^+}$ satisfy the spectral localization condition (3.8). Assume that*

$$\tau_h \gg h^{-1/2},$$

and that, for every $\Lambda \in \mathcal{L}_1$ and every $(x, \xi) \in \mathbb{T}^2 \times (E_{\Lambda^\perp} \setminus \{0\}) \cap \Omega_{E_1, E_2}$, one has

$$G_{\widehat{B}_0, \Lambda}(x, \xi) \neq 0. \quad (5.14)$$

Then every time-dependent configuration-space quantum limit is spatially equidistributed:

$$\nu_t = dx \quad \text{for a.e. } t.$$

In particular, for every $a \in C^\infty(\mathbb{T}^2)$ and every $\psi \in C_c^\infty(\mathbb{R})$, one has

$$\lim_{h \rightarrow 0^+} \int_{\mathbb{R} \times \mathbb{T}^2} \psi(t) a(x) |u_h(t\tau_h)|^2 dt dx = \left(\int_{\mathbb{R}} \psi(t) dt \right) \left(\int_{\mathbb{T}^2} a(x) dx \right).$$

Corollary 5.6. *Let $(u_h)_{h \rightarrow 0^+}$ be a family of solutions to (3.1). Assume that H and R_h satisfy the standing assumptions of Section 2.4, and that the initial data $(u_h^{(0)})_{h \rightarrow 0^+}$ satisfy the spectral localization condition (3.8). Assume that*

$$\tau_h \gg h^{-1/2},$$

and that $\text{Hess}(H)$ is positive definite on Ω_{E_1, E_2} . Then there exists an explicit constant $C_{H, \widehat{B}_0, E_1, E_2} > 0$, depending only on H , $\widehat{B}_0$, E_1 , and E_2 , such that, if

$$\|\partial_x \mathcal{R}\|_{L^\infty(\mathbb{T}^2 \times \Omega_{E_1, E_2})} \leq C_{H, \widehat{B}_0, E_1, E_2}, \quad (5.15)$$

then, for every $a \in C^\infty(\mathbb{T}^2)$ and every $\psi \in C_c^\infty(\mathbb{R})$, one has

$$\lim_{h \rightarrow 0^+} \int_{\mathbb{R}} \psi(t) \int_{\mathbb{T}^2} a(x) |u_h(t\tau_h, x)|^2 dx dt = \left(\int_{\mathbb{R}} \psi(t) dt \right) \left(\int_{\mathbb{T}^2} a(x) dx \right).$$

Proof. Set

$$c_H := \min_{\xi \in \Omega_{E_1, E_2}} |\nabla H(\xi)|, \quad \kappa_H := \min_{\substack{\xi \in \Omega_{E_1, E_2} \\ |v|=1}} \text{Hess}(H)(\xi)v \cdot v \text{ and } M_H := \max_{\xi \in \Omega_{E_1, E_2}} \|\text{Hess}(H)(\xi)\|.$$

By the regularity of the energy window (2.20) and the positive definiteness of $\text{Hess}(H)$, one has

$$c_H > 0, \quad \kappa_H > 0, \quad M_H < +\infty.$$

Fix $\Lambda \in \mathcal{L}_1$ and $\xi \in E_{\Lambda^\perp \setminus \{0\}} \cap \Omega_{E_1, E_2}$. Hence, one has

$$\left| \widehat{B}_0 \text{Hess}(H)(\xi) \frac{\mathbf{e}_\Lambda}{L_\Lambda} \cdot \nabla H(\xi)^\perp \right| \geq |\widehat{B}_0| c_H \kappa_H.$$

Moreover, using (5.15), one has

$$\left| \text{Hess}(H)(\xi) \frac{\mathbf{e}_\Lambda}{L_\Lambda} \cdot \partial_x \mathcal{I}_\Lambda(\mathcal{R})(x, \xi) \right| \leq M_H \|\partial_x \mathcal{R}\|_{L^\infty} \leq M_H C_{H, \widehat{B}_0, E_1, E_2}.$$

Therefore, defining

$$C_{H, \widehat{B}_0, E_1, E_2} := \frac{|\widehat{B}_0| c_H \kappa_H}{2M_H}, \quad (5.16)$$

leads to

$$|G_{\widehat{B}_0, \Lambda}(x, \xi)| \geq \frac{|\widehat{B}_0| c_H \kappa_H}{2} > 0.$$

The conclusion follows from Corollary 5.5. $\square$

5.3 Application to the magnetic Laplacian

We now explain how the preceding criterion applies to the magnetic Laplacian introduced in (2.25). By the torus-adapted magnetic Weyl quantization recalled in (2.26), the magnetic Laplacian can be quantized as

$$h^2(\mathcal{L}^B + V) = \text{Op}_h^B(|\xi|^2 + h \xi \cdot c_1(x) + h^2(c_0(x) + V(x))),$$

where, in the notation of [44, §3.3],

$$c_1(x) = -2A^{\text{per}}(x), \quad c_0(x) = |A^{\text{per}}(x)|^2.$$

Thus the magnetic Laplacian fits into the general definition of $\widehat{H}_h$; see (2.23), with

$$H(\xi) = |\xi|^2, \quad R_h(x, \xi) = \xi \cdot c_1(x) + h(c_0(x) + V(x)).$$

With the notation (2.22), the leading subprincipal contribution is therefore

$$\mathcal{R}(x, \xi) := \xi \cdot c_1(x) = -2 \xi \cdot A^{\text{per}}(x).$$

Using the fact that $\xi \in E_{\Lambda^\perp \setminus \{0\}}$ in the definition of $G_{\widehat{B}_0, \Lambda}$ gives

$$G_{\widehat{B}_0, \Lambda}(x, \xi) = -4 \left\langle \xi, \frac{\mathbf{e}_\Lambda^\perp}{L_\Lambda} \right\rangle \left(\widehat{B}_0 + \frac{\mathbf{e}_\Lambda}{L_\Lambda} \cdot \partial_x \mathcal{I}_\Lambda \left(\frac{\mathbf{e}_\Lambda^\perp}{L_\Lambda} \cdot A^{\text{per}}(x) \right) \right).$$

One can use (2.2) to compute the right term, which leads to

$$\frac{\mathfrak{e}_\Lambda}{L_\Lambda} \cdot \partial_x \mathcal{I}_\Lambda \left(\frac{\mathfrak{e}_\Lambda^\perp}{L_\Lambda} \cdot A^{\text{per}}(x) \right) = \text{rot } \mathcal{I}_\Lambda(A^{\text{per}}) = \mathcal{I}_\Lambda(B - \widehat{B}_0).$$

Since $\xi \in E_{\Lambda^\perp \setminus \{0\}}$, one has

$$G_{\widehat{B}_0, \Lambda}(x, \xi) = -4 \left\langle \xi, \frac{\mathfrak{e}_\Lambda^\perp}{L_\Lambda} \right\rangle \mathcal{I}_\Lambda(B)(x) = \mp 4 |\xi| \mathcal{I}_\Lambda(B)(x). \quad (5.17)$$

Considering the magnetic Schrödinger equation given by

$$\begin{cases} ih \partial_t u_h(t, x) = h^2 (\mathcal{L}^B + V) u_h(t, x), \\ u_h(t = 0, x) = u_h^{(0)}(x), \end{cases} \quad (5.18)$$

under the normalization

$$\|u_h^{(0)}\|_{L^2(\mathbb{T}^2, L)} = 1.$$

Since $H(\xi) = |\xi|^2$, the only critical value of H is 0. Hence any energy window $0 < E_1 < E_2$ automatically satisfies the non-criticality assumption used in Theorem 5.3. We therefore obtain the following corollary.

Corollary 5.7. *Let $(u_h)_{h \rightarrow 0^+}$ be a family of solutions of (5.18) satisfying the spectral localization assumption (3.8). Assume that*

$$\tau_h \gg h^{-1/2},$$

and that, for every $(x, \Lambda) \in \mathbb{T}^2 \times \mathcal{L}_1$,

$$\mathcal{I}_\Lambda(B)(x) \neq 0. \quad (5.19)$$

Then every time-dependent magnetic semiclassical measure is independent of the configuration variable. More precisely, for a.e. $t \in \mathbb{R}$,

$$\nu_t^B(dx, d\xi) = dx \otimes \lambda_t(d\xi), \quad \lambda_t := (\pi_\xi)_* \nu_t^B. \quad (5.20)$$

Consequently, for every $(a, \psi) \in \mathcal{C}^\infty(\mathbb{T}^2) \times \mathcal{C}_c^\infty(\mathbb{R})$, one has

$$\lim_{h \rightarrow 0^+} \int_{\mathbb{R}} \psi(t) \int_{\mathbb{T}^2} a(x) |u_h(t\tau_h, x)|^2 dx dt = \left(\int_{\mathbb{R}} \psi(t) dt \right) \left(\int_{\mathbb{T}^2} a(x) dx \right). \quad (5.21)$$

Remark 5.8. *If the magnetic field satisfies $B > 0$, then (5.19) holds. The case of the magnetic Laplacian therefore recovers exactly the same geometric control condition on B as in [44]. Similar geometric assumptions on magnetic fields already appeared in a classical kinetic setting in the work of Glass and Han-Kwan [23].*

Remark 5.9 (Connection with observability). *The equidistribution result above has a direct interpretation in terms of observability. Indeed, let $\omega \subset \mathbb{T}^2$ be a nonempty open set and let $\psi \in \mathcal{C}_c^\infty(\mathbb{R})$ be nonnegative and not identically zero. Under the assumptions of Theorem 5.3, the conclusion $\nu_t = dx$, combined with the Portmanteau Theorem [9, Chapter 1] applied to the open set $\mathbb{R} \times \omega$, implies*

$$\left(\int_{\mathbb{R}} \psi(t) dt \right) |\omega| \leq \liminf_{h \rightarrow 0^+} \int_{\mathbb{R}} \psi(t) \int_{\omega} |u_h(t\tau_h, x)|^2 dx dt.$$

Moreover, if $|\partial\omega| = 0$, then the preceding $\liminf$ is a limit and one has equality. In particular, no normalized spectrally localized sequence can become asymptotically invisible on ω at the time scale $\tau_h \gg h^{-1/2}$. By the usual contradiction argument based on semiclassical measures, this yields a semiclassical observability estimate in this regime. Such observability statements are, in turn, closely related to controllability properties of the corresponding Schrödinger equation through the HUM method introduced by Lions [36]. We refer for instance to the works of Burq–Zworski [12], Macià [38] and Anantharaman–Macià [5] for related semiclassical approaches to observability and control for Schrödinger equations. For electromagnetic Schrödinger operators on $\mathbb{T}^2$, we also mention the recent work of Le Balc’h, Niu and Sun [33], where a geometric condition involving the magnetic field is shown to play a decisive role in observability.

6 Beyond the Heisenberg time

We now turn to longer time scales and study the remaining dynamics of the limiting measures in the momentum variable. This will conclude the proof of Theorem 1.3. Throughout this section, we assume that

$$\tau_h \gg h^{-1/2},$$

and that the hypotheses of Theorem 5.3 are satisfied. In particular, the configuration-space component of the limiting measure is equidistributed. Hence there exists a measurable family $(\lambda_t)_{t \in \mathbb{R}}$ of probability measures on Ω_{E_1, E_2} such that

$$\nu_t^B(dx, d\xi) = dx \otimes \lambda_t(d\xi), \quad (6.1)$$

where $\lambda_t = (\pi_\xi)_* \nu_t^B$. Let us first recall the behavior below the Heisenberg time scale. By Theorem 5.3, if $h^{-1/2} \ll \tau_h \ll h^{-1}$, the momentum marginal does not evolve: for almost every $t \in \mathbb{R}$,

$$\nu_t^B(dx, d\xi) = dx \otimes \lambda_0(d\xi). \quad (6.2)$$

Hence, after spatial equidistribution has occurred, one cannot expect to observe a non-trivial evolution in the momentum variable before the Heisenberg time scale $\tau_h = h^{-1}$.

The purpose of this section is to describe this remaining evolution of $(\lambda_t)_{t \in \mathbb{R}}$ according to the relative size of τ_h with respect to h^{-1} . In the very long-time regime, we will further identify the conditional measures of λ_t on the energy levels of H . The first step is the following lemma, which gives the effective dynamics of the momentum marginal. It is useful to observe that the effective momentum vector field

$$\tilde{X}_H := -\hat{B}_0 \nabla H^\perp \cdot \partial_\xi$$

satisfies

$$\tilde{X}_H H = -\hat{B}_0 \nabla H^\perp \cdot \nabla H = 0,$$

so that $\tilde{X}_H$ is tangent to the energy levels of H . Consequently, the associated flow Υ_H^t preserves each energy level $\mathbb{S}_E = \{H = E\}$.

At this stage, Theorem 5.3 already yields the product structure (6.1) and proves item (i) of Theorem 1.3. The next lemma proves item (ii) and provides the invariance property needed to establish item (iii).

Lemma 6.1. *Assume that the hypotheses of Theorem 5.3 are satisfied. Then the measure λ_t satisfies the following dynamical properties.*

1. If $\tau_h = h^{-1}$, then the measure λ_t satisfies the following transport equation:

$$\partial_t(\lambda_t) = \widehat{B}_0 \nabla H^\perp(\xi) \cdot \partial_\xi(\lambda_t). \quad (6.3)$$

More precisely, if Υ_H^t denotes the flow on $\mathbb{R}^2$ defined by

$$\dot{\xi}(t) = -\widehat{B}_0 \nabla H(\xi(t))^\perp, \quad (6.4)$$

then

$$\lambda_t = (\Upsilon_H^t)_* \lambda_0 \quad (6.5)$$

for almost every $t \in \mathbb{R}$.

2. If $\tau_h \gg h^{-1}$, then, for almost every $t \in \mathbb{R}$, the measure λ_t is invariant under the flow Υ_H^s , namely

$$\lambda_t = (\Upsilon_H^s)_* \lambda_t, \quad \forall s \in \mathbb{R}. \quad (6.6)$$

Proof. Let $\psi \in C_c^\infty(\mathbb{R})$ and $a = a(\xi) \in C_c^\infty(\mathbb{R}^2)$. As in the previous sections, we study the commutator appearing in the following identity

$$-\frac{ih}{\tau_h} \int_{\mathbb{R}} \psi'(t) \langle \text{Op}_h^B(a(\xi)) u_h(t\tau_h), u_h(t\tau_h) \rangle dt = \int_{\mathbb{R}} \psi(t) \langle [\text{Op}_h^B(a), \text{Op}_h^B(H + hR_h)] u_h(t\tau_h), u_h(t\tau_h) \rangle dt. \quad (6.7)$$

The computation is simpler than in the previous two-microlocal analysis, because the observable a depends only on the momentum variable. By the magnetic commutator expansion (2.15), we obtain

$$[\text{Op}_h^B(a), \text{Op}_h^B(H + hR_h)] = \frac{h}{i} \text{Op}_h^B(\{a, H + hR_h\}) + \frac{\widehat{B}_0 h^2}{i} \text{Op}_h^B(\nabla_\xi(H + hR_h)^\perp \cdot \nabla_\xi a) + h^3 \text{Op}_h^B(r_h), \quad (6.8)$$

where $r_h \in \mathcal{S}^0$ is bounded uniformly in h (and depends on finitely many seminorms of a, H, R_h). Since $a = a(\xi)$, we have $\{a, H\} = 0$ and $\{a, hR_h\} = -h \nabla_\xi a \cdot \partial_x R_h$, so (6.8) becomes

$$[\text{Op}_h^B(a), \text{Op}_h^B(H + hR_h)] = -\frac{h^2}{i} \text{Op}_h^B(\nabla_\xi a \cdot \partial_x R_h) + \frac{\widehat{B}_0 h^2}{i} \text{Op}_h^B(\nabla_\xi H(\xi)^\perp \cdot \nabla_\xi a) + h^3 \text{Op}_h^B(\tilde{r}_h), \quad (6.9)$$

with $\tilde{r}_h \in \mathcal{S}^0$ uniformly bounded. Plugging (6.9) into (6.7) and dividing by $-ih^2$ gives

$$\begin{aligned} \frac{1}{h\tau_h} \int_{\mathbb{R}} \psi'(t) \langle \text{Op}_h^B(a) u_h(t\tau_h), u_h(t\tau_h) \rangle dt &= - \int_{\mathbb{R}} \psi(t) \langle \text{Op}_h^B(\nabla_\xi a \cdot \partial_x R_h) u_h(t\tau_h), u_h(t\tau_h) \rangle dt \\ &\quad + \widehat{B}_0 \int_{\mathbb{R}} \psi(t) \langle \text{Op}_h^B(\nabla_\xi H^\perp \cdot \nabla_\xi a) u_h(t\tau_h), u_h(t\tau_h) \rangle dt \\ &\quad + h \mathcal{O}(1), \end{aligned} \quad (6.10)$$

where $\mathcal{O}(1)$ is uniform in h (for fixed a, ψ) by the Calderón–Vaillancourt Theorem 2.2. Recall that from the definition (2.22) of R_h , one has

$$R_h(x, \xi) = \mathcal{R}(x, \xi) + \mathcal{O}_{\mathcal{S}^m}(h).$$

Since a is compactly supported, the symbol $\nabla_\xi a \cdot \partial_x R_h$ is uniformly bounded in $\mathcal{S}^0$. Therefore, by the Calderón–Vaillancourt theorem,

$$\int_{\mathbb{R}} \psi(t) \langle \text{Op}_h^B(\nabla_\xi a \cdot \partial_x R_h) u_h(t\tau_h), u_h(t\tau_h) \rangle dt = \int_{\mathbb{R}} \psi(t) \langle \text{Op}_h^B(\nabla_\xi a \cdot \partial_x \mathcal{R}) u_h(t\tau_h), u_h(t\tau_h) \rangle dt + h\mathcal{O}(1). \quad (6.11)$$

Finally, combining (6.11) and (6.10) leads to

$$\begin{aligned} \frac{1}{h\tau_h} \int_{\mathbb{R}} \psi'(t) \langle \text{Op}_h^B(a) u_h(t\tau_h), u_h(t\tau_h) \rangle dt &= - \int_{\mathbb{R}} \psi(t) \langle \text{Op}_h^B(\nabla_\xi a \cdot \partial_x \mathcal{R}) u_h(t\tau_h), u_h(t\tau_h) \rangle dt \\ &\quad + \widehat{B}_0 \int_{\mathbb{R}} \psi(t) \langle \text{Op}_h^B(\nabla_\xi H^\perp \cdot \nabla_\xi a) u_h(t\tau_h), u_h(t\tau_h) \rangle dt \\ &\quad + h\mathcal{O}(1). \end{aligned} \quad (6.12)$$

Recall that we place ourselves in the regime $\tau_h \gg h^{-1/2}$ and that from Theorem 5.3 one knows exactly the x -marginal of the measure ν_t^B that is the Lebesgue measure. Moreover, one has

$$\int_{\mathbb{R}} \psi'(t) \langle \text{Op}_h^B(a) u_h(t\tau_h), u_h(t\tau_h) \rangle dt \xrightarrow{h \rightarrow 0^+} \int_{\mathbb{R}} \psi'(t) \int_{\mathbb{R}^2} a(\xi) \lambda_t(d\xi) dt.$$

Therefore, if $\tau_h = h^{-1}$, so that $h\tau_h \rightarrow 1$, the whole left-hand side of (6.12) converges to this limit. If $\tau_h \gg h^{-1}$, then $h\tau_h \rightarrow +\infty$, and the whole left-hand side converges to 0. Moreover, since ν_t^B is Lebesgue in x , the term involving $\partial_x \mathcal{R}$ vanishes in the limit, because $\mathcal{R}$ is periodic in x and therefore $\int_{\mathbb{T}^2} \partial_x \mathcal{R}(x, \xi) dx = 0$. Hence, the right-hand side of (6.12) converges to

$$\widehat{B}_0 \int_{\mathbb{R}} \psi(t) \left[\int_{\mathbb{T}^* \mathbb{T}^2} \nabla_\xi H^\perp(\xi) \cdot \nabla_\xi a(\xi) dx \lambda_t(d\xi) \right] dt = \widehat{B}_0 \int_{\mathbb{R}} \psi(t) \left[\int_{\mathbb{R}^2} \nabla_\xi H^\perp(\xi) \cdot \nabla_\xi a(\xi) \lambda_t(d\xi) \right] dt.$$

In the regime $\tau_h = h^{-1}$, passing to the limit in (6.12) gives

$$\int_{\mathbb{R}} \psi'(t) \int_{\mathbb{R}^2} a(\xi) \lambda_t(d\xi) dt = \widehat{B}_0 \int_{\mathbb{R}} \psi(t) \int_{\mathbb{R}^2} \nabla_\xi H(\xi)^\perp \cdot \nabla_\xi a(\xi) \lambda_t(d\xi) dt. \quad (6.13)$$

This is exactly the weak formulation of (6.3).

In the regime $\tau_h \gg h^{-1}$, the left-hand side of (6.12) converges to zero. Hence, for almost every $t \in \mathbb{R}$,

$$\int_{\mathbb{R}^2} \nabla_\xi H(\xi)^\perp \cdot \nabla_\xi a(\xi) \lambda_t(d\xi) = 0, \quad \forall a \in \mathcal{C}_c^\infty(\mathbb{R}^2).$$

This is the infinitesimal invariance of λ_t under $-\widehat{B}_0 \nabla H^\perp \cdot \partial_\xi$. Equivalently, λ_t is invariant under the associated flow, which gives (6.6). $\square$

We now combine Lemma 6.1 with the energy disintegration obtained in Lemma 3.6. Recall that there exists a probability measure σ^B on $[E_1, E_2]$, depending only on the initial data, such that for almost every $t \in \mathbb{R}$ one can disintegrate

$$\nu_t^B(dx, d\xi) = \nu_{t, \xi, E}^B(dx) \otimes \lambda_{t, E}(d\xi) \otimes \sigma^B(dE),$$

where $\lambda_{t,E}$ is supported on the energy level

$$\mathbb{S}_E := \{\xi \in \mathbb{R}^2 : H(\xi) = E\}.$$

Recall from Lemma 3.6 that the marginal σ^B depends only on the initial data for every time scale. Since we are working under the assumptions of Theorem 5.3, we know that the configuration-space variable is already equidistributed. In other words, for σ^B -almost every E , for $\lambda_{t,E}$ -almost every $\xi \in \mathbb{S}_E$, and for almost every $t \in \mathbb{R}$, one has

$$\nu_{t,\xi,E}^B(dx) = dx.$$

Consequently, one has

$$\nu_t^B(dx, d\xi) = dx \otimes \lambda_{t,E}(d\xi) \otimes \sigma^B(dE). \quad (6.14)$$

We now consider the regime $\tau_h \gg h^{-1}$. By Lemma 6.1, the momentum marginal λ_t is invariant under the flow generated by

$$\tilde{X}_H(\xi) := -\hat{B}_0 \nabla H(\xi)^\perp \cdot \partial_\xi,$$

whose associated flow Υ_H^t preserves each energy level $\mathbb{S}_E$. Therefore, by uniqueness of the disintegration with respect to the energy variable, the conditional measures $\lambda_{t,E}$ satisfy, for σ^B -almost every E and almost every $t \in \mathbb{R}$,

$$(\Upsilon_E^s)_* \lambda_{t,E} = \lambda_{t,E}, \quad \forall s \in \mathbb{R}, \quad (6.15)$$

where Υ_E^s denotes the restriction to $\mathbb{S}_E$ of the flow Υ_H^s generated by $\tilde{X}_H$.

Recall that (2.1) implies $\hat{B}_0 \neq 0$. Since the levels $\mathbb{S}_E$ are regular, ∇H does not vanish on $\mathbb{S}_E$, and hence $\tilde{X}_H$ is a nonvanishing tangent vector field on $\mathbb{S}_E$. Since $\mathbb{S}_E$ is diffeomorphic to a circle, the flow of $\tilde{X}_H$ on $\mathbb{S}_E$ is periodic. Let $L_E > 0$ denote its minimal period. Choosing a base point $\xi_E \in \mathbb{S}_E$, we define

$$\psi_E : \mathbb{R}/L_E\mathbb{Z} \longrightarrow \mathbb{S}_E, \quad \psi_E(\theta) := \Upsilon_E^\theta(\xi_E). \quad (6.16)$$

Then ψ_E is a smooth diffeomorphism and conjugates the translation flow on $\mathbb{R}/L_E\mathbb{Z}$ with the flow Υ_E^s on $\mathbb{S}_E$.

Lemma 6.2. *For every $E \in [E_1, E_2]$, the map ψ_E satisfies*

$$(\psi_E)_*(\partial_\theta) = \tilde{X}_H \quad \text{on } \mathbb{S}_E.$$

Consequently, the flow Υ_E^s is conjugated by ψ_E to the translation flow $\theta \mapsto \theta + s$ on $\mathbb{R}/L_E\mathbb{Z}$.

Proof. Let $\xi \in \mathbb{S}_E$. By definition of ψ_E , there exists $\theta_0 \in \mathbb{R}/L_E\mathbb{Z}$ such that

$$\xi = \psi_E(\theta_0) = \Upsilon_E^{\theta_0}(\xi_E).$$

Then, one has

$$[(\psi_E)_* \partial_\theta](\xi) = \left. \frac{d}{d\theta} \right|_{\theta=0} \psi_E(\theta_0 + \theta) = \left. \frac{d}{d\theta} \right|_{\theta=0} \Upsilon_E^{\theta_0 + \theta}(\xi_E) = \tilde{X}_H(\Upsilon_E^{\theta_0}(\xi_E)) = \tilde{X}_H(\xi).$$

This proves the first statement. The conjugation of the flows follows immediately from the definition of ψ_E . $\square$

Theorem 6.3. *Assume that the hypotheses of Theorem 5.3 are satisfied. Assume that $\tau_h \gg h^{-1}$. Then, on the regular energy window $[E_1, E_2]$, every time-dependent semiclassical measure satisfies, for almost every $t \in \mathbb{R}$,*

$$\nu_t^B(dx, d\xi) = dx \otimes \int_{E_1}^{E_2} (\psi_E)_* \left(\frac{d\theta}{L_E} \right) \sigma^B(dE), \quad (6.17)$$

where σ^B , introduced in Lemma 3.6, depends only on the initial data. Equivalently, for every $\psi \in \mathcal{C}_c^\infty(\mathbb{R})$ and every $a \in \mathcal{C}^\infty(\mathbb{T}^2 \times \Omega_{E_1, E_2})$, one has

$$\lim_{h \rightarrow 0^+} \int_{\mathbb{R}} \psi(t) \left\langle \text{Op}_h^B(a) u_h(t\tau_h), u_h(t\tau_h) \right\rangle dt = \int_{\mathbb{R}} \psi(t) dt \int_{E_1}^{E_2} \int_0^{L_E} \int_{\mathbb{T}^2} a(x, \psi_E(\theta)) dx \frac{d\theta}{L_E} \sigma^B(dE). \quad (6.18)$$

In particular, in this regime, the limiting measure ν_t^B is independent of t on the regular energy window.

Together with Theorem 5.3, Lemma 3.6, and Lemma 6.1, this theorem completes the proof of Theorem 1.3. Indeed, item (i) follows from the time-independence of the momentum marginal below the Heisenberg scale, item (ii) follows from the transport statement in Lemma 6.1, and the present theorem proves item (iii).

Proof. By (6.15), for σ^B -almost every $E \in [E_1, E_2]$ and almost every $t \in \mathbb{R}$, the probability measure $\lambda_{t,E}$ is invariant under the flow Υ_E^s . Fix such an energy E . By Lemma 6.2, the map $\psi_E : \mathbb{R}/L_E\mathbb{Z} \rightarrow \mathbb{S}_E$ conjugates the flow Υ_E^s to the translation flow on the circle $\mathbb{R}/L_E\mathbb{Z}$. Hence the pullback measure

$$\mu_{t,E} := (\psi_E^{-1})_* \lambda_{t,E}$$

is a probability measure on $\mathbb{R}/L_E\mathbb{Z}$ invariant under all translations. The only probability measure with this property is the normalized Lebesgue measure. Thus, one gets

$$\mu_{t,E} = \frac{d\theta}{L_E},$$

or equivalently,

$$\lambda_{t,E} = (\psi_E)_* \left(\frac{d\theta}{L_E} \right).$$

Substituting this identity into the energy disintegration gives

$$\nu_t^B(dx, d\xi) = dx \otimes \int_{E_1}^{E_2} (\psi_E)_* \left(\frac{d\theta}{L_E} \right) \sigma^B(dE),$$

which is (6.17). The formulation (6.18) follows by testing this identity against a and integrating in time against ψ . $\square$

Remark 6.4. *The measure $(\psi_E)_*(d\theta/L_E)$ is the normalized invariant measure of the flow generated by $\tilde{X}_H$ on $\mathbb{S}_E$. If $d\ell_E$ denotes the Euclidean arclength measure on $\mathbb{S}_E$, then this measure can also be written as*

$$(\psi_E)_* \left(\frac{d\theta}{L_E} \right) = \frac{\frac{d\ell_E}{|\nabla H(\xi)|}}{\int_{\mathbb{S}_E} \frac{d\ell_E}{|\nabla H(\xi)|}}.$$

Thus the equidistribution is uniform in the flow parameter, or equivalently with respect to the Liouville measure induced on the energy curve. It is not, in general, the normalized arclength measure unless $|\nabla H|$ is constant on $\mathbb{S}_E$.

We conclude the section, and the article, by spelling out the consequence of the previous theorem in the model case of the magnetic Laplacian. In that case the principal Hamiltonian is $H(\xi) = |\xi|^2$, and the regular energy levels are the Euclidean circles

$$\mathbb{S}_E = \{\xi \in \mathbb{R}^2 : |\xi|^2 = E\}.$$

The effective vector field of Lemma 6.1 is then

$$\tilde{X}_H(\xi) = -\hat{B}_0 \nabla H(\xi)^\perp \cdot \partial_\xi = -2\hat{B}_0 \xi^\perp \cdot \partial_\xi,$$

so its flow is the rotation flow on each circle $\mathbb{S}_E$. Hence the invariant probability measure on $\mathbb{S}_E$ is the normalized angular measure.

Corollary 6.5 (Magnetic Laplacian). *Assume that the hypotheses of Corollary 5.7 are satisfied. If $\tau_h \gg h^{-1}$, then for every $\psi \in \mathcal{C}_c^\infty(\mathbb{R})$ and every $a \in \mathcal{C}^\infty(\mathbb{T}^2 \times \Omega_{E_1, E_2})$, one has*

$$\begin{aligned} \lim_{h \rightarrow 0^+} \int_{\mathbb{R}} \psi(t) \left\langle \text{Op}_h^B(a) u_h(t\tau_h), u_h(t\tau_h) \right\rangle dt \\ = \int_{\mathbb{R}} \psi(t) dt \int_{E_1}^{E_2} \int_{\mathbb{T}^2} \int_0^{2\pi} a(x, \sqrt{E}(\cos \theta, \sin \theta)) \frac{d\theta}{2\pi} dx \sigma^B(dE), \end{aligned} \quad (6.19)$$

where σ^B is the probability measure defined by

$$\forall f \in \mathcal{C}_c^\infty(\mathbb{R}), \quad \lim_{h \rightarrow 0^+} \left\langle f(h^2(\mathcal{L}^B + V)) u_h^{(0)}, u_h^{(0)} \right\rangle = \int_{E_1}^{E_2} f(E) \sigma^B(dE).$$

In particular, in the very long-time regime $\tau_h \gg h^{-1}$, the limiting measure is independent of t , Lebesgue in the position variable, and uniformly distributed in the angular momentum variable on each energy circle. Under the hypotheses of Corollary 5.7, one has $\mathcal{I}_\Lambda(B) \neq 0$ for every $\Lambda \in \mathcal{L}_1$. Therefore Theorem 6.3 applies. This corollary thus completes the description of the successive long-time evolution of the limiting measures associated with admissible initial data for the magnetic Laplacian on the flat two-torus: first, the measure becomes Lebesgue in the position variable; then, beyond the Heisenberg time scale, the remaining momentum component becomes uniformly distributed on each regular energy circle.

A Appendix

A.1 Functional calculus

In the first part of the appendix, we review the elliptic, spectral and functional-calculus properties of the magnetic pseudodifferential operators

$$\hat{H}_h := \text{Op}_h^B(H + hR_h)$$

that are used throughout the paper. In the usual nonmagnetic semiclassical setting, these facts are standard; we refer for instance to Dimassi–Sjöstrand [18, Chapters 7–8] and Zworski [51, §4.7, §14.3 and Appendix C]. The point of this appendix is to record that the same arguments apply to the torus-adapted magnetic Weyl quantization introduced in Section 2. In the proof, the standard adjoint formula, composition formula and Calderón–Vaillancourt theorem need to be replaced by their magnetic counterparts recalled in Morin–Rivière [44, Proposition 3.4 and Theorems 3.5, 3.7] but the overall arguments are the same. For comparison, related elliptic and spectral properties for magnetic pseudodifferential operators in $\mathbb{R}^n$ are established in Iftimie–Măntoiu–Purice [27]. We shall use the Helffer–Sjöstrand formula in the form presented in Dimassi–Sjöstrand [18, Chapter 8] and Zworski [51, §14.3.2]. The original reference is Helffer–Sjöstrand [26].

We first fix the Sobolev conventions used below. For $s \geq 0$, we define

$$\|u\|_{\mathcal{H}^s(\mathbb{T}^2; L)} := \|\mathrm{Op}_1^B(\langle \xi \rangle^s)u\|_{L^2} + \|u\|_{L^2},$$

and

$$\|u\|_{\mathcal{H}_h^s(\mathbb{T}^2; L)} := \|\mathrm{Op}_h^B(\langle \xi \rangle^s)u\|_{L^2} + \|u\|_{L^2}.$$

The corresponding spaces are the spaces of distributions for which these norms are finite and are denoted by $\mathcal{H}^s(\mathbb{T}^2; L)$ and $\mathcal{H}_h^s(\mathbb{T}^2; L)$. Equivalently, one may use any elliptic symbol of order s . For each fixed $h > 0$, the spaces $\mathcal{H}_h^s(\mathbb{T}^2; L)$ and $\mathcal{H}^s(\mathbb{T}^2; L)$ coincide as topological vector spaces, although the equivalence between the corresponding norms is not uniform as $h \rightarrow 0^+$.

Recall that, if $a_h \in \mathcal{S}^\ell$ is uniformly bounded in $\mathcal{S}^\ell$, then

$$\mathrm{Op}_h^B(a_h) : \mathcal{H}_h^s(\mathbb{T}^2; L) \rightarrow \mathcal{H}_h^{s-\ell}(\mathbb{T}^2; L)$$

is uniformly bounded. This follows directly from the magnetic composition formula and the magnetic Calderón–Vaillancourt Theorem 2.2. For even nonnegative integers $s = 2k$, this definition is equivalent to the one obtained from powers of the magnetic Laplacian used in Section 2.

Throughout the appendix, we assume that $m > 0$ and

$$H \in \mathcal{S}^m(\mathbb{R}^2), \quad R_h \in \mathcal{S}^m(\mathbb{T}^2 \times \mathbb{R}^2), \quad (\text{A.1})$$

where the seminorms of $(R_h)_{0 < h \leq h_0}$ are uniformly bounded. We also assume that H and R_h are real-valued and that H is elliptic of order m , namely there exist constants $C > 0$ and $M_0 > 0$ such that

$$|H(\xi)| \geq C\langle \xi \rangle^m, \quad |\xi| \geq M_0. \quad (\text{A.2})$$

Theorem A.1. *Let $m > 0$. Let $H \in \mathcal{S}^m(\mathbb{R}^2)$ be real-valued and elliptic of order m , and let*

$$R_h \in \mathcal{S}^m(\mathbb{T}^2 \times \mathbb{R}^2)$$

be real-valued with seminorms uniformly bounded for $h \in (0, h_0]$. Set

$$a_h = H + hR_h, \quad \hat{H}_h := \mathrm{Op}_h^B(a_h).$$

Then, after possibly reducing $h_0 > 0$, the following properties hold for every $0 < h \leq h_0$:

1. *The operator $\hat{H}_h$ is essentially self-adjoint on $\mathcal{C}^\infty(\mathbb{T}^2; L)$.*

2. Its unique self-adjoint extension has domain

$$\text{Dom}(\widehat{H}_h) = \mathcal{H}^m(\mathbb{T}^2; L).$$

3. For each fixed $h > 0$, the graph norm

$$\|u\|_{L^2} + \|\widehat{H}_h u\|_{L^2}$$

is equivalent to the standard $\mathcal{H}^m$ -norm.

4. For every $z \in \mathbb{C} \setminus \mathbb{R}$, the resolvent

$$(\widehat{H}_h - z)^{-1} : L^2(\mathbb{T}^2; L) \longrightarrow \mathcal{H}^m(\mathbb{T}^2; L)$$

is well defined.

5. For every compact set $K \subset \mathbb{C} \setminus \mathbb{R}$, there exist $h_K > 0$ and $C_K > 0$ such that

$$\|(\widehat{H}_h - z)^{-1}\|_{L^2 \rightarrow \mathcal{H}_h^m} \leq C_K, \quad 0 < h \leq h_K, \quad z \in K.$$

6. Since $\widehat{H}_h$ is self-adjoint, one also has the standard resolvent bound

$$\|(\widehat{H}_h - z)^{-1}\|_{L^2 \rightarrow L^2} \leq \frac{1}{|\Im z|}, \quad z \in \mathbb{C} \setminus \mathbb{R}.$$

Proof. The proof follows the standard elliptic argument for self-adjoint pseudodifferential operators on compact manifolds. Namely, the ellipticity of H , together with the uniform boundedness of R_h in $\mathcal{S}^m$, implies that $a_h = H + hR_h$ is elliptic of order m , uniformly for $0 < h \leq h_0$, after reducing h_0 if necessary. The usual elliptic parametrix construction gives the graph norm equivalence, the closedness of the realization on $\mathcal{H}^m$, and the uniform resolvent estimate away from the real axis. The adjoint formula gives symmetry for real-valued symbols, and the range criterion for closed symmetric operators then gives self-adjointness.

These are the standard arguments in the semiclassical calculus; see Dimassi–Sjöstrand [18, Chapters 7–8] and Zworski [51, §4.7, §14.3 and Appendix C]. In the present magnetic setting, the same proof applies with the magnetic composition formula, adjoint formula and Calderón–Vaillancourt Theorem recalled in Section 2. $\square$

Corollary A.2. *Under the same hypotheses as in Theorem A.1, the operator $\widehat{H}_h = \text{Op}_h^B(a_h)$ has compact resolvent on $L^2(\mathbb{T}^2; L)$. Hence its spectrum is real and purely discrete. More precisely, there exist a sequence of real eigenvalues $(\lambda_j(h))_{j \geq 1}$, counted with multiplicity and satisfying $|\lambda_j(h)| \rightarrow \infty$, and an orthonormal basis of eigenfunctions $(u_j(h))_{j \geq 1}$ of $L^2(\mathbb{T}^2; L)$.*

Proof. By Theorem A.1, the resolvent

$$(\widehat{H}_h - i)^{-1} : L^2(\mathbb{T}^2; L) \longrightarrow \mathcal{H}^m(\mathbb{T}^2; L)$$

is bounded. Since $m > 0$, the embedding

$$\mathcal{H}^m(\mathbb{T}^2; L) \hookrightarrow L^2(\mathbb{T}^2; L)$$

is compact by the Rellich–Kondrachov theorem. Hence $(\widehat{H}_h - i)^{-1}$ is a compact operator on $L^2(\mathbb{T}^2; L)$.

The conclusion follows from the spectral theorem for self-adjoint operators with compact resolvent; see for instance Reed–Simon [46, Chapter VIII, §3]. $\square$

By the spectral theorem, for every $u \in \text{Dom}(\widehat{H}_h)$, one has

$$\widehat{H}_h u = \sum_{j=1}^{\infty} \lambda_j(h) \langle u, u_j(h) \rangle_{L^2} u_j(h), \quad (\text{A.3})$$

with convergence in $L^2(\mathbb{T}^2; L)$. More generally, for every $f \in L^\infty(\mathbb{R})$, we define the bounded operator $f(\widehat{H}_h) : L^2(\mathbb{T}^2; L) \rightarrow L^2(\mathbb{T}^2; L)$ by

$$f(\widehat{H}_h)u := \sum_{j=1}^{\infty} f(\lambda_j(h)) \langle u, u_j(h) \rangle_{L^2} u_j(h), \quad u \in L^2(\mathbb{T}^2; L). \quad (\text{A.4})$$

A.1.1 The Helffer–Sjöstrand formula

We now recall the Helffer–Sjöstrand formula in the form needed in the article. This part is purely functional-analytic and does not use any specific property of the magnetic quantization beyond the self-adjointness of $\widehat{H}_h$.

Let $f \in C_c^\infty(\mathbb{R})$. An almost-analytic extension of f is a function $\widetilde{f} \in C_c^\infty(\mathbb{C})$ such that

$$\widetilde{f}|_{\mathbb{R}} = f$$

and, for every $N \in \mathbb{N}$,

$$\partial_{\bar{z}} \widetilde{f}(z) = \mathcal{O}_N(|\Im z|^N) \quad \text{as } \Im z \rightarrow 0,$$

where

$$\partial_{\bar{z}} := \frac{1}{2}(\partial_x + i\partial_y), \quad z = x + iy.$$

Such extensions are standard; see for instance Zworski [51, §14.3.2].

Proposition A.3 (Helffer–Sjöstrand formula). *Let $f \in C_c^\infty(\mathbb{R})$, and let $\widetilde{f}$ be an almost-analytic extension of f . Then*

$$f(\widehat{H}_h) = \frac{1}{i\pi} \int_{\mathbb{C}} \partial_{\bar{z}} \widetilde{f}(z) (\widehat{H}_h - z)^{-1} L(dz), \quad (\text{A.5})$$

where $L(dz)$ denotes the Lebesgue measure on $\mathbb{C}$. The integral converges in the operator norm on $\mathcal{L}(L^2(\mathbb{T}^2; L))$.

Proof. This is the standard Helffer–Sjöstrand formula for self-adjoint operators. We refer to Helffer–Sjöstrand [26], Dimassi–Sjöstrand [18, Chapter 8] or Zworski [51, §14.3.2]. The convergence follows from the resolvent estimate

$$\|(\widehat{H}_h - z)^{-1}\|_{L^2 \rightarrow L^2} \leq |\Im z|^{-1}$$

and from the almost-analytic decay of $\partial_{\bar{z}} \widetilde{f}(z)$ near the real axis. $\square$

Corollary A.4. *Let $\widehat{H}_h := \text{Op}_h^B(a_h)$, with*

$$a_h(x, \xi) = H(\xi) + hR_h(x, \xi),$$

under the assumptions of Theorem A.1. Let $f \in C_c^\infty(\mathbb{R})$. Then the following holds

$$f(\widehat{H}_h) = \text{Op}_h^B(f \circ a_h) + \mathcal{O}_{L^2 \rightarrow L^2}(h). \quad (\text{A.6})$$

Moreover, one has

$$f(\widehat{H}_h) = f(\text{Op}_h^B(H)) + \mathcal{O}_{L^2 \rightarrow L^2}(h) \quad \text{and} \quad \text{Op}_h^B(f \circ a_h) = f(\text{Op}_h^B(H)) + \mathcal{O}_{L^2 \rightarrow L^2}(h). \quad (\text{A.7})$$

Proof. The first estimate is the standard semiclassical functional calculus for elliptic self-adjoint pseudodifferential operators. In the nonmagnetic case, it follows from the Helffer–Sjöstrand formula and the parameter-dependent symbolic construction of the resolvent; see Dimassi–Sjöstrand [18, Chapter 8] and Zworski [51, §14.3.2, Theorem 14.9]. The same proof applies here with the magnetic Weyl product $\star_h$, using the magnetic composition formula and the magnetic Calderón–Vaillancourt Theorem recalled in Section 2. It gives

$$f(\widehat{H}_h) = \text{Op}_h^B(f \circ a_h) + h \text{Op}_h^B(r_h),$$

where $(r_h)_{0 < h \leq h_0}$ is bounded in $\mathcal{S}^0(\mathbb{T}^2 \times \mathbb{R}^2)$. Hence, by the magnetic Calderón–Vaillancourt theorem,

$$f(\widehat{H}_h) = \text{Op}_h^B(f \circ a_h) + \mathcal{O}_{L^2 \rightarrow L^2}(h).$$

This proves (A.6).

It remains to compare $f \circ a_h$ with $f \circ H$. Since $a_h = H + hR_h$, Taylor’s formula gives

$$f(a_h(x, \xi)) - f(H(\xi)) = hR_h(x, \xi) \int_0^1 f'(H(\xi) + shR_h(x, \xi)) \, ds.$$

Set

$$q_h(x, \xi) := R_h(x, \xi) \int_0^1 f'(H(\xi) + shR_h(x, \xi)) \, ds.$$

Since $H + shR_h$ is elliptic of order m , uniformly for $s \in [0, 1]$ and $0 < h \leq h_0$, the condition $H(\xi) + shR_h(x, \xi) \in \text{supp } f'$ forces ξ to remain in a fixed compact set. Hence all derivatives of $f'(H + shR_h)$ are uniformly bounded, and q_h is bounded in $\mathcal{S}^0$ and

$$f \circ a_h = f \circ H + hq_h.$$

By the magnetic Calderón–Vaillancourt theorem,

$$\text{Op}_h^B(f \circ a_h) = \text{Op}_h^B(f \circ H) + \mathcal{O}_{L^2 \rightarrow L^2}(h). \quad (\text{A.8})$$

Applying the first part of the proof to the h -independent symbol $H(\xi)$ also gives

$$f(\text{Op}_h^B(H)) = \text{Op}_h^B(f \circ H) + \mathcal{O}_{L^2 \rightarrow L^2}(h).$$

Combining this estimate with (A.6) and (A.8) proves (A.7). $\square$

A.2 Disintegration of Radon measures

We shall also use the following standard disintegration theorem. We recall it for completeness in the form needed in the paper. We refer to Baccelli–Błaszczyszyn–Karray [7, Appendix 14.D, Theorem 14.D.10] for a proof, and also to Kallenberg [31, Chapter 5] for a classical account of conditioning and disintegration.

Theorem A.5 (Disintegration of measures). *Let X and Y be locally compact Polish spaces and let W be a positive Radon measure on $X \times Y$. Assume that the marginal*

$$\omega := (\pi_Y)_* W$$

is σ -finite. Then there exists an ω -measurable family

$$(\nu_y)_{y \in Y}$$

of probability measures on X such that, for every nonnegative Borel function $F : X \times Y \rightarrow [0, +\infty]$,

$$\int_{X \times Y} F(x, y) W(dx, dy) = \int_Y \left(\int_X F(x, y) \nu_y(dx) \right) \omega(dy). \quad (\text{A.9})$$

Equivalently,

$$W(dx, dy) = \nu_y(dx) \omega(dy). \quad (\text{A.10})$$

Moreover, the family $(\nu_y)_{y \in Y}$ is unique for ω -almost every y .

Remark A.6. The measurability of the family $(\nu_y)_{y \in Y}$ means that, for every bounded Borel function $\varphi : X \rightarrow \mathbb{R}$, the map

$$y \mapsto \int_X \varphi(x) \nu_y(dx)$$

is Borel measurable. In the paper, the theorem is applied only to finite-dimensional manifolds or Borel subsets thereof, so the above assumptions are automatically satisfied.

References

- [1] N. Anantharaman, *Entropy and the localization of eigenfunctions*, Ann. of Math. (2) **168** (2008), no. 2, 435–475.
- [2] N. Anantharaman, C. Fermanian-Kammerer, and F. Macià, *Semiclassical completely integrable systems: long-time dynamics and observability via two-microlocal Wigner measures*, Amer. J. Math. **137** (2015), no. 3, 577–638.
- [3] N. Anantharaman, M. Léautaud, and F. Macià, *Wigner measures and observability for the Schrödinger equation on the disk*, Invent. Math. **206** (2016), no. 2, 485–599.
- [4] N. Anantharaman and F. Macià, *The dynamics of the Schrödinger flow from the point of view of semiclassical measures*, Amer. Math. Soc., Providence, RI, 2012, 93–116.
- [5] N. Anantharaman and F. Macià, *Semiclassical measures for the Schrödinger equation on the torus*, J. Eur. Math. Soc. (JEMS) **16** (2014), no. 6, 1253–1288.
- [6] N. Anantharaman and S. Nonnenmacher, *Half-delocalization of eigenfunctions for the Laplacian on an Anosov manifold*, Ann. Inst. Fourier (Grenoble) **57** (2007), no. 7, 2465–2523.
- [7] F. Baccelli, B. Błaszczyszyn and M. K. Karray, *Random Measures, Point Processes, and Stochastic Geometry*, HAL preprint, 2020.
- [8] C. Bardos, G. Lebeau, and J. Rauch, *Sharp sufficient conditions for the observation, control, and stabilization of waves from the boundary*, SIAM J. Control Optim. **30** (1992), no. 5, 1024–1065.
- [9] P. Billingsley, *Convergence of Probability Measures*, 2nd ed., Wiley Series in Probability and Statistics, John Wiley & Sons, 1999.

- [10] G. Boil and S. Vũ Ngoc, *Long-time dynamics of coherent states in strong magnetic fields*, American Journal of Mathematics **143** (2021), no. 6, 1747–1789.
- [11] J. Bourgain, N. Burq, and M. Zworski, *Control for Schrödinger operators on 2-tori: rough potentials*, J. Eur. Math. Soc. (JEMS) **15** (2013), no. 5, 1597–1628.
- [12] N. Burq and M. Zworski, *Geometric control in the presence of a black box*, J. Amer. Math. Soc. **17** (2004), no. 2, 443–471.
- [13] N. Burq and M. Zworski, *Control for Schrödinger operators on tori*, Math. Res. Lett. **19** (2012), no. 2, 309–324.
- [14] L. Charles, *Quantization of compact symplectic manifolds*, J. Geom. Anal. **26** (2016), no. 4, 2664–2710.
- [15] L. Charles and T. Lefeuvre, *Semiclassical defect measures of magnetic Laplacians on hyperbolic surfaces*, J. Éc. polytech. Math. **13** (2026), 593–627.
- [16] Y. Colin de Verdière, *Ergodicité et fonctions propres du laplacien*, Comm. Math. Phys. **102** (1985), no. 3, 497–502.
- [17] R. Cooke, *A Cantor–Lebesgue theorem in two dimensions*, Proc. Amer. Math. Soc. **30** (1971), 547–550.
- [18] M. Dimassi and J. Sjöstrand, *Spectral Asymptotics in the Semi-Classical Limit*, London Mathematical Society Lecture Note Series, vol. 268, Cambridge University Press, Cambridge, 1999.
- [19] S. Dyatlov and L. Jin, *Semiclassical measures on hyperbolic surfaces have full support*, Acta Math. **220** (2018), no. 2, 297–339.
- [20] S. Dyatlov, L. Jin and S. Nonnenmacher, *Control of eigenfunctions on surfaces of variable curvature*, J. Amer. Math. Soc. **35** (2022), no. 2, 361–465.
- [21] C. Fermanian-Kammerer, *Équation de la chaleur et mesures semi-classiques*, PhD thesis, Université Paris-Sud, Orsay, 1995.
- [22] C. Fermanian-Kammerer, *Mesures semi-classiques 2-microlocales*, C. R. Acad. Sci. Paris Sér. I Math. **331** (2000), no. 7, 515–518.
- [23] O. Glass and D. Han-Kwan, *On the controllability of the Vlasov–Poisson system in the presence of external force fields*, J. Differential Equations **252** (2012), no. 10, 5453–5491.
- [24] A. Hassell, *Ergodic billiards that are not quantum unique ergodic*, Ann. of Math. (2) **171** (2010), no. 1, 605–618.
- [25] B. Helffer and R. Purice, *Magnetic calculus and semiclassical trace formulas*, J. Phys. A Math. Theor. **43** (2010), no. 47, 474028.
- [26] B. Helffer and J. Sjöstrand, *Équation de Schrödinger avec champ magnétique et équation de Harper*, in *Schrödinger Operators*, Lecture Notes in Physics, vol. 345, Springer, Berlin, 1989, pp. 118–197.

- [27] V. Iftimie, M. Măntoiu, and R. Purice, *Magnetic pseudodifferential operators*, Publ. Res. Inst. Math. Sci. **43** (2007), no. 3, 585–623.
- [28] V. Iftimie, M. Măntoiu, and R. Purice, *Quantum observables as magnetic PsDO*, Rev. Roumaine Math. Pures Appl. **64** (2019), no. 2–3, 197–223.
- [29] S. Jaffard, *Contrôle interne exact des vibrations d’une plaque rectangulaire*, Portugal. Math. **47** (1990), no. 4, 423–429.
- [30] D. Jakobson, *Quantum limits on flat tori*, Ann. of Math. (2) **145** (1997), no. 2, 235–266.
- [31] O. Kallenberg, *Foundations of Modern Probability*, 2nd ed., Probability and its Applications, Springer, New York, 2002.
- [32] E. Kim and N. Miller, *Semiclassical measures on hyperbolic manifolds*, Preprint, arXiv:2503.01528 (2025).
- [33] K. Le Bal’c’h, J. Niu, and C. Sun, *Geometric condition for the observability of electromagnetic Schrödinger operators on $\mathbb{T}^2$* , J. London Math. Soc., 113(5):e70555, 2026.
- [34] G. Lebeau, *Contrôle de l’équation de Schrödinger*, J. Math. Pures Appl. (9) **71** (1992), no. 3, 267–291.
- [35] E. Lindenstrauss, *Invariant measures and arithmetic quantum unique ergodicity*, Ann. of Math. (2) **163** (2006), no. 1, 165–219.
- [36] J.-L. Lions, *Contrôlabilité exacte, perturbations et stabilisation de systèmes distribués. Tome 1*, Recherches en Mathématiques Appliquées, vol. 8, Masson, Paris, 1988.
- [37] F. Macià, *Semiclassical measures and the Schrödinger flow on Riemannian manifolds*, Nonlinearity **22** (2009), no. 5, 1003–1020.
- [38] F. Macià, *High-frequency propagation for the Schrödinger equation on the torus*, J. Funct. Anal. **258** (2010), no. 3, 933–955.
- [39] F. Macià and G. Rivière, *Concentration and non-concentration for the Schrödinger evolution on Zoll manifolds*, Comm. Math. Phys. **345** (2016), no. 3, 1019–1054.
- [40] F. Macià and G. Rivière, *Two-microlocal regularity of quasimodes on the torus*, Anal. PDE **11** (2018), no. 8, 2111–2136.
- [41] F. Macià and G. Rivière, *Observability and quantum limits for the Schrödinger equation on $\mathbb{S}^d$* , in *Probabilistic methods in geometry, topology and spectral theory*, Contemp. Math., Amer. Math. Soc., Providence, RI, 2019, 139–153.
- [42] M. Măntoiu and R. Purice, *The magnetic Weyl calculus*, J. Math. Phys. **45** (2004), no. 4, 1394–1417.
- [43] J. Marklof and Z. Rudnick, *Almost all eigenfunctions of a rational polygon are uniformly distributed*, J. Spectr. Theory **2** (2012), no. 1, 107–113.
- [44] L. Morin and G. Rivière, *Quantum unique ergodicity for magnetic Laplacians on $\mathbb{T}^2$* , to appear in Duke Math. J., arXiv:2411.18449 (2024).

-
- [45] F. Nier, *A semi-classical picture of quantum scattering*, Ann. Sci. École Norm. Sup. (4) **29** (1996), no. 2, 149–183.
 - [46] M. Reed and B. Simon, *Methods of Modern Mathematical Physics. I: Functional Analysis*, 2nd ed., Academic Press, New York, 1980.
 - [47] Z. Rudnick and P. Sarnak, *The behaviour of eigenstates of arithmetic hyperbolic manifolds*, Comm. Math. Phys. **161** (1994), no. 1, 195–213.
 - [48] A. I. Shnirel'man, *Ergodic properties of eigenfunctions*, Uspekhi Mat. Nauk **29** (1974), no. 6(180), 181–182.
 - [49] S. Zelditch, *Uniform distribution of eigenfunctions on compact hyperbolic surfaces*, Duke Math. J. **55** (1987), no. 4, 919–941.
 - [50] S. Zelditch, *On a “Quantum Chaos” theorem of R. Schrader and M. Taylor*, J. Funct. Anal. **109** (1992), no. 1, 1–21.
 - [51] M. Zworski, *Semiclassical Analysis*, Graduate Studies in Mathematics, vol. 138, American Mathematical Society, Providence, RI, 2012.
 - [52] A. Zygmund, *On Fourier coefficients and transforms of functions of two variables*, Studia Math. **50** (1974), no. 2, 189–201.